

(12) **United States Patent**
McBride et al.

(10) **Patent No.:** **US 12,404,858 B2**
(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **CATHETER BLOOD PUMP HEART ASSIST DEVICE**

(71) Applicants: **THE PENN STATE RESEARCH FOUNDATION**, University Park, PA (US); **TC1 LLC**, Pleasanton, CA (US)

(72) Inventors: **Mark W. McBride**, Bellefonte, PA (US); **Thomas M. Mallison**, State College, PA (US); **Gregory P. Dillon**, State College, PA (US); **Robert L. Campbell**, Port Matilda, PA (US); **David A. Boger**, Jenkintown, PA (US); **Stephen A. Hambric**, State College, PA (US); **Robert F. Kunz**, State College, PA (US); **James P. Runt**, State College, PA (US); **Justin M. Walsh**, Spring Mills, PA (US); **Boris Leschinsky**, Mahwah, NJ (US)

(73) Assignees: **THE PENN STATE RESEARCH FOUNDATION**, University Park, PA (US); **TC1 LLC**, Pleasanton, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 161 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/047,876**

(22) Filed: **Oct. 19, 2022**

(65) **Prior Publication Data**
US 2023/0125276 A1 Apr. 27, 2023

Related U.S. Application Data

(63) Continuation of application No. 17/084,896, filed on Oct. 30, 2020, now Pat. No. 11,708,833, which is a (Continued)

(51) **Int. Cl.**
A61M 60/13 (2021.01)
A61M 60/148 (2021.01)
(Continued)

(52) **U.S. Cl.**
CPC *F04D 3/00* (2013.01); *A61M 60/13* (2021.01); *A61M 60/148* (2021.01);
(Continued)

(58) **Field of Classification Search**
CPC A61M 1/101; A61M 1/122; A61M 1/1024; A61M 1/1034; A61M 2205/0266;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

264,134 A 9/1882 Brown et al.
1,902,418 A 3/1933 Pilgrim
(Continued)

FOREIGN PATENT DOCUMENTS

AU 2013220350 A1 9/2014
CA 2256427 A1 10/1998
(Continued)

OTHER PUBLICATIONS

Abiomed, "Impella 5.0 with the Impella Console, Circulatory Support System, Instructions for Use & Clinical Reference Manual," Jun. 2010, in 122 pages.

(Continued)

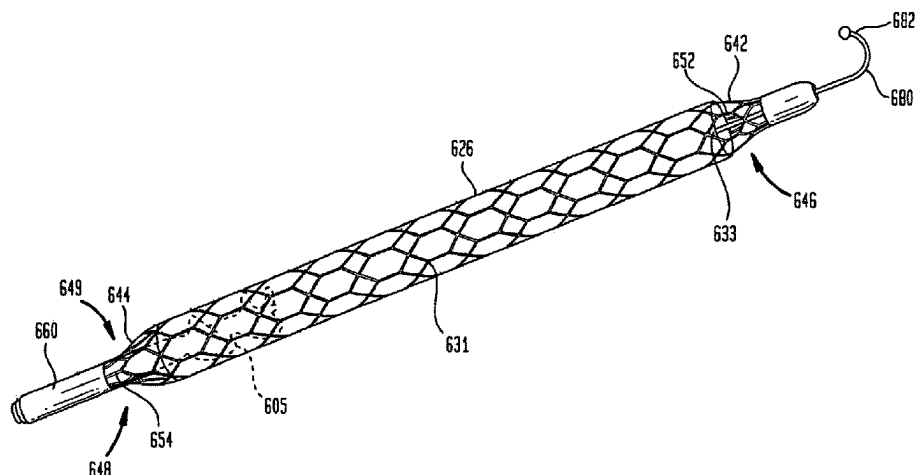
Primary Examiner — Brian T Gedeon

(74) *Attorney, Agent, or Firm* — WORKMAN NYDEGGER

(57) **ABSTRACT**

A catheter blood pump includes a cannula with an expandable central portion having a proximal end, a distal end and a blood flow conduit therebetween. A flexible inlet portion at the distal end is provided with a plurality of spaced apart inlet struts configured to prevent an obstruction from enter-

(Continued)



ing, and a flexible outlet portion is provided with a plurality of spaced apart outlet struts operative to reduce a swirl velocity of blood, wherein the plurality of spaced apart inlet struts and the plurality of spaced apart outlet struts are fabricated from a collapsible shape memory material.

19 Claims, 22 Drawing Sheets

Related U.S. Application Data

continuation of application No. 16/181,963, filed on Nov. 6, 2018, now Pat. No. 10,864,309, which is a continuation of application No. 15/633,189, filed on Jun. 26, 2017, now Pat. No. 10,149,932, which is a continuation of application No. 15/176,620, filed on Jun. 8, 2016, now Pat. No. 9,717,833, which is a continuation of application No. 13/618,071, filed on Sep. 14, 2012, now Pat. No. 9,364,593, which is a continuation of application No. 12/945,594, filed on Nov. 12, 2010, now Pat. No. 9,364,592, which is a continuation of application No. 11/728,051, filed on Mar. 23, 2007, now Pat. No. 7,841,976.

- (60) Provisional application No. 60/785,299, filed on Mar. 23, 2006, provisional application No. 60/785,531, filed on Mar. 23, 2006.

(51) Int. Cl.

A61M 60/174 (2021.01)
A61M 60/237 (2021.01)
A61M 60/414 (2021.01)
A61M 60/808 (2021.01)
A61M 60/824 (2021.01)
F04D 3/00 (2006.01)
F04D 3/02 (2006.01)
F04D 29/18 (2006.01)
F04D 29/24 (2006.01)
F04D 29/52 (2006.01)
F04D 29/54 (2006.01)
A61M 60/829 (2021.01)
B33Y 80/00 (2015.01)

(52) U.S. Cl.

CPC *A61M 60/174* (2021.01); *A61M 60/237* (2021.01); *A61M 60/414* (2021.01); *A61M 60/808* (2021.01); *A61M 60/824* (2021.01); *F04D 3/02* (2013.01); *F04D 29/181* (2013.01); *F04D 29/247* (2013.01); *F04D 29/528* (2013.01); *F04D 29/542* (2013.01); *A61M 60/829* (2021.01); *A61M 2205/0266* (2013.01); *B33Y 80/00* (2014.12); *F05B 2240/311* (2013.01); *F05B 2280/5001* (2013.01)

(58) Field of Classification Search

CPC *F04D 29/026*; *F04D 29/247*; *F04D 29/181*; *F04D 29/382*
 See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

2,356,659 A 8/1944 De Paiva
 2,649,052 A 8/1953 Weyer
 2,664,050 A 12/1953 Abresch
 2,684,035 A 7/1954 Kemp
 2,776,161 A 1/1957 Dingman et al.

2,789,511 A 4/1957 Doble
 2,896,926 A 7/1959 Chapman
 2,935,068 A 5/1960 Donaldson
 3,012,079 A 12/1961 Bruson et al.
 3,025,647 A 3/1962 Moody
 3,080,824 A 3/1963 Boyd
 3,135,943 A 6/1964 Richard
 3,455,540 A 7/1969 Marcmann
 3,510,229 A 5/1970 Smith
 3,812,812 A 5/1974 Hurwitz
 3,860,713 A 1/1975 Shema et al.
 3,860,968 A 1/1975 Shapiro
 3,904,901 A 9/1975 Renard et al.
 3,995,617 A 12/1976 Watkins et al.
 4,066,556 A 1/1978 Vaillancourt
 4,105,016 A 8/1978 Donovan, Jr.
 4,115,040 A 9/1978 Knorr
 4,129,129 A 12/1978 Amrine
 4,135,253 A 1/1979 Reich et al.
 4,143,425 A 3/1979 Runge
 4,149,535 A 4/1979 Volder
 4,155,040 A 5/1979 Ackerman et al.
 4,304,524 A 12/1981 Coxon
 D264,134 S 4/1982 Xanthopoulos
 4,382,199 A 5/1983 Isaacson
 4,392,836 A 7/1983 Sugawara
 4,458,366 A 7/1984 MacGregor
 4,537,561 A 8/1985 Xanthopoulos
 4,540,402 A 9/1985 Aigner
 4,560,375 A 12/1985 Schulte et al.
 4,589,822 A 5/1986 Clausen et al.
 4,625,712 A 12/1986 Wampler
 4,655,745 A 4/1987 Corbett
 4,673,334 A 6/1987 Allington et al.
 4,675,361 A 6/1987 Ward, Jr.
 4,682,015 A 7/1987 Quan
 4,686,982 A 8/1987 Nash
 4,696,667 A 9/1987 Masch
 4,704,121 A 11/1987 Moise
 4,728,319 A 3/1988 Masch
 4,753,221 A 6/1988 Kensley et al.
 4,769,006 A 9/1988 Papantonakos
 4,777,951 A 10/1988 Cribier et al.
 4,790,315 A 12/1988 Mueller et al.
 4,817,586 A 4/1989 Wampler
 4,819,653 A 4/1989 Marks
 4,846,152 A 7/1989 Wampler et al.
 4,895,557 A 1/1990 Moise et al.
 4,900,227 A 2/1990 Trouplin
 4,902,272 A 2/1990 Milder et al.
 4,906,229 A 3/1990 Wampler
 4,908,012 A 3/1990 Moise et al.
 4,919,647 A 4/1990 Nash
 4,927,407 A 5/1990 Dorman
 4,930,341 A 6/1990 Euteneuer
 4,944,722 A 7/1990 Carriker et al.
 4,944,748 A 7/1990 Bramm et al.
 4,955,856 A 9/1990 Phillips
 4,957,504 A 9/1990 Chardack
 4,964,864 A 10/1990 Summers et al.
 4,968,293 A 11/1990 Nelson
 4,969,865 A 11/1990 Hwang et al.
 4,976,270 A 12/1990 Parl et al.
 4,985,014 A 1/1991 Orejola
 4,994,017 A 2/1991 Yozu
 4,995,857 A 2/1991 Arnold
 5,000,177 A 3/1991 Hoffmann et al.
 5,021,048 A 6/1991 Buckholtz
 5,044,902 A 9/1991 Malbec
 5,045,072 A 9/1991 Castillo et al.
 5,049,134 A 9/1991 Golding et al.
 5,059,174 A 10/1991 Vaillancourt
 5,061,256 A 10/1991 Wampler
 5,074,756 A 12/1991 Davis
 5,089,016 A 2/1992 Millner et al.
 5,092,844 A 3/1992 Schwartz et al.
 5,098,256 A 3/1992 Smith
 5,106,368 A 4/1992 Uldall et al.
 5,112,200 A 5/1992 Isaacson et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

5,112,292 A	5/1992	Hwang et al.	5,807,311 A	9/1998	Palestrant
5,112,349 A	5/1992	Summers et al.	5,814,011 A	9/1998	Corace
5,129,883 A	7/1992	Black	5,824,070 A	10/1998	Jarvik
5,142,155 A	8/1992	Mauze et al.	5,851,174 A	12/1998	Jarvik et al.
5,145,637 A	9/1992	Richardson et al.	5,859,482 A	1/1999	Crowell et al.
5,147,186 A	9/1992	Buckholtz	5,868,702 A	2/1999	Stevens et al.
5,147,187 A	9/1992	Ito et al.	5,868,703 A	2/1999	Bertolero et al.
5,153,910 A	10/1992	Mickelson et al.	5,888,241 A	3/1999	Jarvik
5,163,910 A	11/1992	Schwartz et al.	5,888,242 A	3/1999	Antaki et al.
5,169,378 A	12/1992	Figuera	5,895,557 A	4/1999	Kronzer
5,171,212 A	12/1992	Buck et al.	5,904,668 A	5/1999	Hyman et al.
5,190,528 A	3/1993	Fonger et al.	5,911,685 A	6/1999	Siess et al.
5,195,960 A	3/1993	Hossain et al.	5,921,913 A	7/1999	Siess
5,201,679 A	4/1993	Velte, Jr. et al.	5,927,956 A	7/1999	Lim et al.
5,211,546 A	5/1993	Isaacson et al.	5,941,813 A	8/1999	Sievers et al.
5,221,270 A	6/1993	Parker	5,951,263 A	9/1999	Taylor et al.
5,234,407 A	8/1993	Teirstein et al.	5,957,941 A	9/1999	Ream
5,234,416 A	8/1993	Macaulay et al.	5,964,694 A	10/1999	Siess et al.
5,282,787 A	2/1994	Worrich	5,993,420 A	11/1999	Hyman et al.
5,290,227 A	3/1994	Pasque	6,007,478 A	12/1999	Siess et al.
5,300,112 A	4/1994	Barr	6,007,479 A	12/1999	Rottenberg et al.
5,308,354 A	5/1994	Zacca et al.	6,015,272 A	1/2000	Antaki et al.
5,312,341 A	5/1994	Turi	6,015,434 A	1/2000	Yamane
5,324,177 A	6/1994	Golding et al.	6,018,208 A	1/2000	Maher et al.
5,344,443 A	9/1994	Palma et al.	6,022,363 A	2/2000	Walker et al.
5,346,458 A	9/1994	Affeld	6,027,863 A	2/2000	Donadio, III
5,346,568 A	9/1994	Gsellmann	6,053,705 A	4/2000	Schob et al.
5,360,317 A	11/1994	Clausen et al.	6,056,705 A	5/2000	Stigar-Brown
5,364,342 A	11/1994	Beuchat et al.	6,056,719 A	5/2000	Mickley
5,376,114 A	12/1994	Jarvik	6,058,593 A	5/2000	Siess
5,393,197 A	2/1995	Lemont et al.	6,059,760 A	5/2000	Sandmore et al.
5,393,207 A	2/1995	Maher et al.	6,068,610 A	5/2000	Ellis et al.
5,397,222 A	3/1995	Moss et al.	6,071,093 A	6/2000	Hart
5,405,341 A	4/1995	Martin	6,083,260 A	7/2000	Aboul-Hosn
5,405,383 A	4/1995	Barr	6,086,527 A	7/2000	Talpade
5,415,637 A	5/1995	Khosravi	6,086,570 A	7/2000	Aboul-Hosn et al.
5,437,541 A	8/1995	Vainrub	6,093,001 A	7/2000	Burgreen et al.
5,449,342 A	9/1995	Hirose et al.	6,106,494 A	8/2000	Saravia et al.
5,458,459 A	10/1995	Hubbard et al.	6,109,895 A	8/2000	Ray et al.
5,490,763 A	2/1996	Abrams et al.	6,113,536 A	9/2000	Aboul-Hosn et al.
5,505,701 A	4/1996	Anaya Fernandez de Lomana	6,123,659 A	9/2000	le Blanc et al.
5,514,154 A	5/1996	Lau et al.	6,123,725 A	9/2000	Aboul-Hosn
5,527,159 A	6/1996	Bozeman, Jr. et al.	6,132,363 A	10/2000	Freed et al.
5,533,957 A	7/1996	Aldea	6,135,943 A	10/2000	Yu
5,534,287 A	7/1996	Lukic	6,136,025 A	10/2000	Barbut et al.
5,554,114 A	9/1996	Wallace et al.	6,139,487 A	10/2000	Siess
5,586,868 A	12/1996	Lawless et al.	6,152,704 A	11/2000	Aboul-Hosn et al.
5,588,812 A	12/1996	Taylor et al.	6,162,194 A	12/2000	Shipp
5,601,418 A	2/1997	Ohara et al.	6,176,822 B1	1/2001	Nix et al.
5,601,420 A	2/1997	Warner et al.	6,176,848 B1	1/2001	Rau et al.
5,609,574 A	3/1997	Kaplan et al.	6,178,922 B1	1/2001	Denesuk et al.
5,613,476 A	3/1997	Oi et al.	6,183,412 B1	2/2001	Benkowski et al.
5,613,935 A	3/1997	Jarvik	6,186,665 B1	2/2001	Maher et al.
5,643,226 A	7/1997	Cosgrove et al.	6,190,304 B1	2/2001	Downey et al.
5,678,306 A	10/1997	Bozeman, Jr. et al.	6,190,357 B1	2/2001	Ferrari et al.
5,692,882 A	12/1997	Bozeman, Jr. et al.	6,190,537 B1	2/2001	Kanataev et al.
5,702,418 A	12/1997	Ravenscroft	6,200,260 B1	3/2001	Bolling
5,704,926 A	1/1998	Sutton	6,203,528 B1	3/2001	Deckert et al.
5,707,218 A	1/1998	Maher et al.	6,210,133 B1	4/2001	Aboul-Hosn et al.
5,722,930 A	3/1998	Larson, Jr. et al.	6,210,318 B1	4/2001	Lederman
5,725,513 A	3/1998	Ju et al.	6,210,397 B1	4/2001	Aboul-Hosn et al.
5,725,570 A	3/1998	Heath	6,214,846 B1	4/2001	Elliott
5,730,628 A	3/1998	Hawkins	6,217,541 B1	4/2001	Yu
5,735,897 A	4/1998	Buirge	6,221,100 B1	4/2001	Strecker
5,738,649 A	4/1998	Macoviak	6,227,797 B1	5/2001	Watterson et al.
5,741,234 A	4/1998	Aboul-Hosn	6,228,063 B1	5/2001	Aboul-Hosn
5,741,429 A	4/1998	Donadio, III et al.	6,234,960 B1	5/2001	Aboul-Hosn et al.
5,746,709 A	5/1998	Rom et al.	6,234,995 B1	5/2001	Peacock, III
5,749,855 A	5/1998	Reitan	6,244,835 B1	6/2001	Antaki et al.
5,755,784 A	5/1998	Jarvik	6,245,007 B1	6/2001	Bedingham et al.
5,775,190 A	7/1998	Terai	6,245,026 B1	6/2001	Campbell et al.
5,776,111 A	7/1998	Tesio	6,247,892 B1	6/2001	Kazatchkov et al.
5,776,161 A	7/1998	Globerman	6,248,091 B1	6/2001	Voelker
5,776,190 A	7/1998	Jarvik	6,254,359 B1	7/2001	Aber
5,779,721 A	7/1998	Nash	6,254,564 B1	7/2001	Wilk et al.
			6,287,319 B1	9/2001	Aboul-Hosn et al.
			6,287,336 B1	9/2001	Globerman et al.
			6,295,877 B1	10/2001	Aboul-Hosn et al.
			6,299,635 B1	10/2001	Frantzen

(56)

References Cited

U.S. PATENT DOCUMENTS

6,305,962	B1	10/2001	Maier et al.	7,070,555	B2	7/2006	Siess
6,387,037	B1	5/2002	Bolling et al.	7,122,019	B1	10/2006	Kesten et al.
6,395,026	B1	5/2002	Aboul-Hosn et al.	7,125,376	B2	10/2006	Viole et al.
6,413,222	B1	7/2002	Pantages et al.	7,144,365	B2	12/2006	Bolling et al.
6,422,990	B1	7/2002	Prem	7,150,711	B2	12/2006	Nusser et al.
6,425,007	B1	7/2002	Messinger	7,160,243	B2	1/2007	Medvedev
6,428,464	B1	8/2002	Bolling	7,172,551	B2	2/2007	Leasure
6,447,441	B1	9/2002	Yu et al.	7,175,588	B2	2/2007	Morello
6,454,775	B1	9/2002	Demarais et al.	7,214,038	B2	5/2007	Saxer et al.
6,468,298	B1	10/2002	Pelton	7,229,258	B2	6/2007	Wood et al.
6,494,694	B2	12/2002	Lawless et al.	7,238,010	B2	7/2007	Hershberger et al.
6,503,224	B1	1/2003	Forman et al.	7,241,257	B1	7/2007	Ainsworth et al.
6,508,777	B1	1/2003	Macoviak et al.	7,262,531	B2	8/2007	Li et al.
6,508,787	B2	1/2003	Erbel et al.	7,264,606	B2	9/2007	Jarvik et al.
6,517,315	B2	2/2003	Belady	7,267,667	B2	9/2007	Houde et al.
6,517,528	B1	2/2003	Pantages et al.	7,284,956	B2	10/2007	Nose et al.
6,527,699	B1	3/2003	Goldowsky	7,288,111	B1	10/2007	Holloway et al.
6,532,964	B2	3/2003	Aboul-Hosn et al.	7,290,929	B2	11/2007	Smith et al.
6,533,716	B1	3/2003	Schmitz-Rode et al.	7,329,236	B2	2/2008	Kesten et al.
6,544,216	B1	4/2003	Sammler et al.	7,331,921	B2	2/2008	Viole et al.
6,547,519	B2	4/2003	deBlanc et al.	7,335,192	B2	2/2008	Keren et al.
6,565,588	B1	5/2003	Clement et al.	7,341,570	B2	3/2008	Keren et al.
6,565,598	B1	5/2003	Lootz	7,381,179	B2	6/2008	Aboul-Hosn et al.
6,572,349	B2	6/2003	Sorensen et al.	7,393,181	B2	7/2008	McBride et al.
6,609,883	B2	8/2003	Woodard et al.	7,393,189	B2	7/2008	Davis et al.
6,610,004	B2	8/2003	Viole et al.	7,469,716	B2	12/2008	Parrino et al.
6,613,008	B2	9/2003	Aboul-Hosn et al.	7,478,999	B2	1/2009	Limoges
6,616,323	B2	9/2003	McGill	7,491,163	B2	2/2009	Viole et al.
6,623,420	B2	9/2003	Reich et al.	7,534,258	B2	5/2009	Gomez et al.
6,623,475	B1	9/2003	Siess	7,547,200	B2	6/2009	O'Mahony et al.
6,641,093	B2	11/2003	Coudrais	7,589,441	B2	9/2009	Kalsi et al.
6,641,558	B1	11/2003	Aboul-Hosn et al.	7,605,298	B2	10/2009	Bechert et al.
6,645,241	B1	11/2003	Strecker	7,619,560	B2	11/2009	Penna et al.
6,652,548	B2	11/2003	Evans et al.	7,632,079	B2	12/2009	Hershberger et al.
6,660,014	B2	12/2003	Demarais et al.	7,633,193	B2	12/2009	Masoudipour et al.
6,673,105	B1	1/2004	Chen	7,645,225	B2	1/2010	Medvedev et al.
6,692,318	B2	2/2004	McBride	7,657,324	B2	2/2010	Westlund et al.
6,702,830	B1	3/2004	Demarais et al.	7,682,673	B2	3/2010	Houston et al.
6,709,418	B1	3/2004	Aboul-Hosn et al.	7,722,568	B2	5/2010	Lenker et al.
6,716,189	B1	4/2004	Jarvik et al.	7,731,675	B2	6/2010	Aboul-Hosn et al.
6,749,598	B1	6/2004	Keren et al.	7,736,296	B2	6/2010	Siess et al.
6,776,578	B2	8/2004	Belady	7,744,566	B2	6/2010	Pirovano et al.
6,776,794	B1	8/2004	Hong et al.	7,758,521	B2	7/2010	Morris et al.
6,783,328	B2	8/2004	Lucke et al.	7,759,521	B2	7/2010	Bleuel
6,790,171	B1	9/2004	Grundeman et al.	7,766,892	B2	8/2010	Keren et al.
6,794,784	B2	9/2004	Takahashi et al.	7,780,628	B1	8/2010	Keren et al.
6,794,789	B2	9/2004	Siess et al.	7,785,246	B2	8/2010	Aboul-Hosn et al.
6,814,713	B2	11/2004	Aboul-Hosn et al.	7,811,279	B2	10/2010	John
6,817,836	B2	11/2004	Nose et al.	7,819,833	B2	10/2010	Ainsworth et al.
6,818,001	B2	11/2004	Wulfman et al.	7,820,205	B2	10/2010	Takakusagi et al.
6,835,049	B2	12/2004	Ray	7,828,710	B2	11/2010	Shifflette
6,860,713	B2	3/2005	Hoover	7,841,976	B2	11/2010	McBride et al.
6,866,625	B1	3/2005	Ayre et al.	7,842,976	B2	11/2010	Fujii et al.
6,866,805	B2	3/2005	Hong et al.	7,852,996	B2	12/2010	Lemke
6,887,215	B2	5/2005	McWeeney	7,855,316	B2	12/2010	Meyer et al.
6,889,082	B2	5/2005	Bolling et al.	7,878,967	B1	2/2011	Khanal
6,901,289	B2	5/2005	Dahl et al.	7,918,828	B2	4/2011	Lundgaard et al.
6,926,662	B1	8/2005	Aboul-Hosn et al.	7,927,068	B2	4/2011	McBride et al.
6,935,344	B1	8/2005	Aboul-Hosn et al.	7,934,912	B2	5/2011	Voltenburg et al.
6,939,377	B2	9/2005	Jayaraman et al.	7,935,102	B2	5/2011	Breznock et al.
6,942,611	B2	9/2005	Siess	7,942,804	B2	5/2011	Khaw
6,949,066	B2	9/2005	Bearnson et al.	7,942,844	B2	5/2011	Moberg et al.
6,962,488	B2	11/2005	Davis et al.	7,955,365	B2	6/2011	Doty
6,966,748	B2	11/2005	Woodard et al.	7,993,259	B2	8/2011	Kang et al.
6,972,956	B2	12/2005	Franz et al.	7,998,054	B2	8/2011	Bolling
6,974,436	B1	12/2005	Aboul-Hosn et al.	7,998,190	B2	8/2011	Gharib et al.
6,981,942	B2	1/2006	Khaw et al.	8,012,079	B2	9/2011	Delgado, III
6,984,392	B2	1/2006	Bechert et al.	8,025,647	B2	9/2011	Siess et al.
7,010,954	B2	3/2006	Siess et al.	8,052,399	B2	11/2011	Stemple et al.
7,011,620	B1	3/2006	Siess	8,062,008	B2	11/2011	Voltenburg et al.
7,014,417	B2	3/2006	Salomon	8,079,948	B2	12/2011	Shifflette
7,018,182	B2	3/2006	O'Mahony et al.	8,083,503	B2	12/2011	Voltenburg et al.
7,022,100	B1	4/2006	Aboul-Hosn et al.	8,110,267	B2	2/2012	Houston et al.
7,027,875	B2	4/2006	Siess et al.	8,114,008	B2	2/2012	Hidaka et al.
7,037,069	B2	5/2006	Arnold et al.	8,123,669	B2	2/2012	Siess et al.
				8,142,400	B2	3/2012	Rotem et al.
				8,177,703	B2	5/2012	Smith et al.
				8,206,350	B2	6/2012	Mann et al.
				8,209,015	B2	6/2012	Glenn

(56)

References Cited

U.S. PATENT DOCUMENTS

8,216,122 B2	7/2012	Kung	10,195,323 B2	2/2019	Tiller et al.
8,235,943 B2	8/2012	Breznock et al.	10,520,025 B1	12/2019	Peterson et al.
8,236,040 B2	8/2012	Mayberry et al.	11,033,728 B2	6/2021	Schenck et al.
8,236,044 B2	8/2012	Robaina	11,058,865 B2	7/2021	Fitzgerald et al.
8,255,050 B2	8/2012	Mohl	11,654,276 B2	5/2023	Fitzgerald et al.
8,257,054 B2	9/2012	Voltenburg et al.	11,660,441 B2	5/2023	Fitzgerald et al.
8,257,312 B2	9/2012	Duffy	11,925,795 B2	3/2024	Muller et al.
8,262,619 B2	9/2012	Chebator et al.	2001/0016676 A1	8/2001	Williams et al.
8,277,470 B2	10/2012	Demarais et al.	2001/0016729 A1	8/2001	Divino et al.
8,317,715 B2	11/2012	Belleville et al.	2002/0010487 A1	1/2002	Evans et al.
8,329,913 B2	12/2012	Murata et al.	2002/0047435 A1	4/2002	Takahashi et al.
8,333,687 B2	12/2012	Farnan et al.	2002/0094287 A1	7/2002	Davis
8,348,991 B2	1/2013	Weber et al.	2002/0107506 A1	8/2002	McGuckin, Jr. et al.
8,364,278 B2	1/2013	Pianca et al.	2002/0111663 A1	8/2002	Dahl
8,371,832 B2	2/2013	Rotem et al.	2002/0115933 A1	8/2002	Duchon et al.
8,376,707 B2	2/2013	McBride et al.	2002/0151761 A1	10/2002	Viole et al.
8,382,818 B2	2/2013	Davis et al.	2002/0169413 A1	11/2002	Keren et al.
8,388,565 B2	3/2013	Shifflette	2003/0018380 A1	1/2003	Craig et al.
8,388,582 B2	3/2013	Eubanks et al.	2003/0023201 A1	1/2003	Aboul-Hosn et al.
8,409,128 B2	4/2013	Ferrari	2003/0028118 A1	2/2003	Dupree
8,414,645 B2	4/2013	Dwork et al.	2003/0093086 A1	5/2003	Briggs et al.
8,439,859 B2	5/2013	Pfeffer et al.	2003/0100816 A1	5/2003	Siess
8,449,443 B2	5/2013	Rodefeld et al.	2003/0135086 A1	7/2003	Khaw
8,485,961 B2	7/2013	Campbell et al.	2003/0187322 A1	10/2003	Siess
8,489,190 B2	7/2013	Pfeffer et al.	2003/0205233 A1	11/2003	Aboul-Hosn et al.
8,491,285 B2	7/2013	Haser et al.	2003/0208097 A1	11/2003	Aboul-Hosn et al.
8,535,211 B2	9/2013	Campbell et al.	2003/0212430 A1	11/2003	Bose et al.
8,540,615 B2	9/2013	Aboul-Hosn et al.	2003/0225366 A1	12/2003	Morgan et al.
8,545,379 B2	10/2013	Marseille et al.	2003/0231959 A1	12/2003	Snider
8,545,380 B2	10/2013	Farnan et al.	2004/0010229 A1	1/2004	Houde et al.
8,579,858 B2	11/2013	Reitan et al.	2004/0019251 A1	1/2004	Viole et al.
8,585,572 B2	11/2013	Mehmanesh	2004/0044266 A1	3/2004	Siess
8,591,110 B2	11/2013	Smith et al.	2004/0101406 A1	5/2004	Hoover
8,591,393 B2	11/2013	Walters et al.	2004/0113502 A1	6/2004	Li et al.
8,597,110 B2	12/2013	Kammler et al.	2004/0116862 A1	6/2004	Ray
8,597,170 B2	12/2013	Walters et al.	2004/0152944 A1	8/2004	Medvedev et al.
8,608,635 B2	12/2013	Yomtov et al.	2004/0236173 A1	11/2004	Viole et al.
8,617,239 B2	12/2013	Reitan	2004/0253129 A1	12/2004	Sorensen et al.
8,618,239 B2	12/2013	Gray	2005/0013698 A1	1/2005	Davis
8,684,904 B2	4/2014	Campbell et al.	2005/0027281 A1	2/2005	Lennox
8,690,749 B1	4/2014	Nunez	2005/0038408 A1	2/2005	von Segesser
8,721,516 B2	5/2014	Scheckel	2005/0049664 A1	3/2005	Harris et al.
8,721,517 B2	5/2014	Zeng et al.	2005/0049696 A1	3/2005	Siess et al.
8,727,959 B2	5/2014	Reitan et al.	2005/0085683 A1	4/2005	Bolling et al.
8,734,331 B2	5/2014	Evans et al.	2005/0090883 A1	4/2005	Westlund
8,734,334 B2	5/2014	Haramaty et al.	2005/0095124 A1	5/2005	Arnold et al.
8,784,441 B2	7/2014	Rosenbluth et al.	2005/0113631 A1	5/2005	Bolling et al.
8,790,236 B2	7/2014	LaRose et al.	2005/0135924 A1	6/2005	Prasad et al.
8,795,576 B2	8/2014	Tao et al.	2005/0135942 A1	6/2005	Wood
8,801,590 B2	8/2014	Mohl	2005/0137680 A1	6/2005	Ortiz et al.
8,814,776 B2	8/2014	Hastie et al.	2005/0165269 A9	7/2005	Aboul-Hosn et al.
8,814,933 B2	8/2014	Siess	2005/0165466 A1	7/2005	Morris et al.
8,849,398 B2	9/2014	Evans	2005/0218022 A1	10/2005	Cervantes
8,944,748 B2	2/2015	Liebing	2005/0250975 A1	11/2005	Carrier et al.
8,992,406 B2	3/2015	Corbett	2005/0277912 A1	12/2005	John
8,998,792 B2	4/2015	Scheckel	2006/0005886 A1	1/2006	Parrino
9,028,216 B2	5/2015	Schumacher et al.	2006/0008349 A1	1/2006	Khaw
9,089,634 B2	7/2015	Schumacher et al.	2006/0018943 A1	1/2006	Bechert et al.
9,089,670 B2	7/2015	Scheckel	2006/0036127 A1	2/2006	Delgado
9,162,017 B2	10/2015	Evans et al.	2006/0058869 A1	3/2006	Olson et al.
9,217,442 B2	12/2015	Wiessler et al.	2006/0062672 A1	3/2006	McBride
9,308,302 B2	4/2016	Zeng	2006/0063965 A1	3/2006	Aboul-Hosn et al.
9,314,558 B2	4/2016	Er	2006/0089521 A1	4/2006	Chang
9,327,067 B2	5/2016	Zeng et al.	2006/0155158 A1	7/2006	Aboul-Hosn
9,328,741 B2	5/2016	Liebing	2006/0167404 A1	7/2006	Pirovano et al.
9,358,329 B2	6/2016	Fitzgerald et al.	2006/0195005 A1	8/2006	Sakai
9,358,330 B2	6/2016	Schumacher	2006/0222533 A1	10/2006	Reeves et al.
9,381,288 B2	7/2016	Schenck	2006/0264695 A1	11/2006	Viole et al.
9,421,311 B2	8/2016	Tanner	2006/0270894 A1	11/2006	Viole et al.
9,446,179 B2	9/2016	Keenan	2007/0005010 A1	1/2007	Mori et al.
9,675,739 B2	6/2017	Tanner et al.	2007/0100314 A1	5/2007	Keren et al.
9,717,833 B2	8/2017	McBride	2007/0142785 A1	6/2007	Lundgaard et al.
9,872,947 B2	1/2018	Keenan et al.	2007/0156006 A1	7/2007	Smith et al.
10,086,121 B2	10/2018	Fitzgerald et al.	2007/0203442 A1	8/2007	Bechert et al.
10,117,980 B2	11/2018	Keenan et al.	2007/0208298 A1	9/2007	Ainsworth et al.
			2007/0212240 A1	9/2007	Voyeux et al.
			2007/0217932 A1	9/2007	Voyeux et al.
			2007/0217933 A1	9/2007	Haser et al.
			2007/0233270 A1	10/2007	Weber et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2007/0237739	A1	10/2007	Doty	2011/0218516	A1	9/2011	Grigorov
2007/0248477	A1	10/2007	Nazarifar et al.	2011/0236210	A1	9/2011	McBride et al.
2007/0282417	A1	12/2007	Houston et al.	2011/0237863	A1	9/2011	Ricci et al.
2008/0004645	A1	1/2008	To et al.	2011/0238172	A1	9/2011	Akdis
2008/0004690	A1	1/2008	Robaina	2011/0257462	A1	10/2011	Rodefelf
2008/0011640	A1	1/2008	Cervantes	2011/0270182	A1	11/2011	Breznock
2008/0015506	A1	1/2008	Davis	2011/0275884	A1	11/2011	Scheckel
2008/0031953	A1	2/2008	Takakusagi et al.	2011/0300010	A1	12/2011	Jarnagin et al.
2008/0093764	A1	4/2008	Ito et al.	2011/0311383	A1	12/2011	White
2008/0103442	A1	5/2008	Kesten et al.	2012/0004495	A1	1/2012	Bolling et al.
2008/0103516	A1	5/2008	Wulfman et al.	2012/0004496	A1	1/2012	Farnan et al.
2008/0103591	A1	5/2008	Siess	2012/0029265	A1	2/2012	LaRose et al.
2008/0114339	A1	5/2008	McBride et al.	2012/0045352	A1	2/2012	Lawyer et al.
2008/0119943	A1	5/2008	Armstrong et al.	2012/0059213	A1	3/2012	Spence
2008/0132748	A1	6/2008	Shifflette	2012/0059460	A1	3/2012	Reitan
2008/0167679	A1	7/2008	Papp	2012/0083740	A1	4/2012	Chebator et al.
2008/0168796	A1	7/2008	Masoudipour et al.	2012/0093628	A1	4/2012	Liebing
2008/0200878	A1	8/2008	Davis et al.	2012/0109172	A1	5/2012	Schmitz et al.
2008/0275290	A1	11/2008	Viole et al.	2012/0142994	A1	6/2012	Toellner
2008/0306327	A1	12/2008	Shifflette	2012/0172654	A1	7/2012	Bates
2009/0018567	A1	1/2009	Escudero et al.	2012/0172655	A1	7/2012	Campbell et al.
2009/0023975	A1	1/2009	Marseille et al.	2012/0172656	A1	7/2012	Walters et al.
2009/0024085	A1	1/2009	To et al.	2012/0178985	A1	7/2012	Walters et al.
2009/0053085	A1	2/2009	Thompson et al.	2012/0178986	A1	7/2012	Campbell et al.
2009/0060743	A1	3/2009	McBride et al.	2012/0184803	A1	7/2012	Simon et al.
2009/0062597	A1	3/2009	Shifflette	2012/0203056	A1	8/2012	Corbett
2009/0073037	A1	3/2009	Penna et al.	2012/0220854	A1	8/2012	Messerly et al.
2009/0087325	A1	4/2009	Voltenburg et al.	2012/0224970	A1	9/2012	Schumacher et al.
2009/0093764	A1	4/2009	Pfeffer et al.	2012/0226097	A1	9/2012	Smith et al.
2009/0093765	A1	4/2009	Glenn	2012/0234411	A1	9/2012	Scheckel
2009/0093796	A1	4/2009	Pfeffer et al.	2012/0237357	A1	9/2012	Schumacher et al.
2009/0099638	A1	4/2009	Grew	2012/0245404	A1	9/2012	Smith et al.
2009/0112312	A1	4/2009	LaRose et al.	2012/0255657	A1	10/2012	Carlson
2009/0118567	A1	5/2009	Siess	2012/0264523	A1	10/2012	Liebing
2009/0163864	A1	6/2009	Breznock et al.	2012/0265002	A1	10/2012	Roehn et al.
2009/0167679	A1	7/2009	Klier et al.	2012/0323318	A1	12/2012	Yusuf et al.
2009/0171137	A1	7/2009	Farnan et al.	2013/0031936	A1	2/2013	Goncalves et al.
2009/0182188	A1	7/2009	Marseille et al.	2013/0039465	A1	2/2013	Okuno
2009/0234378	A1	9/2009	Escudero et al.	2013/0041202	A1	2/2013	Toellner
2009/0259089	A1	10/2009	Gelbart et al.	2013/0053622	A1	2/2013	Corbett
2009/0306588	A1	12/2009	Nguyen et al.	2013/0053623	A1	2/2013	Evans et al.
2010/0004595	A1	1/2010	Nguyen et al.	2013/0053693	A1	2/2013	Breznock et al.
2010/0016960	A1	1/2010	Bolling	2013/0066140	A1	3/2013	McBride et al.
2010/0030161	A1	2/2010	Duffy	2013/0085318	A1	4/2013	Toellner
2010/0030186	A1	2/2010	Stivland	2013/0085319	A1	4/2013	Evans et al.
2010/0041939	A1	2/2010	Siess	2013/0096364	A1	4/2013	Reichenbach et al.
2010/0047099	A1	2/2010	Miyazaki et al.	2013/0103063	A1	4/2013	Escudero et al.
2010/0087773	A1	4/2010	Ferrari	2013/0106212	A1	5/2013	Nakazumi et al.
2010/0094089	A1	4/2010	Litscher et al.	2013/0129503	A1	5/2013	McBride et al.
2010/0127871	A1	5/2010	Pontin	2013/0138205	A1	5/2013	Kushwaha et al.
2010/0137802	A1	6/2010	Yodfat et al.	2013/0204362	A1	8/2013	Toellner et al.
2010/0174239	A1	7/2010	Yodfat et al.	2013/0209292	A1	8/2013	Baykut et al.
2010/0191035	A1	7/2010	Kang et al.	2013/0237744	A1	9/2013	Pfeffer et al.
2010/0197994	A1	8/2010	Mehmanesh	2013/0245360	A1	9/2013	Schumacher
2010/0210895	A1	8/2010	Aboul-Hosn et al.	2013/0303830	A1	11/2013	Zeng et al.
2010/0245523	A1	9/2010	Howell	2013/0303831	A1	11/2013	Evans
2010/0268017	A1	10/2010	Siess	2013/0303969	A1	11/2013	Keenan et al.
2010/0274330	A1	10/2010	Burwell et al.	2013/0303970	A1	11/2013	Keenan et al.
2010/0286210	A1	11/2010	Murata et al.	2013/0331639	A1	12/2013	Campbell et al.
2010/0286791	A1	11/2010	Goldsmith	2013/0338559	A1	12/2013	Franano et al.
2011/0004045	A1	1/2011	Larsen et al.	2013/0345492	A1	12/2013	Pfeffer et al.
2011/0004046	A1	1/2011	Campbell et al.	2014/0005467	A1	1/2014	Farnan et al.
2011/0004291	A1	1/2011	Davis et al.	2014/0010686	A1	1/2014	Tanner et al.
2011/0009687	A1	1/2011	Mohl	2014/0012065	A1	1/2014	Fitzgerald et al.
2011/0015610	A1	1/2011	Plahey et al.	2014/0039465	A1	2/2014	Schulz
2011/0021865	A1	1/2011	Aboul-Hosn et al.	2014/0051908	A1	2/2014	Khanal et al.
2011/0034874	A1	2/2011	Reitan et al.	2014/0066691	A1	3/2014	Siebenhaar
2011/0071338	A1	3/2011	McBride et al.	2014/0067057	A1	3/2014	Callaway et al.
2011/0076439	A1	3/2011	Zeilon	2014/0088455	A1	3/2014	Christensen et al.
2011/0098805	A1	4/2011	Dwork et al.	2014/0128659	A1	5/2014	Heuring et al.
2011/0106004	A1	5/2011	Eubanks et al.	2014/0148638	A1	5/2014	LaRose et al.
2011/0106115	A1	5/2011	Haselby et al.	2014/0163664	A1	6/2014	Goldsmith
2011/0152831	A1	6/2011	Rotem et al.	2014/0188086	A1	7/2014	Govari et al.
2011/0152906	A1	6/2011	Escudero et al.	2014/0205434	A1	7/2014	Graichen
2011/0152907	A1	6/2011	Escudero et al.	2014/0255176	A1	9/2014	Bredenbreuker et al.
				2014/0275725	A1	9/2014	Schenck et al.
				2014/0275726	A1	9/2014	Zeng
				2014/0276948	A1	9/2014	Zirps
				2014/0301822	A1	10/2014	Scheckel

(56)	References Cited			EP	3800357	B1	5/2024
	U.S. PATENT DOCUMENTS			FR	2267800	A1	11/1975
				GB	0886219	A	1/1962
				GB	2239675	A	7/1991
2014/0303596	A1	10/2014	Schumacher et al.	JP	S4823295	U	3/1973
2015/0025558	A1	1/2015	Wulfman et al.	JP	S58190448	A	11/1983
2015/0031936	A1	1/2015	LaRose et al.	JP	02-211169	A	8/1990
2015/0051435	A1	2/2015	Siess et al.	JP	H06114101	A	4/1994
2015/0051436	A1	2/2015	Spanier et al.	JP	08-500512	A	1/1996
2015/0080743	A1	3/2015	Siess	JP	H08501466	A	2/1996
2015/0087890	A1	3/2015	Spanier et al.	JP	H08196624	A	8/1996
2015/0141738	A1	5/2015	Toellner et al.	JP	09-114101	A	5/1997
2015/0141739	A1	5/2015	Hsu et al.	JP	10-099440	A	4/1998
2015/0151032	A1	6/2015	Voskoboynikov et al.	JP	H1099447	A	4/1998
2015/0176582	A1	6/2015	Liebing	JP	2001-079093	A	3/2001
2015/0209498	A1	7/2015	Franano et al.	JP	3208454	B2	9/2001
2015/0224970	A1	8/2015	Yasui et al.	JP	2002505168	A	2/2002
2015/0250935	A1	9/2015	Anderson et al.	JP	2004514506	A	5/2004
2015/0290371	A1	10/2015	Muller et al.	JP	2005-514085	A	5/2005
2015/0290372	A1	10/2015	Muller et al.	JP	2007-252960	A	10/2007
2015/0343179	A1	12/2015	Schumacher et al.	JP	2009-530041	A	8/2009
2016/0123098	A1	5/2016	Marr	JP	2011-000620	A	1/2011
2016/0184500	A1	6/2016	Leng	JP	2011-505902	A	3/2011
2016/0213826	A1	7/2016	Tanner et al.	JP	2011-157961	A	8/2011
2016/0213827	A1	7/2016	Tanner et al.	JP	2012-531975	A	12/2012
2016/0250399	A1	9/2016	Tiller et al.	JP	2016-515000	A	5/2016
2016/0250400	A1	9/2016	Schumacher	JP	6114101	B2	4/2017
2016/0256620	A1	9/2016	Scheckel et al.	TW	500877	B2	9/2002
2016/0279310	A1	9/2016	Scheckel et al.	WO	89/04644	A1	6/1989
2016/0319846	A1	11/2016	Liebing	WO	89/05164	A1	6/1989
2017/0014562	A1	1/2017	Liebing	WO	89/05668	A2	6/1989
2017/0087287	A1	3/2017	Keenan et al.	WO	94/06486	A1	3/1994
2018/0021495	A1	1/2018	Muller et al.	WO	9405347	A1	3/1994
2018/0055979	A1	3/2018	Corbett et al.	WO	95/14500	A1	6/1995
2018/0064862	A1	3/2018	Keenan et al.	WO	9526695	A2	10/1995
2018/0243489	A1	8/2018	Haddadi	WO	9715228	A1	5/1997
2018/0296742	A1	10/2018	Toellner	WO	97/37698	A1	10/1997
2019/0030228	A1	1/2019	Keenan et al.	WO	9737694	A1	10/1997
2019/0148346	A1	5/2019	Feichtinger et al.	WO	98/11349	A1	3/1998
2019/0254909	A1	8/2019	Lee et al.	WO	99/00368	A1	1/1999
2021/0046230	A1	2/2021	Fitzgerald et al.	WO	99/02204	A1	1/1999
2021/0077690	A1	3/2021	Schenck et al.	WO	99/16387	A1	4/1999
2021/0113827	A1	4/2021	McBride et al.	WO	99/37352	A1	7/1999
2021/0187270	A1	6/2021	Schenck et al.	WO	99/44651	A1	9/1999
2022/0372989	A1	11/2022	McBride et al.	WO	99/44670	A1	9/1999
2023/0302271	A1	9/2023	Spanier et al.	WO	99/59652	A1	11/1999
				WO	99/65546	A1	12/1999
				WO	12148	A2	3/2000
				WO	18448	A2	4/2000
				WO	19097	A1	4/2000
CA	2322012	A1	9/1999	WO	37139	A1	6/2000
CA	2367469	A1	10/2000	WO	38591	A2	7/2000
CA	2407938	A1	11/2001	WO	41612	A2	7/2000
CA	2480467	A1	8/2003	WO	43053	A1	7/2000
CA	2701810	A1	4/2009	WO	43062	A1	7/2000
CA	2701870	A1	4/2009	WO	45874	A1	8/2000
CN	101820933	A	9/2010	WO	61207	A1	10/2000
CN	211584537	U	9/2020	WO	69489	A1	11/2000
DE	19613565	C1	7/1997	WO	01/19444	A1	3/2001
DE	10059714	C1	5/2002	WO	117581	A2	3/2001
DE	112004001809	T5	11/2006	WO	01/24897	A1	4/2001
EP	0193762	A2	9/1986	WO	124867	A1	4/2001
EP	364293	A2	4/1990	WO	01/78807	A1	10/2001
EP	453234	A1	10/1991	WO	01/83016	A2	11/2001
EP	533432	A1	3/1993	WO	02/43791	A1	6/2002
EP	1207934	A2	5/2002	WO	2070039	A2	9/2002
EP	0691108	B1	12/2002	WO	02/81919	A1	10/2002
EP	1393762	A1	3/2004	WO	03/48582	A1	6/2003
EP	1591079	A1	11/2005	WO	03/54660	A2	7/2003
EP	2151257	A1	2/2010	WO	03/68303	A2	8/2003
EP	2263732	A2	12/2010	WO	03/70299	A1	8/2003
EP	2298374	A1	3/2011	WO	3103745	A2	12/2003
EP	2399639	A1	12/2011	WO	2005/030296	A2	4/2005
EP	2427230	A1	3/2012	WO	2005089674	A1	9/2005
EP	2662099	A1	11/2013	WO	2005123158	A1	12/2005
EP	3115070	A1	1/2017	WO	2006/034158	A2	3/2006
EP	3453234	A1	3/2019	WO	2006/046779	A1	5/2006
EP	3520834	A1	8/2019	WO	2006/051023	A1	5/2006
EP	3533432	A2	9/2019	WO	2007/112033	A2	10/2007
EP	3808405	A1	4/2021	WO			

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	2008/034068	A2	3/2008
WO	2009073037	A1	6/2009
WO	2009076460	A2	6/2009
WO	2010/042546	A1	4/2010
WO	2010/063494	A1	6/2010
WO	2010/127871	A1	11/2010
WO	2010133567	A1	11/2010
WO	2010149393	A1	12/2010
WO	2011/003043	A1	1/2011
WO	2011/035927	A1	3/2011
WO	2011035926	A1	3/2011
WO	2011035929	A2	3/2011
WO	2011039091	A1	4/2011
WO	2011076439	A1	6/2011
WO	2011089022	A1	7/2011
WO	2011/126895	A2	10/2011
WO	2012007140	A1	1/2012
WO	2012007141	A1	1/2012
WO	2012/094525	A2	7/2012
WO	2012/094534	A2	7/2012
WO	2013/032849	A1	3/2013
WO	2013/120957	A1	8/2013
WO	2013148697	A1	10/2013
WO	2013160407	A1	10/2013
WO	2013/173239	A1	11/2013
WO	2013/173245	A1	11/2013
WO	2014/008102	A1	1/2014
WO	2014019274	A1	2/2014
WO	2014/174914	A1	10/2014
WO	2015/055515	A1	4/2015
WO	2015063277	A2	5/2015
WO	2015/160942	A1	10/2015
WO	2016/028644	A1	2/2016
WO	2016/116608	A2	7/2016
WO	2016/118777	A1	7/2016
WO	2016/118781	A2	7/2016
WO	2016183468	A1	11/2016
WO	2017192775	A1	11/2017

OTHER PUBLICATIONS

Abiomed—Recovering Hearts. Saving Lives., Impella 2.5 System, Instructions for Use, Jul. 2007, in 86 sheets.

Aboul-Hosn et al., “The Hemopump: Clinical Results and Future Applications”, *Assisted Circulation* 4, 1995, in 14 pages.

Barras et al., “Nitinol-Its Use in Vascular Surgery and Other Applications,” *Eur. J. Vasc. Endovasc. Surg.*, 2000, pp. 564-569; vol. 19.

Biscarini et al., “Enhanced Nitinol Properties for Biomedical Applications,” *Recent Patents on Biomedical Engineering*, 2008, pp. 180-196, vol. 1(3).

Cardiovascular Diseases (CVDs) Fact Sheet No. 317; World Health Organization [Online], Sep. 2011. <http://www.who.int/mediacentre/factsheets/fs317/en/index.html>, accessed on Aug. 29, 2012.

Compendium of Technical and Scientific Information for the HemoPump Temporary Cardiac Assist System, Johnson & Johnson Interventional Systems, 1988, in 15 pages.

Dekker et al., “Efficacy of a New Intraaortic Propeller Pump vs the Intraaortic Balloon Pump, an Animal Study”, *Chest*, Jun. 2003, vol. 123, No. 6, pp. 2089-2095.

Duerig et al., “An Overview of Nitinol Medical Applications,” *Materials Science Engineering*, 1999, pp. 149-160; vol. A273.

European Search Report received in European Patent Application No. 05799883.3, dated May 10, 2011, in 4 pages.

Extended European Search Report received in European Patent Application No. 07753903.9, dated Oct. 8, 2012, in 7 pages.

Extended European Search Report received in European Patent Application No. 13813687.4, dated Feb. 24, 2016, in 6 pages.

Extended European Search Report received in European Patent Application No. 13813867.2, dated Feb. 26, 2016, in 6 pages.

Extended European Search Report received in European Patent Application No. 14779928.2, dated Oct. 7, 2016, in 6 pages.

Extended European Search Report received in European Patent Application No. 14764392.8, dated Oct. 27, 2016, n 7 pages.

Federal and Drug Administration 510(k) Summary for Predicate Device Impella 2.5 (K112892), prepared Sep. 5, 2012.

Grech, “Percutaneous Coronary Intervention. I: History and Development,” *BMJ*, May 17, 2003, pp. 1080-1082, vol. 326.

Hsu et al., “Review of Recent Patents on Foldable Ventricular Assist Devices,” *Recent Patents on Biomedical Engineering*, 2012, pp. 208-222, vol. 5.

Ide et al., “Evaluation of the Pulsatility of a New Pulsatile Left Ventricular Assist Device—the Integrated Cardioassist Catheter-in Dogs,” *J. of Thorac and Cardiovasc Sur*, Feb. 1994, pp. 569-575, vol. 107(2).

Ide et al., “Hemodynamic Evaluation of a New Left Ventricular Assist Device: An Integrated Cardioassist Catheter as a Pulsatile Left Ventricle-Femoral Artery Bypass,” *Blackwell Scientific Publications, Inc.*, 1992, pp. 286-290, vol. 16(3).

Impella CP®—Instructions for Use & Clinical Reference Manual (United States only), Abiomed, Inc., Jul. 2014, 148 pages, www.abiomed.com.

Impella LD® with the Impella® Controller—Circulatory Support System—Instructions for Use & Clinical Reference Manual (United States only), Abiomed, Inc., Sep. 2010, 132 pages, www.abiomed.com.

International Preliminary Examination Report received in International Patent Application No. PCT/US2003/04853, mailed on Jul. 26, 2004, in 5 pages.

International Preliminary Examination Report received in International Patent Application No. PCT/US2003/04401, dated May 18, 2004, in 4 pages.

International Preliminary Report on Patentability and Written Opinion of the International Searching Authority received in International Patent Application No. PCT/US2005/033416, mailed on Mar. 20, 2007, in 7 pages.

International Preliminary Report on Patentability and Written Opinion of the International Searching Authority received in International Patent Application No. PCT/US2007/007313, mailed on Sep. 23, 2008, in 6 pages.

International Preliminary Report on Patentability and Written Opinion received in International Patent Application No. PCT/US2014/020878, mailed Sep. 15, 2015, in 8 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2005/033416, mailed on Dec. 11, 2006, in 8 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2007/007313, mailed on Mar. 4, 2008, in 6 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2012/020382, mailed on Jul. 31, 2012, in 11 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2012/020369, mailed on Jul. 30, 2012, in 10 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2012/020553, mailed on Aug. 17, 2012, in 8 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2012/020383, mailed on Aug. 17, 2012, in 9 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2013/040798, mailed Aug. 21, 2013, in 16 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2013/040799, mailed Aug. 21, 2013, in 19 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2013/040809, mailed Sep. 2, 2013, in 25 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2013/048332, mailed Oct. 16, 2013, in 17 pages.

(56)

References Cited**OTHER PUBLICATIONS**

International Search Report and Written Opinion received in International Patent Application No. PCT/US2013/048343, mailed Oct. 11, 2013, in 15 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2014/020878, mailed May 7, 2014, in 13 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2015/026013, mailed Oct. 22, 2015, in 12 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2015/026014, mailed Oct. 22, 2015, in 13 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2015/026025, mailed Oct. 22, 2015, in 12 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2015/025959, mailed Oct. 22, 2015, in 9 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2015/025960, mailed Oct. 22, 2015, in 11 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2015/045370, mailed Feb. 25, 2016, in 10 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2016/014379, mailed Jul. 29, 2016, in 17 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2016/014391, mailed Jul. 28, 2016, in 15 pages.

International Search Report received in International Patent Application No. PCT/US2003/004401, mailed on Jan. 22, 2004, in 7 pages.

International Search Report received in International Patent Application No. PCT/US2003/004853, mailed on Nov. 10, 2003, in 5 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2010/040847, mailed on Jan. 6, 2011, in 15 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2014/020790, mailed on Oct. 9, 2014, in 9 pages.

JOMED Reitan Catheter Pump RCP, Percutaneous Circulatory Support, in 10 pages, believed to be published prior to Oct. 15, 2003.

JOMED Reitan Catheter Pump RCP, Feb. 18, 2003, in 4 pages.

Krishnamani et al., "Emerging Ventricular Assist Devices for Long-Term Cardiac Support," *National Review, Cardiology*, Feb. 2010, pp. 71-76, vol. 7.

Kunst et al., "Integrated unit for programmable control of the 21F Hemopump and registration of physiological signals," *Medical & Biological Engineering & Computing*, Nov. 1994, pp. 694-696.

Mihaylov et al., "Development of a New Introduction Technique for the Pulsatile Catheter Pump," *Artificial Organs*, 1997, pp. 425-427; vol. 21(5).

Mihaylov et al., "Evaluation of the Optimal Driving Mode During Left Ventricular Assist with Pulsatile Catheter Pump in Calves," *Artificial Organs*, 1999, pp. 1117-1122; vol. 23(12).

Minimally Invasive Cardiac Assist JOMED Catheter PumpTM, in 6 pages, believed to be published prior to Jun. 16, 1999.

Morgan, "Medical Shape Memory Alloy Applications—The Market and its Products," *Materials Science and Engineering*, 2004, pp. 16-23, vol. A 378.

Morsink et al., "Numerical Modelling of Blood Flow Behaviour in the Valved Catheter of the PUCA-Pump, a LVAD," *The International Journal of Artificial Organs*, 1997, pp. 277-284; vol. 20(5).

Nishimura et al., "The Enabler Cannula Pump: A Novel Circulatory Support System," *The International Journal of Artificial Organs*, 1999, pp. 317-323; vol. 22(5).

Nullity Action against the owner of the German part DE 50 2007 005 015.6 of European patent EP 2 047 872 B1, dated Jul. 13, 2015, in 61 pages.

Petrini et al., "Biomedical Applications of Shape Memory Alloys," *Journal of Metallurgy*, 2011, pp. 1-15.

Raess et al., "Impella 2.5," *J. Cardiovasc. Transl. Res.*, 2009, pp. 168-172, vol. 2(2).

Rakhorst et al., "In Vitro Evaluation of the Influence of Pulsatile Intraventricular Pumping on Ventricular Pressure Patterns," *Artificial Organs*, 1994, pp. 494-499, vol. 18(7).

Reitan, Evaluation of a New Percutaneous Cardiac Assist Device, Department of Cardiology, Faculty of Medicine, Lund University, Sweden, 2002, in 172 pages.

Reitan et al., "Hemodynamic Effects of a New Percutaneous Circulatory Support Device in a Left Ventricular Failure Model," *ASAIO Journal*, 2003, pp. 731-736, vol. 49.

Reitan et al., "Hydrodynamic Properties of a New Percutaneous Intra-Aortic Axial Flow Pump," *ASAIO Journal* 2000, pp. 323-328.

Rothman, "The Reitan Catheter Pump: A New Versatile Approach for Hemodynamic Support", London Chest Hospital Barts & The London NHS Trust, Oct. 22-27, 2006 (TCT 2006: Transcatheter Cardiovascular Therapeutics 18th Annual Scientific Symposium, Final Program), in 48 pages.

Schmitz-Rode et al., "An Expandable Percutaneous Catheter Pump for Left Ventricular Support," *Journal of the American College of Cardiology*, 2005, pp. 1856-1861, vol. 45(11).

Schmitz-Rode et al., "Axial flow catheter pump for circulatory support," *Biomedizinische Technik*, 2002, Band 47, *Erganzungsband* 1, Teil 1, pp. 142-143.

Shabari et al., "Improved Hemodynamics with a Novel Miniaturized Intra-Aortic Axial Flow Pump in a Porcine Model of Acute Left Ventricular Dysfunction," *ASAIO Journal*, 2013, pp. 240-245; vol. 59.

Sharony et al., "Cardiopulmonary Support and Physiology—The Intra-Aortic Cannula Pump: A Novel Assist Device for the Acutely Failing Heart," *The Journal of Thoracic and Cardiovascular Surgery*, Nov. 1992, pp. 924-929, vol. 118(5).

Sharony et al., "Right Heart Support During Off-Pump Coronary Artery Surgery—A Multi-Center Study," *The Heart Surgery Forum*, 2002, pp. 13-16, vol. 5(1).

Sieß et al., "Hydraulic refinement of an intraarterial microaxial blood pump", *The International Journal of Artificial Organs*, 1995, vol. 18, No. 5, pp. 273-285.

Sieß, "Systemanalyse und Entwicklung intravasaler Rotationspumpen zur Herzunterstützung", *Helmholtz-Institut für Biomedizinische Technik an der RWTH Aachen*, Jun. 24, 1998, in 105 pages.

Siess et al., "Basic design criteria for rotary blood pumps," H. Masuda, *Rotary Blood Pumps*, Springer, Japan, 2000, pp. 69-83.

Siess et al., "Concept, realization, and first in vitro testing of an intraarterial microaxial blood pump," *Artificial Organs*, 1995, pp. 644-652, vol. 19, No. 7, Blackwell Science, Inc., Boston, International Society for Artificial Organs.

Siess et al., "From a lab type to a product: A retrospective view on Impella's assist technology," *Artificial Organs*, 2001, pp. 414-421, vol. 25, No. 5, Blackwell Science, Inc., International Society for Artificial Organs.

Siess et al., "System analysis and development of intravascular rotation pumps for cardiac assist," *Dissertation*, Shaker Verlag, Aachen, 1999, 39 pages.

Smith et al., "First-In-Man Study of the Reitan Catheter Pump for Circulatory Support in Patients Undergoing High-Risk Percutaneous Coronary Intervention," *Catheterization and Cardiovascular Interventions*, 2009, pp. 859-865, vol. 73(7).

Sokolowski et al., "Medical Applications of Shape Memory Polymers," *Biomed. Mater.* 2007, pp. S23-S27, vol. 2.

"Statistical Analysis and Clinical Experience with the Recover® Pump Systems", *Impella CardioSystems GmbH*, Sep. 2005, 2 sheets.

Stoeckel et al., "Self-Expanding Nitinol Stents—Material and Design Considerations," *European Radiology*, 2003, in 13 sheets.

Stolinski et al., "The heart-pump interaction: effects of a microaxial blood pump," *International Journal of Artificial Organs*, 2002, pp. 1082-1088, vol. 25, Issue 11.

(56)

References Cited**OTHER PUBLICATIONS**

Supplemental European Search Report received from the European Patent Office in EP Application No. EP 05799883 dated Mar. 19, 2010, 3 pages.

Takagaki et al., "A Novel Miniature Ventricular Assist Device for Hemodynamic Support," *ASAIO Journal*, 2001, pp. 412-416; vol. 47.

Throckmorton et al., "Flexible Impeller Blades in an Axial Flow Pump for Intravascular Cavopulmonary Assistance of the Fontan Physiology," *Cardiovascular Engineering and Technology*, Dec. 2010, pp. 244-255, vol. 1(4).

Throckmorton et al., "Uniquely shaped cardiovascular stents enhance the pressure generation of intravascular blood pumps," *The Journal of Thoracic and Cardiovascular Surgery*, Sep. 2012, pp. 704-709, vol. 133, No. 3.

Verkerke et al., "Numerical Simulation of the PUCA Pump, a Left Ventricular Assist Device," *Abstracts of the XIXth ESAO Congress, The International Journal of Artificial Organs*, 1992, p. 543, vol. 15(9).

Verkerke et al., "Numerical Simulation of the Pulsating Catheter Pump: A Left Ventricular Assist Device," *Artificial Organs*, 1999, pp. 924-931, vol. 23(10).

Verkerke et al., "The PUCA Pump: A Left Ventricular Assist Device," *Artificial Organs*, 1993, pp. 365-368, vol. 17(5).

Wampler et al., "The Sternotomy Hemopump, a Second Generation Intraarterial Ventricular Assist Device," *ASAIO Journal*, 1993, pp. M218-M223, vol. 39.

Weber et al., "Principles of Impella Cardiac Support," *Supplemental to Cardiac Interventions Today*, Aug./Sep. 2009.

Written Opinion received in International Patent Application No. PCT/US2003/04853, dated Feb. 25, 2004, 5 pages.

Extended European Search Report received in European Patent Application No. 19161643.2, dated Jun. 24, 2019, in 9 pages.

International Search Report and Written Opinion in International Patent Application No. PCT/US2019/015680, mailed Apr. 4, 2019, 15 pages.

Definitions of "undulating" and "undulation," *Shorter Oxford English Dictionary*, 5th edition, p. 3436 (2002) (4 pages).

1st Auxiliary Application dated Oct. 11, 2013, European Application No. 07019657.1, 23 pages.

Andrew J. Carter et al., (1998), *Progressive Vascular Remodeling and Reduced Neointimal Formation After Placement of a Thermoelastic Self-Expanding Nitinol Stent in an Experimental Model, Catheterization and Cardiovascular Diagnosis*, (44), 193-201.

Applicant's Submission in Application No. 20204996.1, dated Oct. 7, 2021.

Arvand et al.: "A Validated Computational Fluid Dynamics Model to Estimate Hemolysis in a Rotary Blood Pump", *Artificial Organs*, vol. 29, No. 7, 2005, pp. 531-540.

Combined Search and Examination Report for Great Britain Application No. 1308544.4, dated Nov. 13, 2013, 6 pages.

Combined Search and Examination Report for Great Britain Application No. 1414709.4, dated Dec. 16, 2014, 5 pages.

Copending U.S. Appl. No. 12/829,359, filed Jul. 1, 2010.

Copending U.S. Appl. No. 18/054,482 (Year: 2022).

Decision on Rejection of the objection, dated Oct. 1, 2014, European Application No. 04763480.3, 3 pages.

Decision rejecting the opposition (EPC Art. 101(2)), dated Oct. 1, 2014, European Application No. 07 019 657.1, 13 pages.

European Search Opinion in Application No. 20204996.1, dated Feb. 19, 2021.

European Search Report for App. No. 21183329.8, dated Oct. 22, 2021, 10 pgs.

European Search Report for App. No. 21183329.8, dated 22OCT2021, 10 pgs.

European Search Report for Patent Application No. 20187258.7, dated Apr. 9, 2021 (16 pages).

Extended EP Search Report, dated Dec. 13, 2019, for EP patent application No. EP 19195969.1 (4 pgs.).

Extended EP Search Report, dated Mar. 15, 2018, for related EP patent application No. EP 15833166.0, in 7 pages.

Extended European Search Report for European Patent Application No. 21156867.0, dated Jun. 10, 2021, 6 pages.

Extended European Search Report for Patent Application No. EP24170611.8 dated Jul. 23, 2024; 10 pp.

Extended European Search Report received in European Patent Application No. 13790890.1, dated Jan. 7, 2016, in 6 pages.

Extended European Search Report received in European Patent Application No. 13791118.6, dated Jan. 7, 2016, in 6 pages.

Extended European Search Report received in European Patent Application No. 20176135.0, dated Aug. 31, 2020, 7 pages.

Extended European Search Report received in European Patent Application No. 22195112.2, dated Jan. 2, 2023, 12 pgs.

Extended European Search Report received in Patent Application No. 20205009.2, dated Mar. 16, 2021 (8 pages).

Fact and Arguments from Hoffmann Eitle, Opposition, EP 2 234 658 81, Proprietor: AIS GmbH Aachen Innovative Solutions (DE), Opponent: Dr. Niels Holder (DE), dated Feb. 3, 2012; 29 pages.

Fact and Arguments from Hoffmann Eitle, Opposition, EP 2 047 872 81, Proprietor: AIS GmbH Aachen Innovative Solutions (DE), Opponent: Dr. Niels Holder (DE), dated Jun. 8, 2011; 32 pages.

Facts and Ground for the Opposition dated Oct. 17, 2012, European Application No. 04763480.3, 43 pages.

Facts of the Case and Petitions, dated Feb. 7, 2014, European Application No. 04763480.3, 13 pages.

Facts of the Case and Petitions, dated Oct. 1, 2014, European Application No. 04763480.3, 16 pages.

Fuentes et al. "Phase Change Behavior of Nitinol Shape Memory Alloys," *Advanced Engineering Materials*, 2002, 4, No. 7, 437-451.

Garonfarinas: "Fast Three Dimensional Numerical Hemolysis Approximation", *Artificial Organs*, vol. 28, No. 11, 2004, pp. 1016-1025.

Giersiepen et al.: "Estimation of Shear Stress-related Blood Damage in Heart Valve Prostheses—In vitro Comparison of 25 Aortic Valves", *International Journal of Artificial Organs*, vol. 13, No. 5, 1990, pp. 300-306.

Gu et al.: "Evaluation of Computational Models for Hemolysis Estimation", *ASAIO Journal*, 2005, pp. 202-207.

In response to the Proprietor's letter of Jul. 18, 2012 from Hoffmann Eitle dated Oct. 24, 2012, Opposition, EP 2 234 658 81, Proprietor: AIS GmbH Aachen Innovative Solutions (DE), Opponent Dr. Niels Holder (DE), 7 pages.

International Preliminary Report on Patentability for International Patent Application No. PCT/US2017/042803, mailed Jan. 31, 2019, 10 pages.

International Search Report and Written Opinion for International Application No. PCT/US2019/028482, mailed Jul. 25, 2019, 15 pages.

International Search Report and Written Opinion for International Patent Application No. PCT/US2017/042803, mailed Oct. 5, 2017, 18 pages.

International Search Report and Written Opinion for International Patent Application No. PCT/US2017/042810, mailed Sep. 28, 2017, 18 pages.

International Search Report and Written Opinion in International Patent Application No. PCT/US2016/014371, dated May 2, 2016, in 18 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2015/025950, mailed Sep. 3, 2015, in 15 pages.

International Search Report and Written Opinion received in International Patent Application No. PCT/US2016/051553, dated Feb. 8, 2017, in 15 pages.

International Search Report received in International Patent Application No. PCT/US2003/004353, dated Jul. 3, 2003, in 3 pages.

International Search Report received in PCT Application No. PCT/US2005/033416, dated Dec. 11, 2006, 4 pages.

JP Notice of Allowance, dated Apr. 22, 2019 for related JP patent application No. 2016-500668.

Motion to dismiss the objection by Dr. Niels Holder dated Jan. 17, 2012 to EPO in European Patent No. 2 04 7 872 B 1, 12 pages.

(56)

References Cited

OTHER PUBLICATIONS

Notice of Reasons for Refusal and Search Report received in Japanese Patent Application No. 2015-512724, dated Mar. 28, 2017, 24 pages.
Office Action dated Jun. 24, 2020 for U.S. Appl. No. 16/296,952 (pp. 1-13).
Office Action issued in European Application No. 19732754.7, dated Oct 20, 2021, 8 pages.
Office Action Received in German Patent Application No. 102013008158.0, dated Feb. 15, 2019, 14 pages.
Opinion on behalf of the Opponent dated Aug. 26, 2013, filed with the European Patent Office in European Application No. 04763480.3 (EP Patent No. 1 651 290 81), 23 pages.
Opposition by Dr. Niels Holder dated Jul. 18, 2012 to EPO in European Patent No. 2 234 658 81, 14 pages.
Opposition Opinion of EP 2 234 658, dated Jan. 20, 2014; 3 pages.
Partial EP search report, dated Dec. 1, 2020, for related EP patent application No. 20187258.7 (12 pgs.).
Reitan, Oyvind, et al., Hydrodynamic Properties of a New Percutaneous Intra-aortic Axial Flow Pump. ASAIO Journal 2000. pp. 323-329.

Reply to the Objection by Thoratec Corporation of Oct. 17, 2012, from the European Patent Office dated Mar. 22, 2013 , European Patent No. 1 651 290, 14 pages.
Response to Memorandum of Aug. 26, 2013 with the invitation to an oral hearing, dated Oct. 11, 2013, European Patent No. 2 234 658, 28 pages.
Response to the Summons dated Jun. 14, 2013: from Fish & Richardson P.C., dated Oct. 7, 2013, Opposition against EP 2 047 872 81, 12 pages.
Responsive to the Summons dated Aug. 26, 2013: from Fish & Richardson P.C., dated Oct. 7, 2013, Opposition against EP 2 234 658 81, 9 pages.
Spini et al. "Transition temperature range of thermally activated nickel-titanium archwires", J. Appl. Oral Sci. 2014;22 (2):109-117.
Statement of Appeal, dated Feb. 6, 2015, European Patent No. 1 651 290, Opponent and Appellant Thoratec Corporation, 30 pages.
Synopsis zu Ansprüchen 1 bis 5 der EP 2 047 872, in 11 pages.
U.S. Appl. No. 12/565,651, filed Sep. 23, 2009.
U.S. Appl. No. 12/772,810, filed May 3, 2010.
Wikipedia, "Ball Bearing," accessed Feb. 28, 2025, wikipedia.com, https://en.wikipedia.org/wiki/Ball_bearing (Year: 2025).
wikipedia.org, "Bearing (mechanical)", accessed Jan. 4, 2023, [https://en.wikipedia.org/wiki/Bearing_\(mechanical\)](https://en.wikipedia.org/wiki/Bearing_(mechanical)). (Year: 2023).

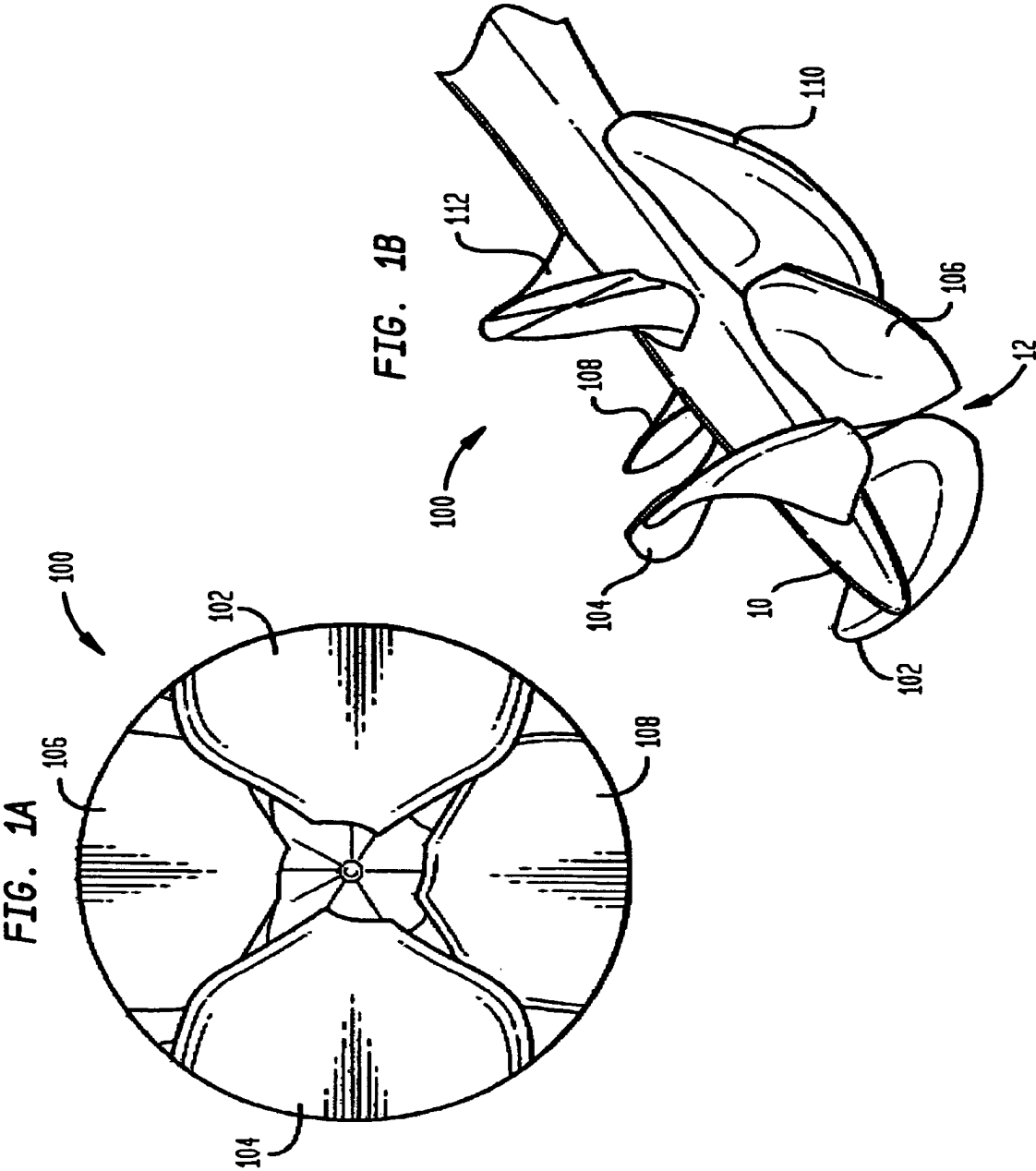


FIG. 2A

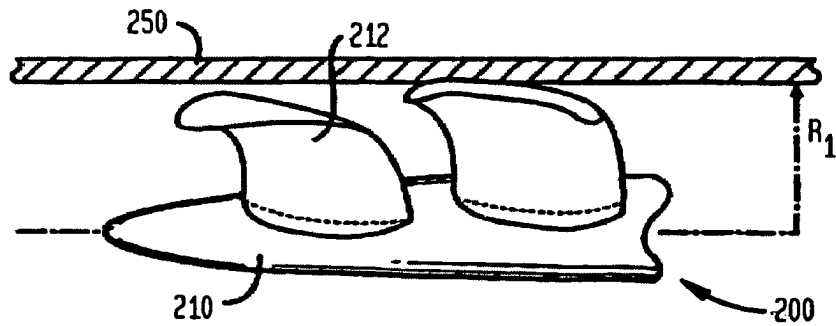


FIG. 2B

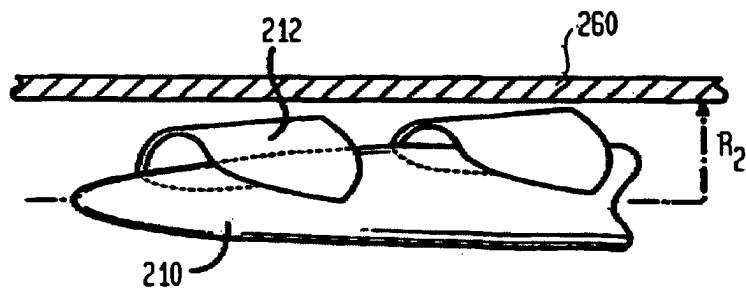


FIG. 3

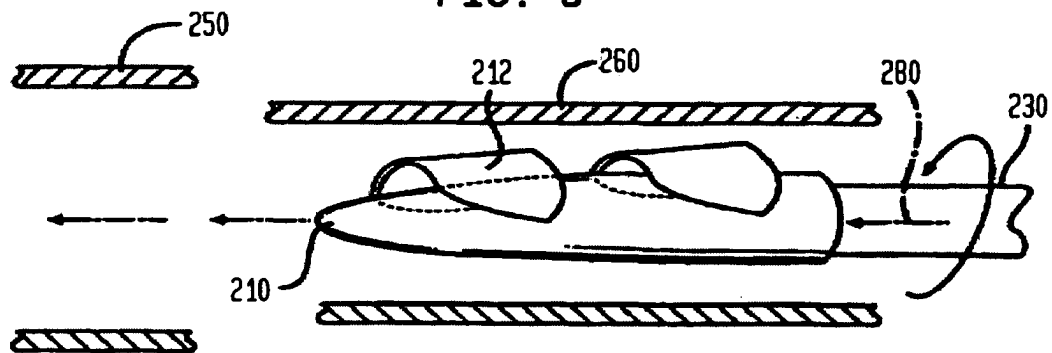


FIG. 4A

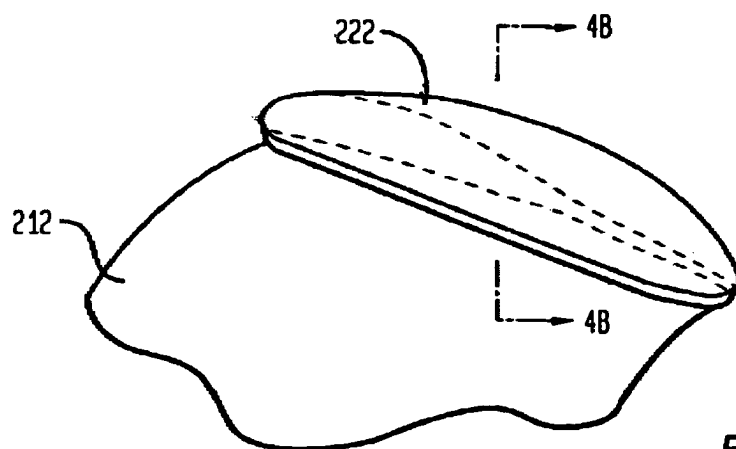


FIG. 4B

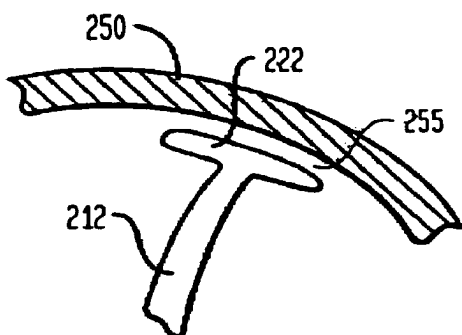


FIG. 5A

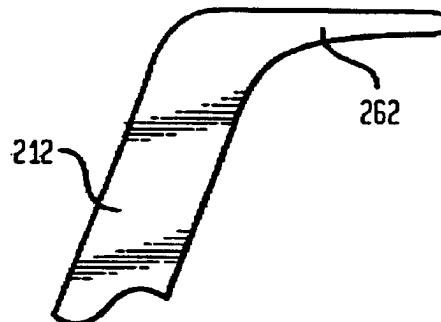


FIG. 5B

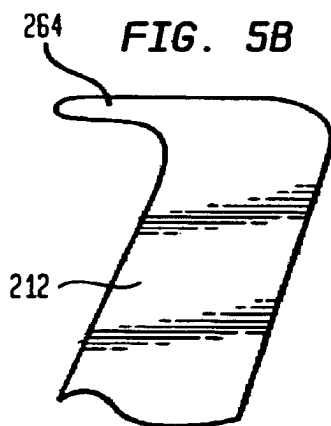


FIG. 5C

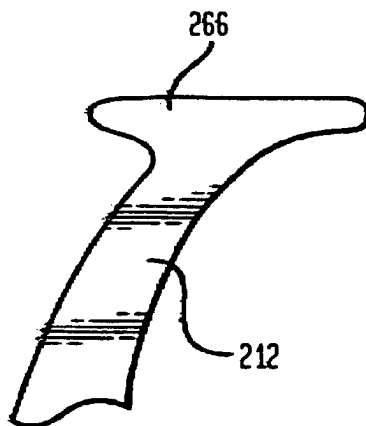
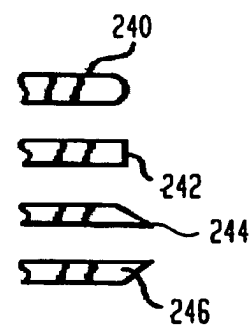


FIG. 5D



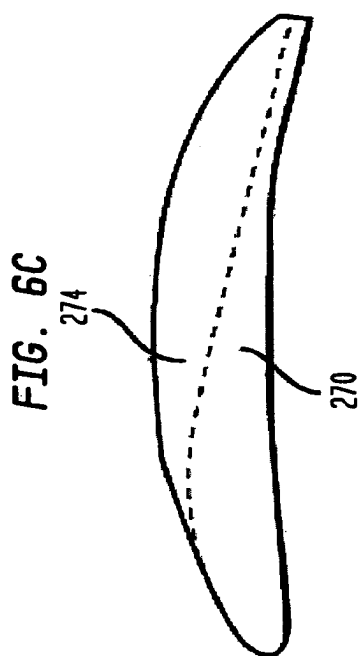
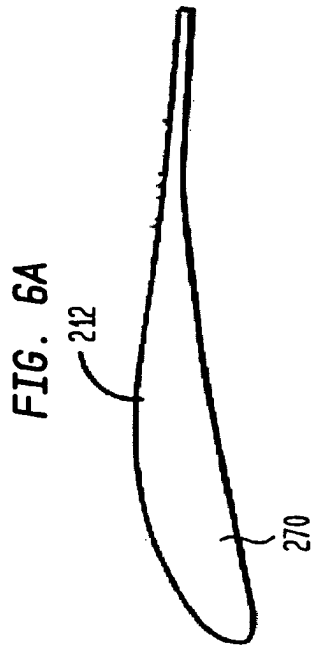
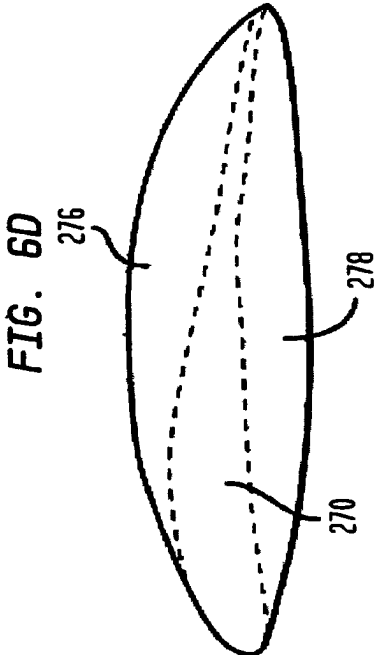
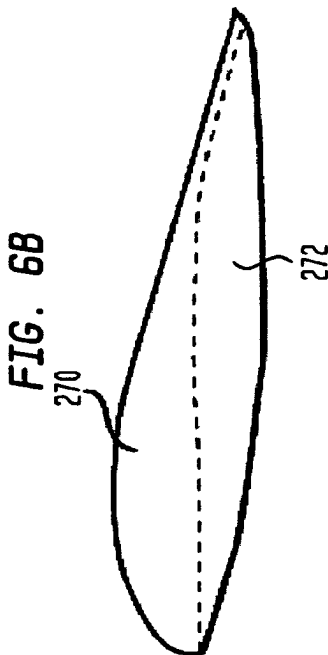


FIG. 7A

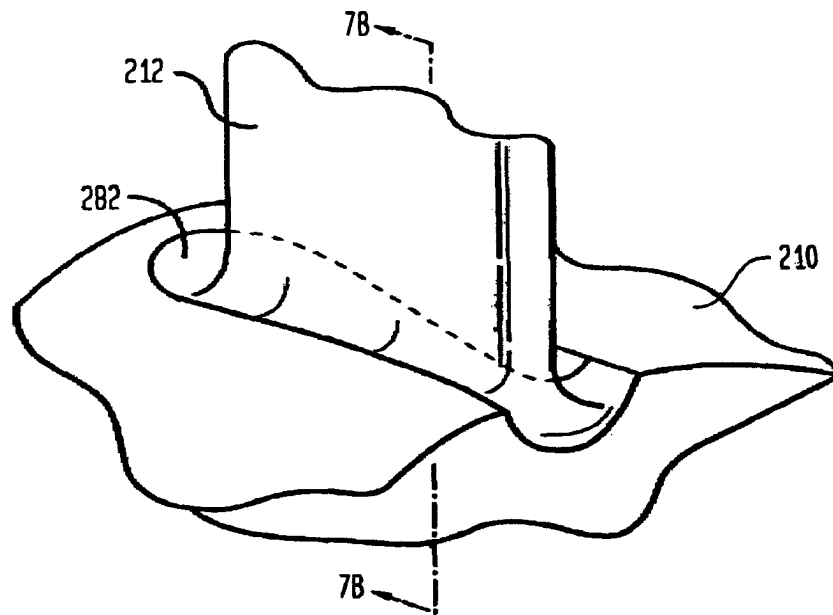


FIG. 7B

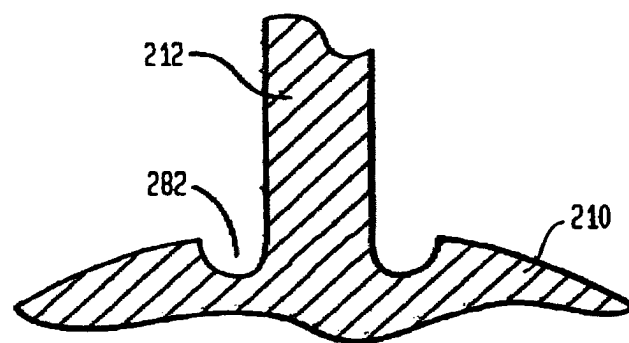
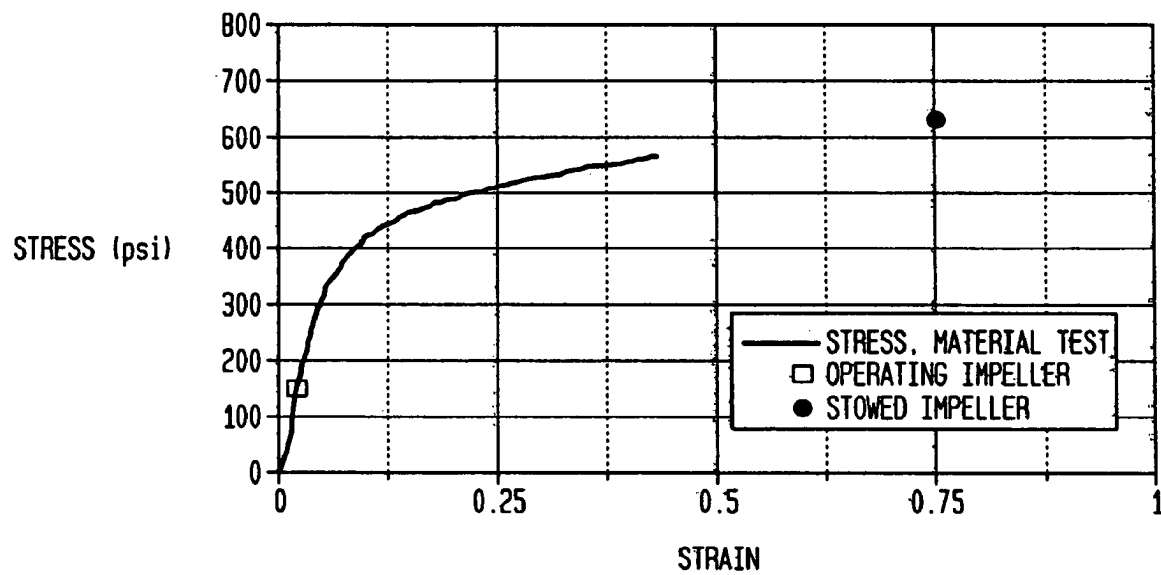
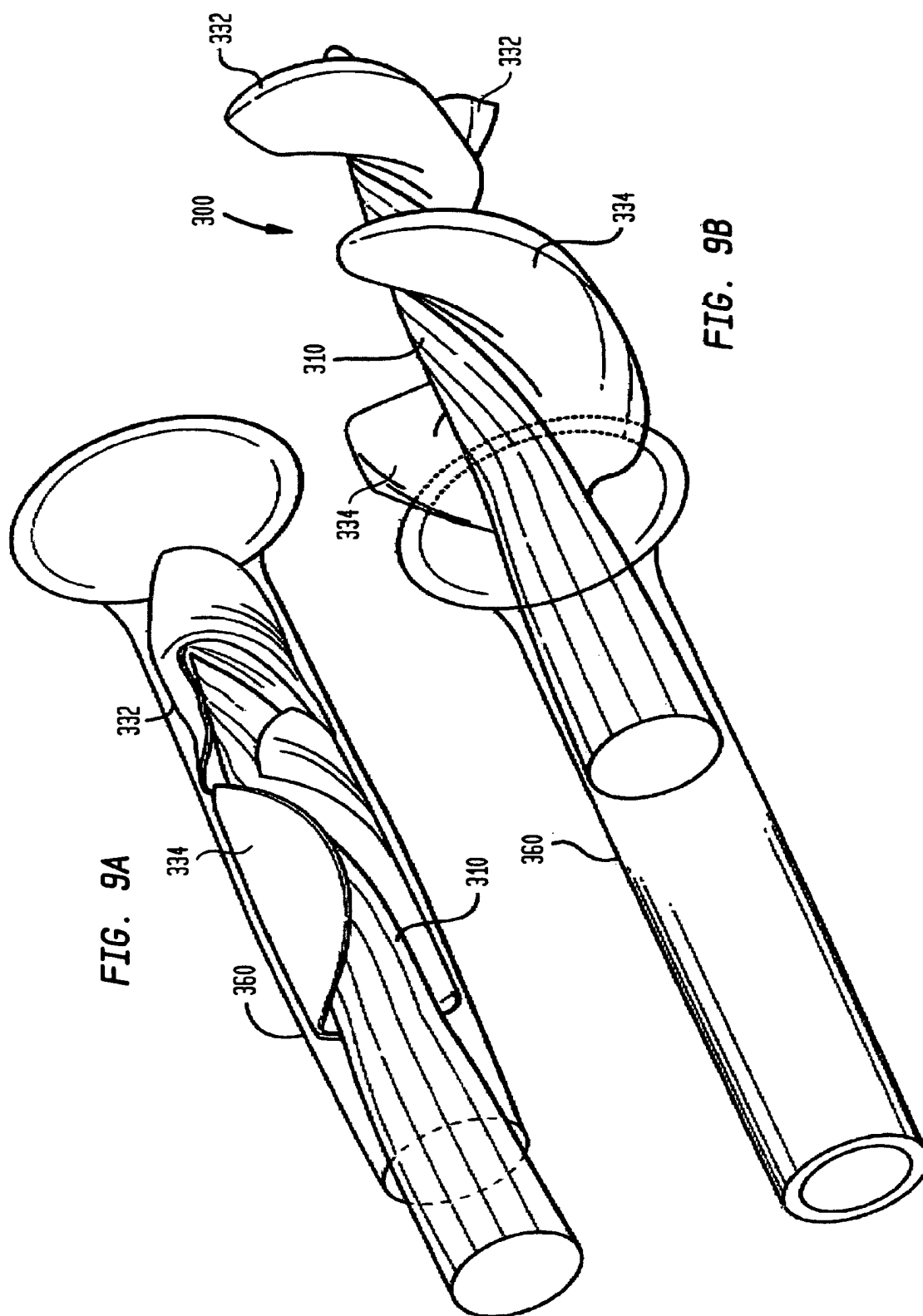


FIG. 8

10 ksi IMPELLER MATERIAL





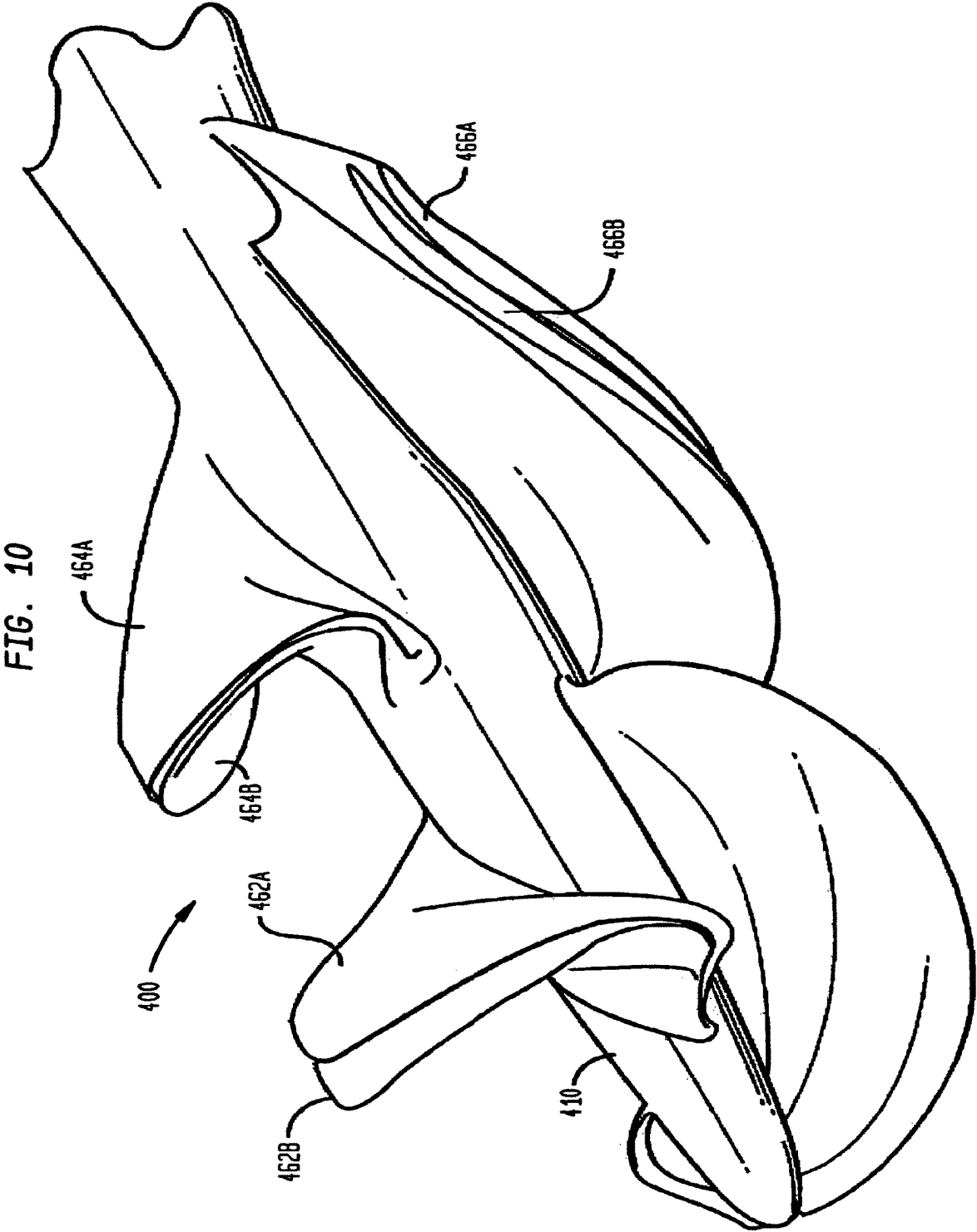
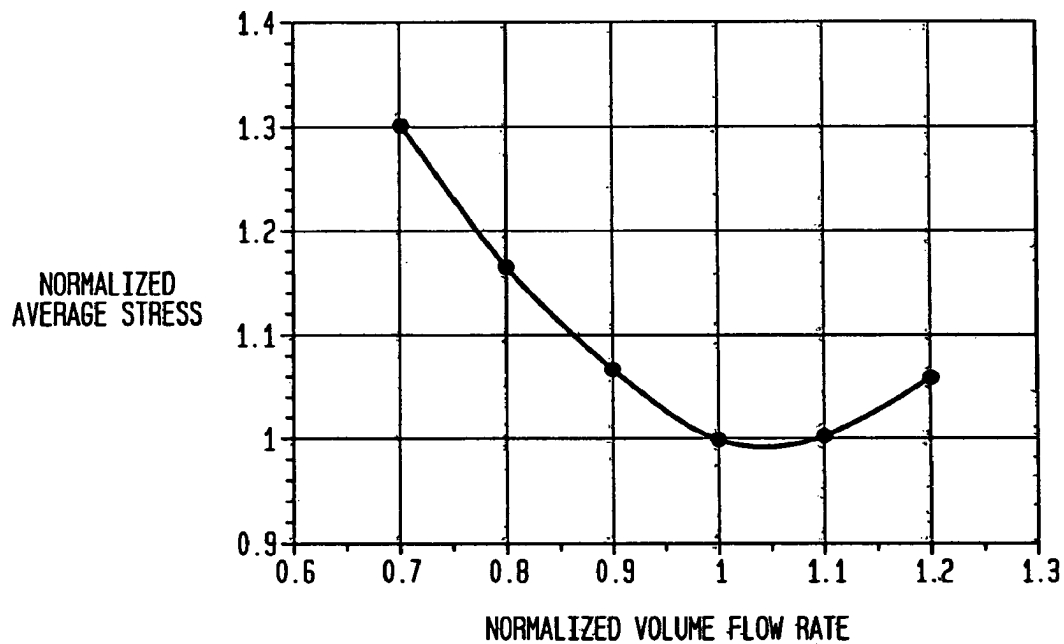
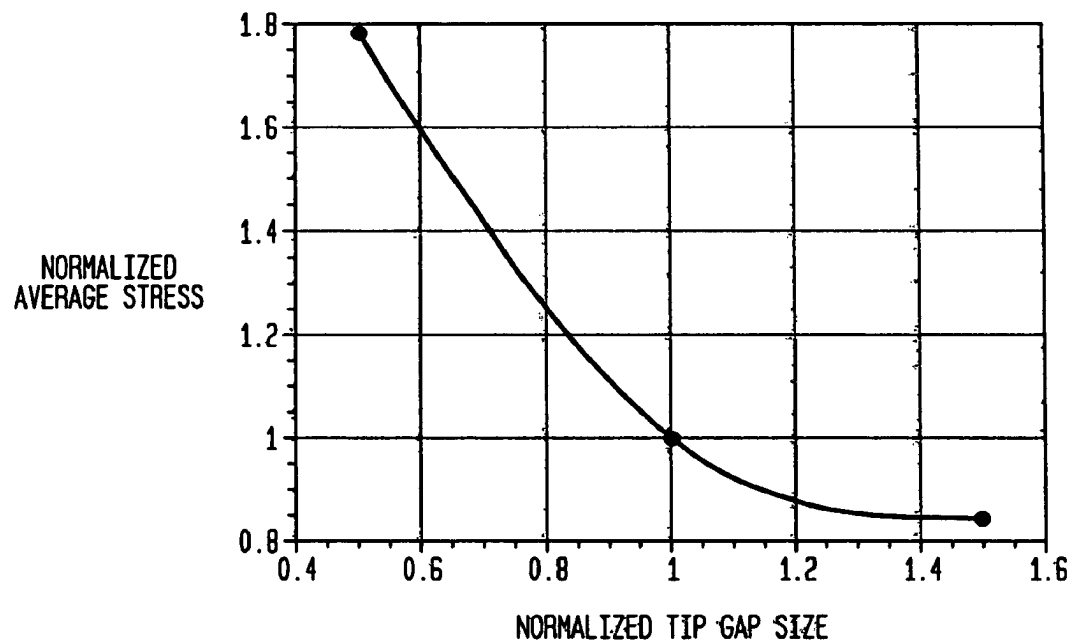


FIG. 11**FIG. 12**

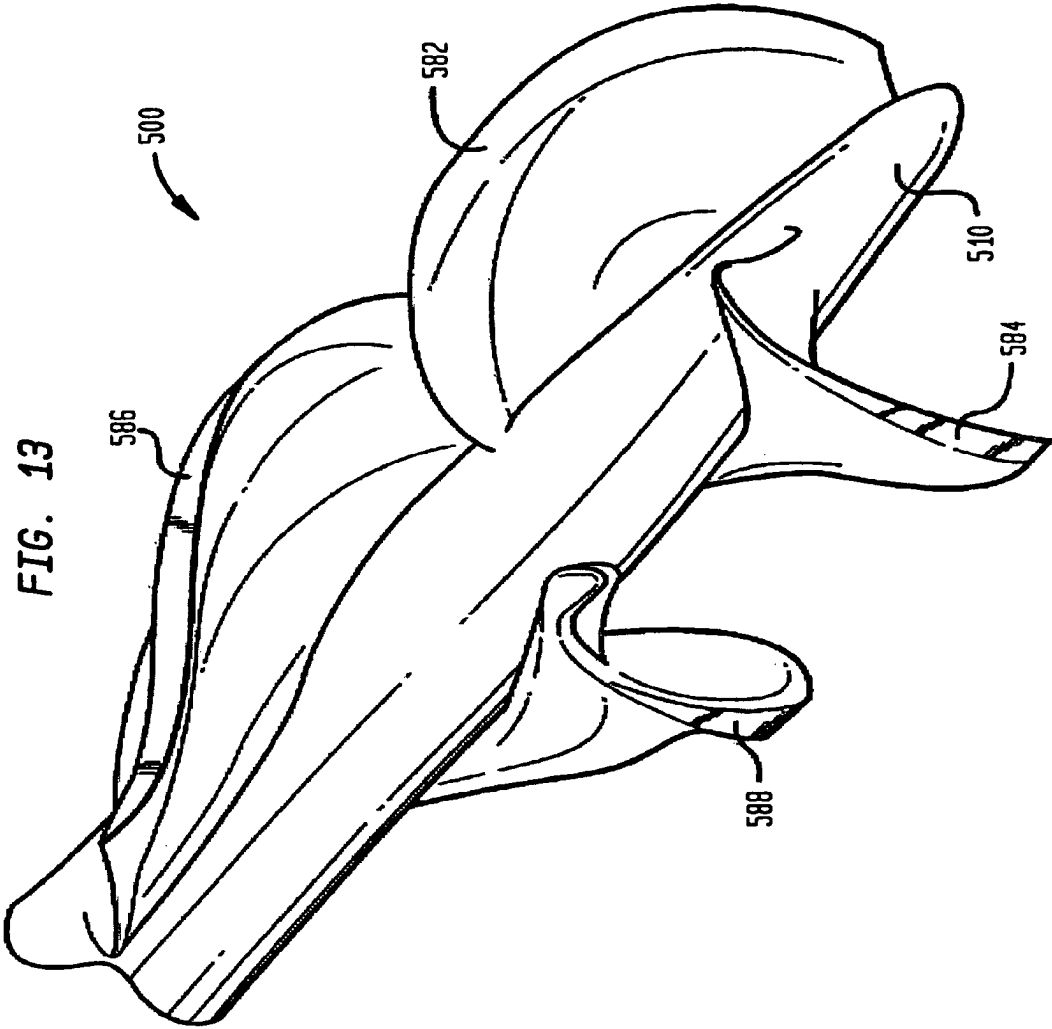


FIG. 14

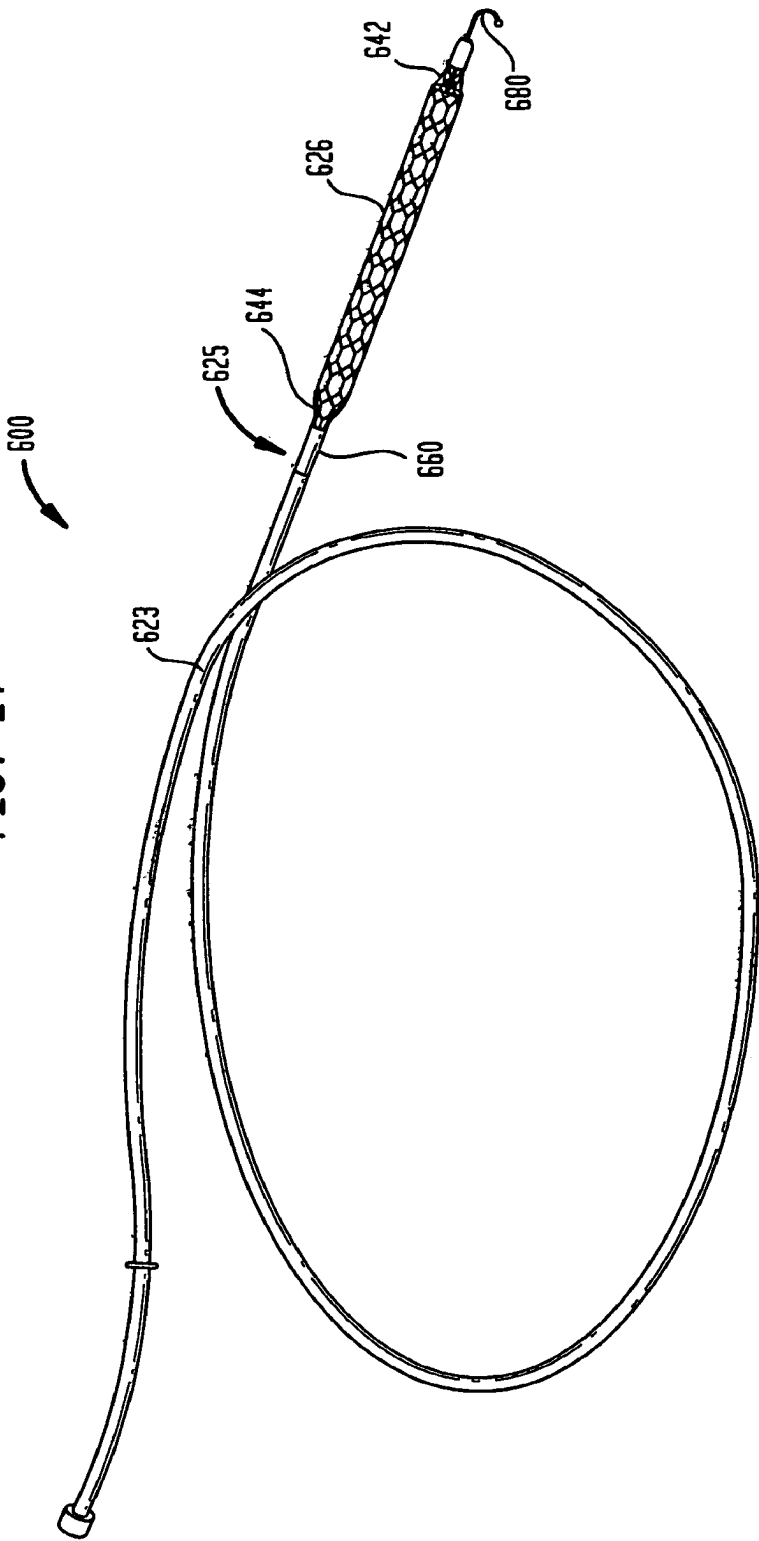


FIG. 15A

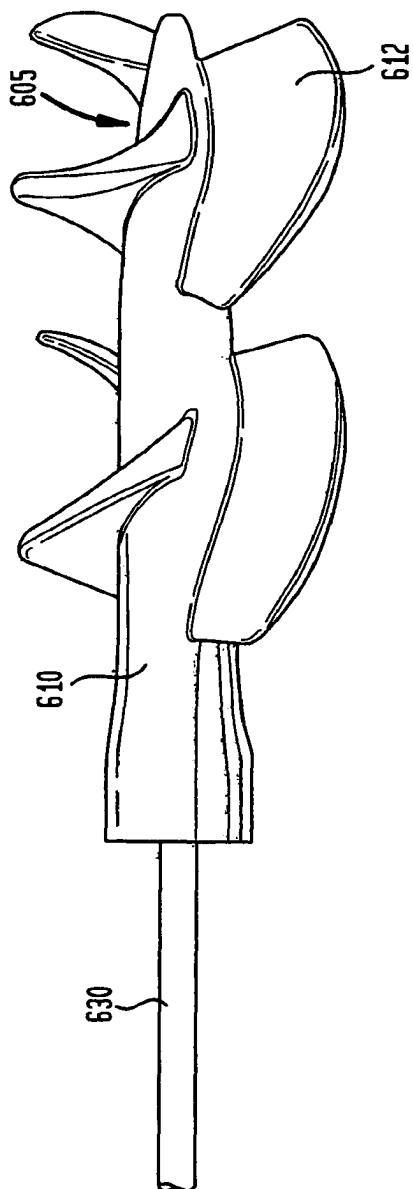


FIG. 15B

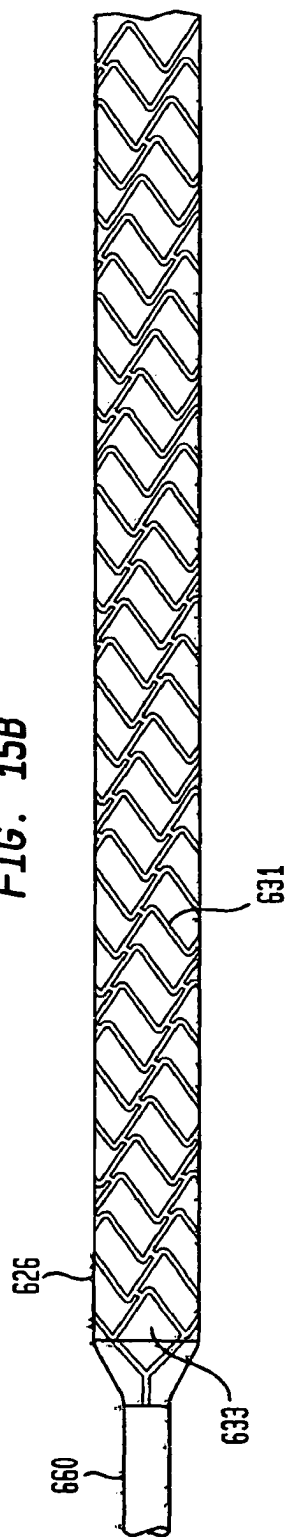


FIG. 15C

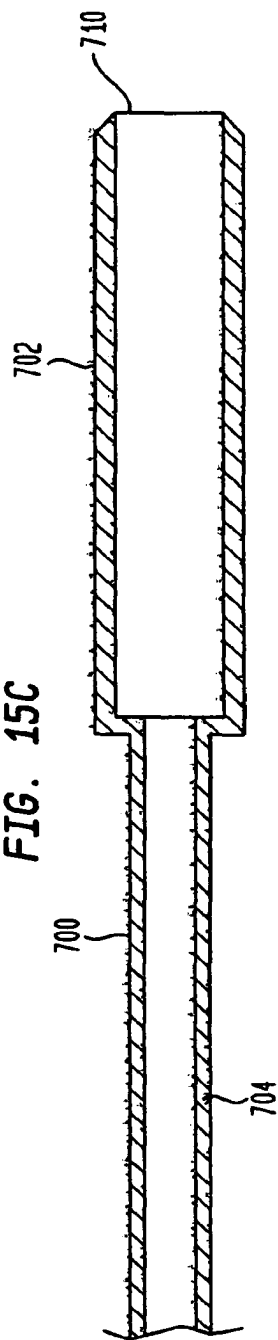
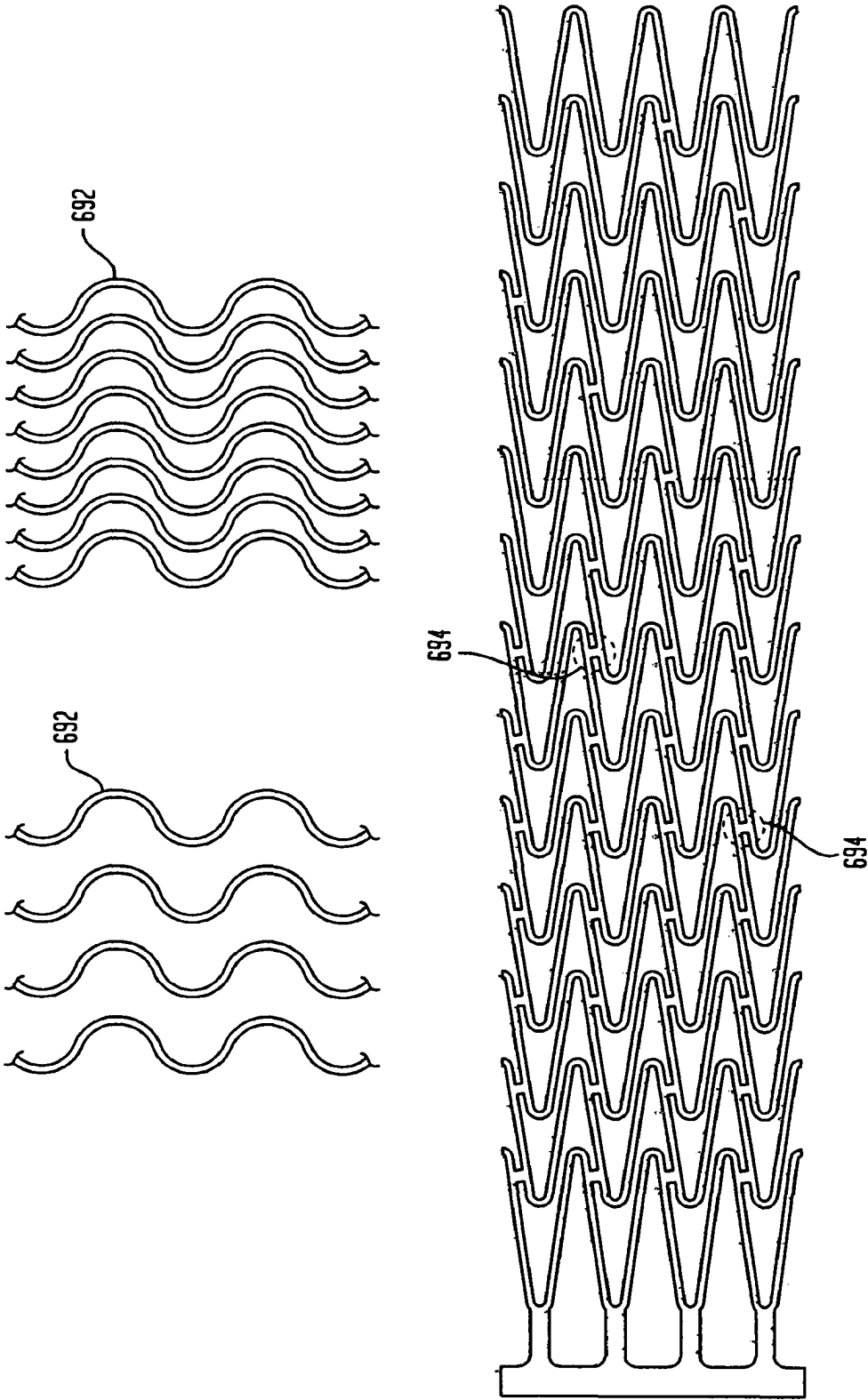


FIG. 16



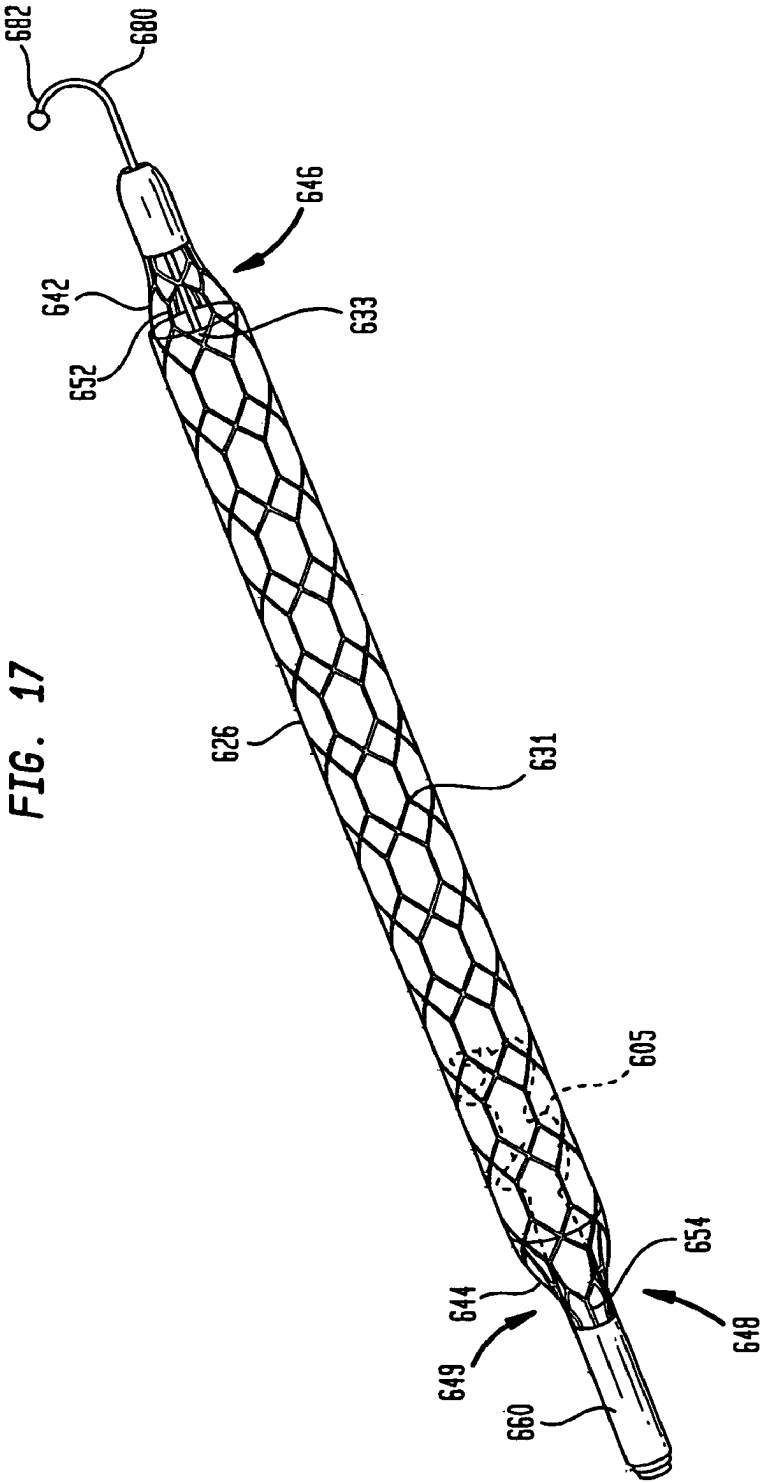


FIG. 18

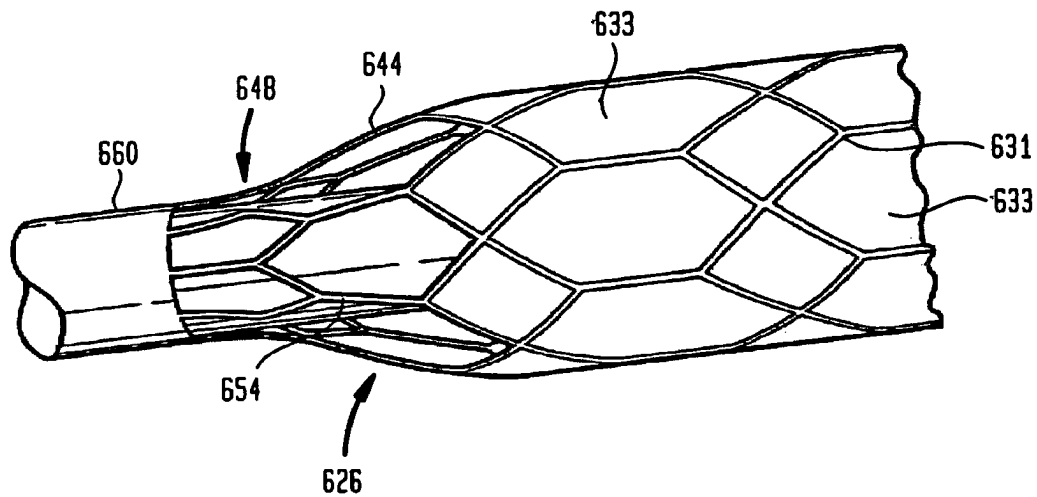


FIG. 19

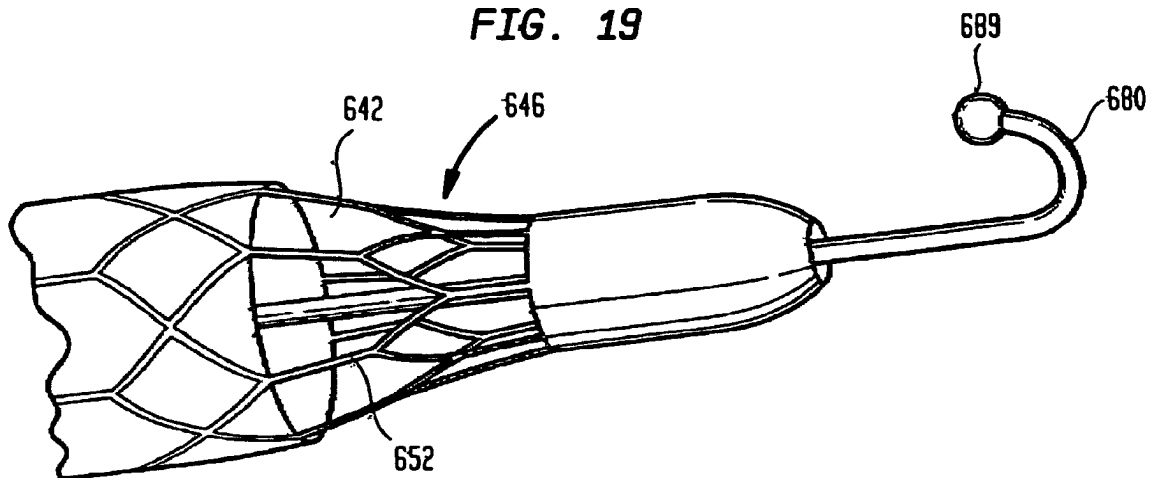
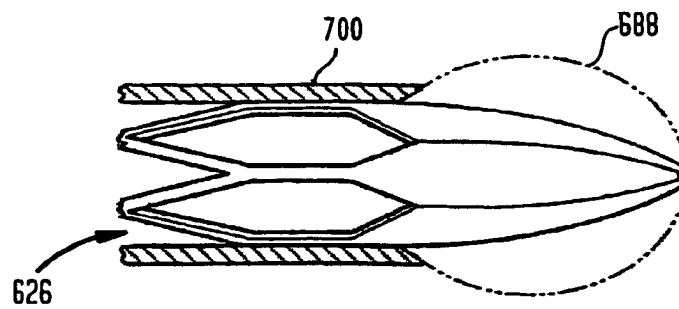
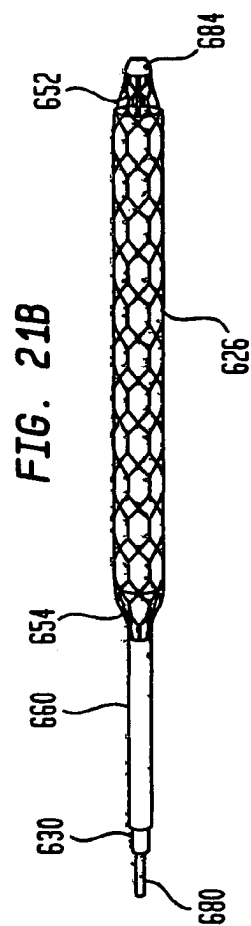
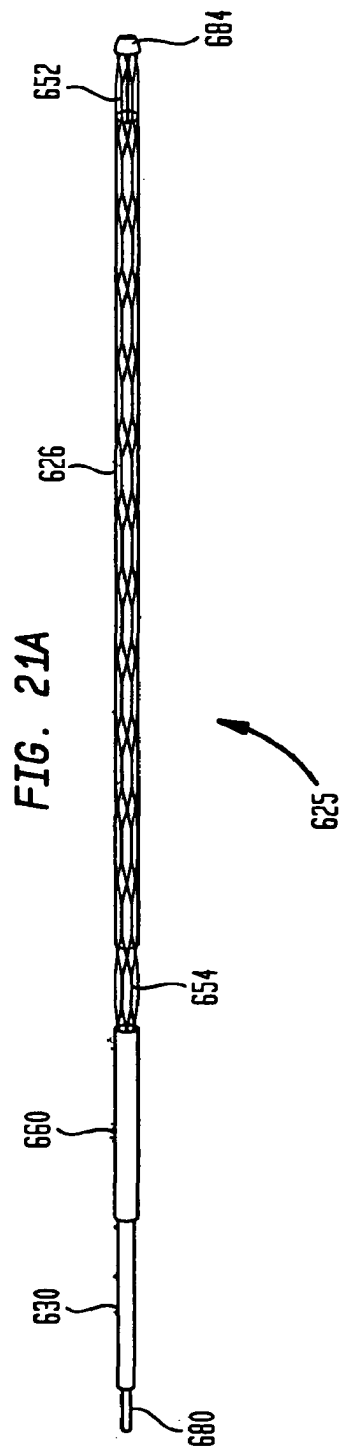


FIG. 20





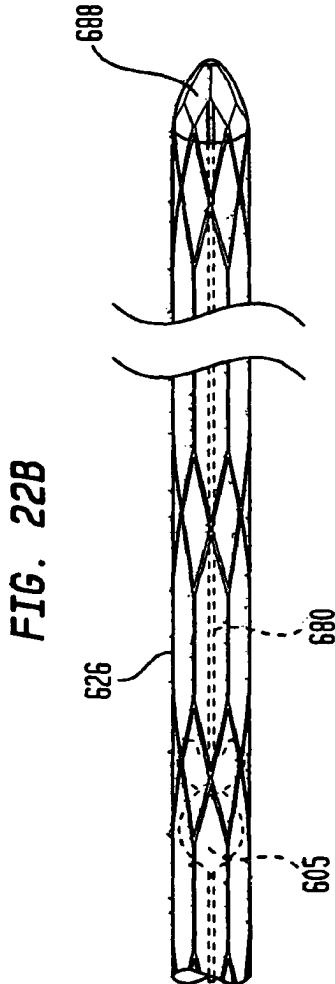
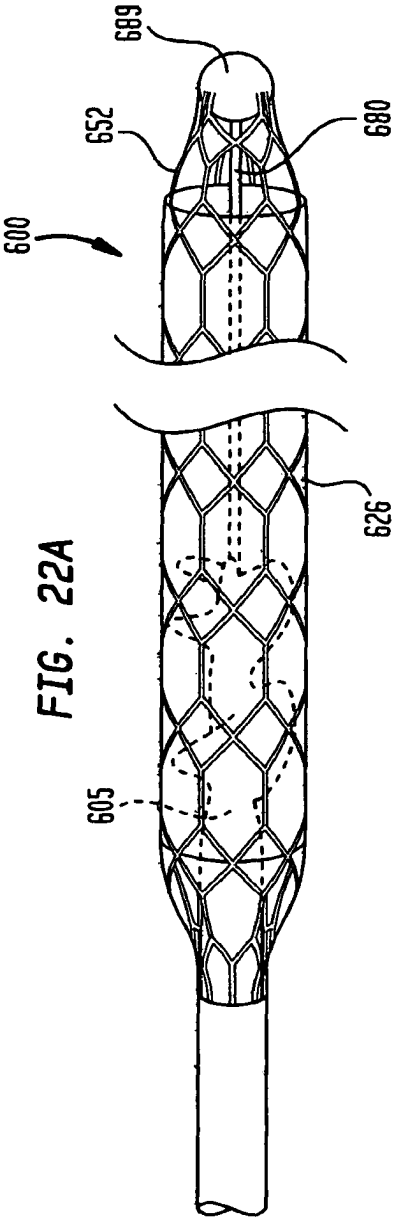


FIG. 23A

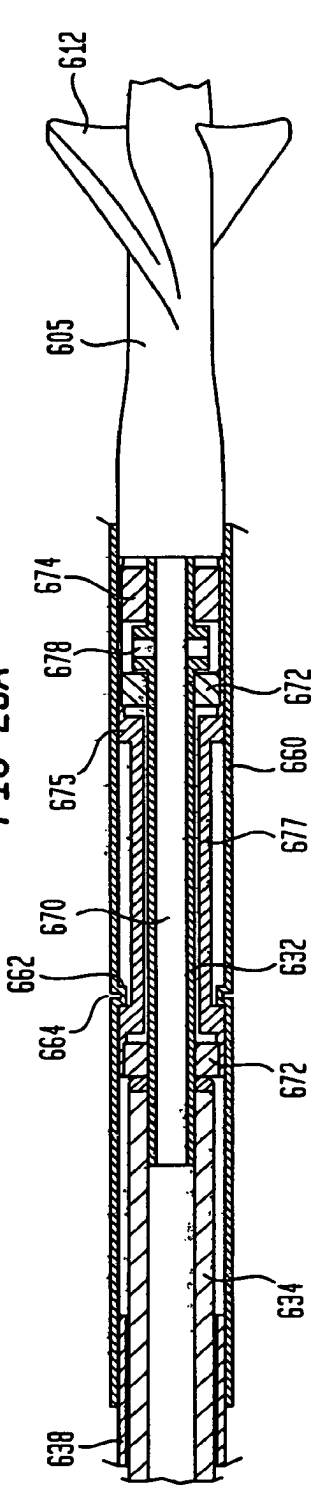


FIG. 23B

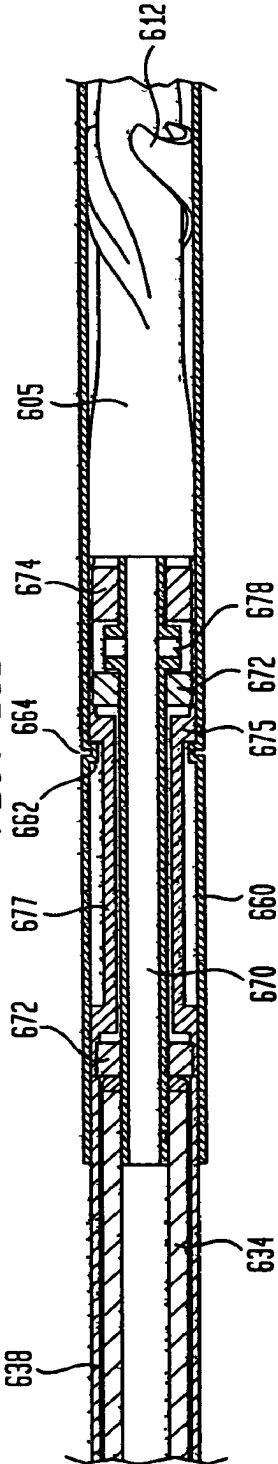


FIG. 23C

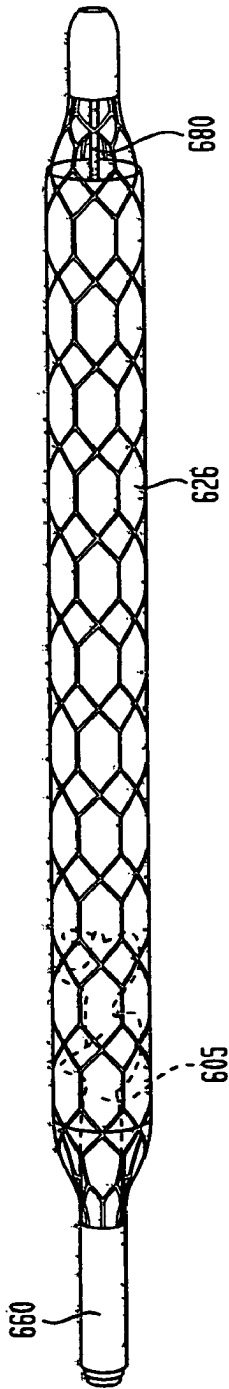


FIG. 24

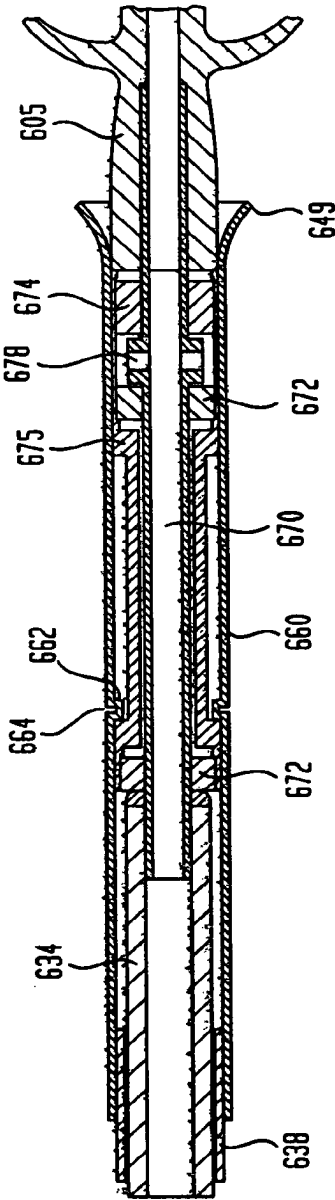


FIG. 25

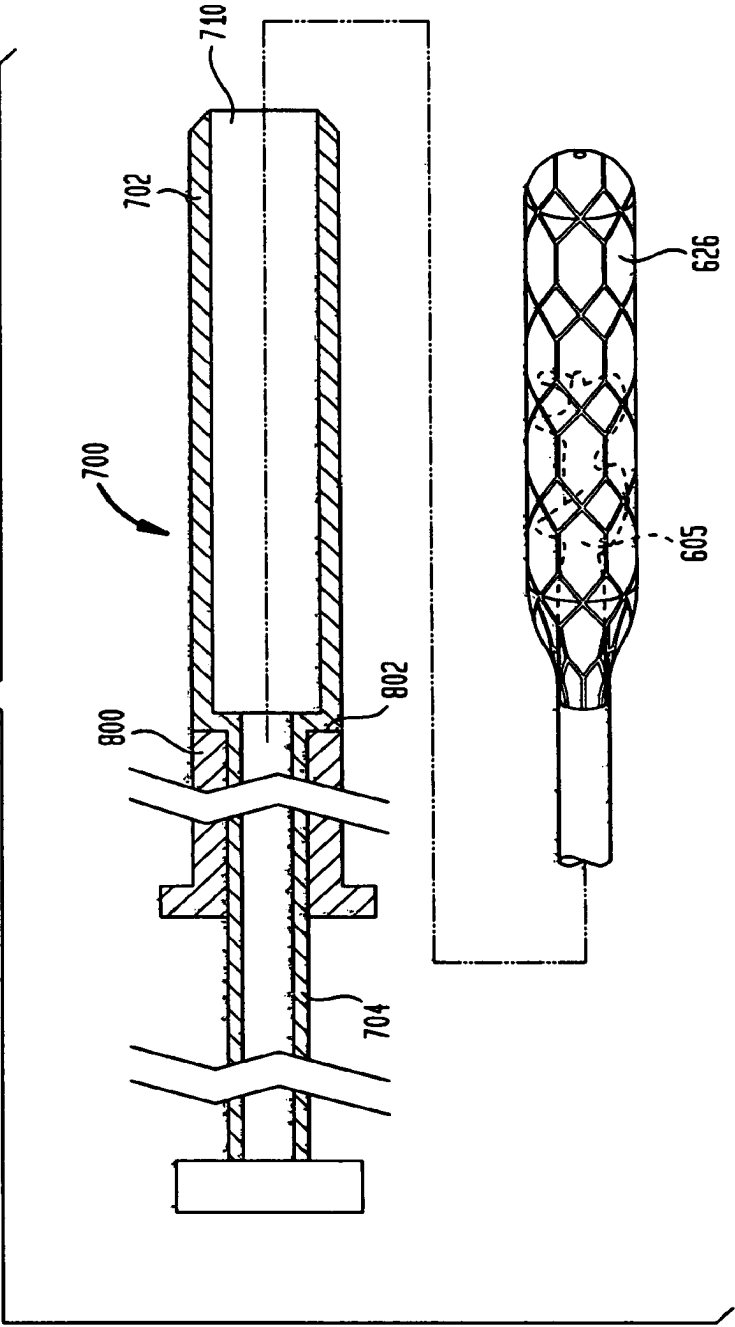
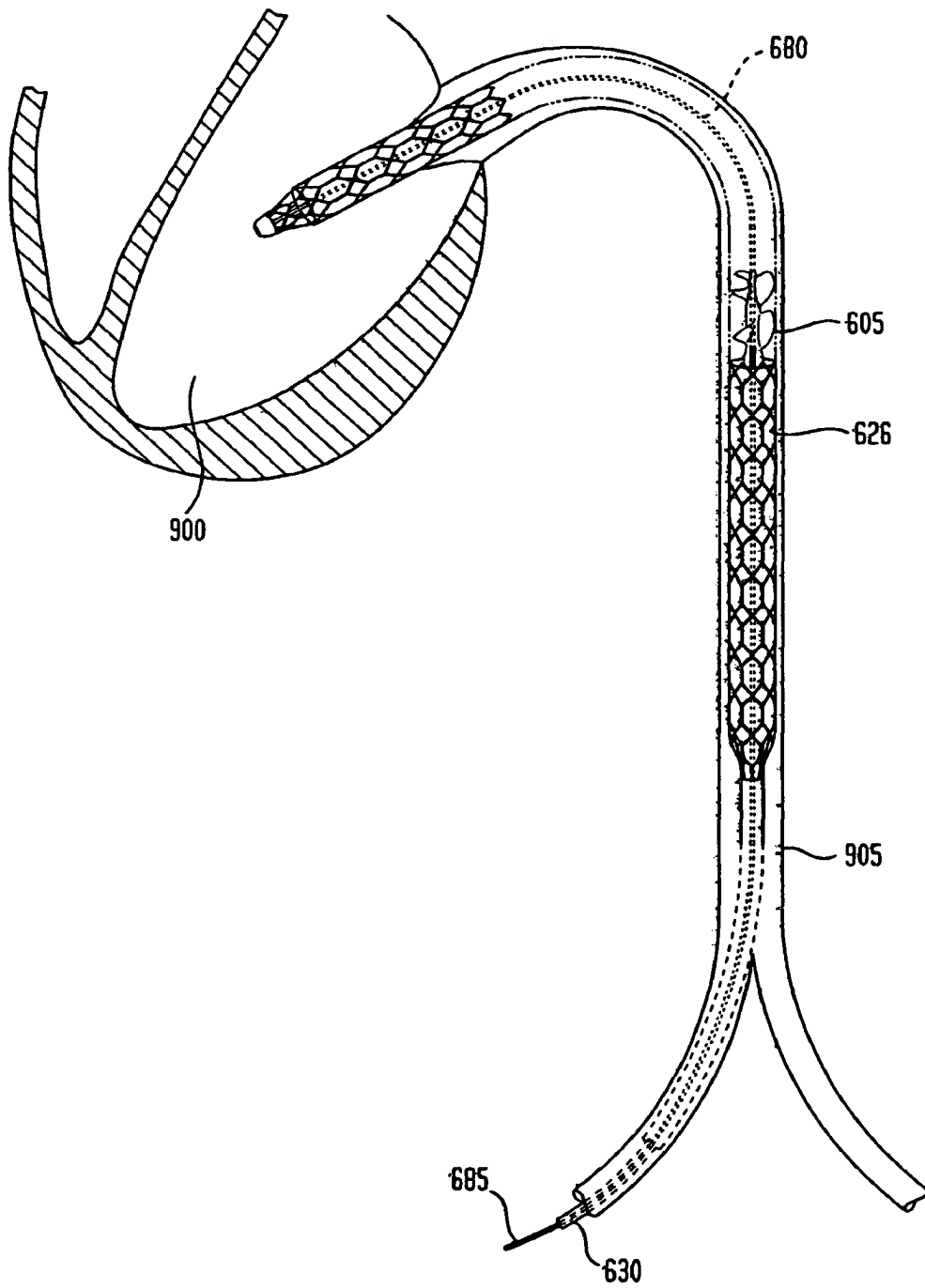


FIG. 26



1

CATHETER BLOOD PUMP HEART ASSIST DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/084,896 filed Oct. 30, 2020, which is a continuation of U.S. application Ser. No. 16/181,963, filed Nov. 6, 2018 and now issued U.S. Pat. No. 10,864,309, which is a continuation of U.S. application Ser. No. 15/633,189, filed Jun. 26, 2017, now U.S. Pat. No. 10,149,932, which is a continuation of U.S. application Ser. No. 15/176,620, filed Jun. 8, 2016, now U.S. Pat. No. 9,717,833, which is a continuation of U.S. application Ser. No. 13/618,071, filed Sep. 14, 2012, now U.S. Pat. No. 9,364,593, which is a continuation of U.S. application Ser. No. 12/945,594, filed on Nov. 12, 2010, now U.S. Pat. No. 9,364,592, which is a continuation of U.S. application Ser. No. 11/728,051, filed on Mar. 23, 2007, now U.S. Pat. No. 7,841,976, which claims the benefit of the filing dates of U.S. Provisional Application No. 60/785,299, filed on Mar. 23, 2006 and 60/785,531, filed on Mar. 23, 2006, the disclosures of which are hereby incorporated by reference herein in their entirety for all purposes. Further, the entire disclosures of U.S. Provisional Application No. 60/610,938, filed on Sep. 17, 2004, and U.S. patent application Ser. No. 11/227,277, filed on Sep. 15, 2005, are hereby incorporated by reference herein in their entirety for all purposes.

FIELD OF THE INVENTION

The present invention relates to fluid pumping impellers, more particularly to expandable fluid pumping impellers. Still more particularly, the present invention relates to blood pumps such as left or right ventricular assist devices with an expandable impeller for treatment of heart disease.

BACKGROUND OF THE INVENTION

Heart disease is a major problem in society, and claims many lives per year. After a heart attack, only a small number of patients can be treated successfully and non-invasively using medicines, such as pharmaceuticals. However, with sufficient mechanical assistance to the heart function, a majority of patients may recover from a heart attack, including even those with cardiogenic shock.

In a conventional approach, a blood pump having a fixed cross-section is surgically inserted within the left ventricle of the heart and the aortic arch to assist the heart in its function. Surgical placement is required, since it is presently impractical or impossible to insert a pump of the size needed for sustaining adequate blood flow percutaneously. The object of the surgically inserted pump is to reduce the load on the heart muscle for a period of time, which may be as long as a week, allowing the affected heart muscle to recover while healing in a resting mode.

Surgical insertion, however, can cause additional serious stresses in heart failure cases. Percutaneous insertion of a left ventricular assist device ("LVAD") therefore is desired. However, the conventional fixed cross-sectional diameter of such an LVAD cannot fit through the femoral artery of the leg in which it must travel to be positioned into the left ventricle. The maximum diameter of such a fixed diameter LVAD would have to be limited to approximately four millimeters for practical percutaneous insertion. This would limit the maximum pumped blood flow rate to approxi-

2

mately two liters per minute, approximately one-half the desired sustaining blood flow value for many cases. While the pumping rate can be increased by increasing the diameter of the device, particularly the diameter of the impeller, the size of the femoral artery is a limiting factor for percutaneous insertion. Hence, there is an urgent need for a pumping device that can be implanted through percutaneous insertion and yet provide the sustaining blood flow rates that conventional surgically implanted pumps provide.

SUMMARY OF THE INVENTION

The present invention may be used as an LVAD, a right ventricular assist device ("RVAD") or in other situations that may benefit from a blood pump that is expandable in situ after being inserted into the body of a patient. The blood pump has an impeller design that allows compression and expansion of the impeller at the discretion of the operator. This compression/expansion feature allows for increased blood flow through the blood pump due to an increase, by expansion, of the impeller size, thereby producing a blood flow capable of sustaining human life without the need for significant contribution by the heart muscle. The blood flow provided is typically at least 4 liters per minute of blood, the flow that is usually sufficient to sustain life.

The difference in using the blood pump as an LVAD as opposed to an RVAD is the location of the pump in the patient and the flow direction of blood through the pump.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is an end view of an embodiment of an impeller having three rows of blades (blade rows);

FIG. 1B is a perspective view of the impeller of FIG. 1A;

FIG. 2A is a highly schematic side elevational view of an embodiment of an impeller according to the present invention in its deployed configuration;

FIG. 2B is a highly schematic side elevational view of the impeller of FIG. 2A in its stored configuration;

FIG. 3 is a highly schematic side elevational view schematically illustrating the deployment of the impeller of FIG. 2A;

FIG. 4A is an enlarged perspective view of a portion of a blade having an embodiment of a winglet;

FIG. 4B is a cross-sectional view of the blade of FIG. 4A within a portion of a vessel;

FIGS. 5A-5C are cross-sectional views of blades having exemplary winglet configurations;

FIG. 5D shows cross-sectional views of exemplary winglet edge geometries

FIGS. 6A-6D are end views of an impeller blade, further illustrating possible winglet configurations;

FIG. 7A is an enlarged perspective view of a portion of an impeller according to the present invention having an indentation in the hub surrounding the proximate end of the blade;

FIG. 7B is a cross-sectional view of the portion of the impeller shown in FIG. 7A;

FIG. 8 is a stress-strain graph for a polyurethane material used to form an impeller blade;

FIG. 9A is a perspective view of an embodiment of an impeller in a stored configuration within a storage housing;

FIG. 9B is a perspective view of the impeller of FIG. 9A after emergence from the storage housing;

FIG. 10 is a perspective view superimposing deployed and operational configurations of an embodiment of an impeller;

FIG. 11 is a graph showing normalized average fluid shearing stresses as a function of normalized volume flow rates;

FIG. 12 is a graph showing normalized average fluid shearing stresses as a function of tip gap sizes;

FIG. 13 is a side view of another embodiment of an impeller according to the present invention;

FIG. 14 is a perspective view of a blood pump according to the present invention;

FIG. 15A is a side elevational view of the impeller portion of the blood pump of FIG. 14;

FIG. 15B is a side elevational view of a cannula in which the impeller of the blood pump of FIG. 14 operates;

FIG. 15C is a partial longitudinal cross-sectional view of the retainer sheath for use with the blood pump of FIG. 14;

FIG. 16 shows highly schematic side views of different mesh designs; and

FIG. 17 is a perspective view of the expandable portion of the cannula shown in FIG. 14 in the deployed state;

FIG. 18 is an enlarged perspective view of the discharge or proximal end of the expanded cannula having a hexagonal mesh;

FIG. 19 is an enlarged perspective view of the inlet or distal end of the expanded cannula showing a guide wire having a distal tip;

FIG. 20 is an enlarged longitudinal cross-sectional view of an alternate embodiment of the distal end of the cannula forming a dilator;

FIGS. 21A and 21B are side elevational views of the expandable portion of the cannula in stored and deployed configurations, respectively;

FIGS. 22A and 22B are longitudinal highly schematic views of the blood pump of the present invention in the deployed and stored configurations, respectively, showing system components;

FIG. 23A is a longitudinal cross-sectional view of the blood pump of FIG. 14 in its deployed configuration;

FIG. 23B is a longitudinal cross-sectional view of the blood pump of FIG. 14 in its retracted position;

FIG. 23C is a side elevational view of a cannula in which the impeller of the blood pump of FIG. 14 operates;

FIG. 24 is a longitudinal cross-sectional view of an alternate embodiment of a blood pump in its deployed configuration;

FIG. 25 is a side elevational view in partial cross-section showing a pre-parked sheath embodiment of the present invention; and

FIG. 26 is a highly schematic view showing the blood pump deployment in a patient.

DETAILED DESCRIPTION

An impeller according to embodiments of the present invention includes a hub, and at least one blade supported by the hub. The impeller may have a deployed configuration in which the blade extends away from the hub, and a stored configuration in which the impeller is radially compressed, for example by folding the blade towards the hub.

In some embodiments, the outer edge of a blade may have a winglet. The impeller also may have a trench or indentation proximate to a blade root to facilitate folding of the blade and/or to reduce shear stresses in the fluid flow induced by rotation of the impeller.

Some embodiments of the present invention include impellers that do not radially compress, but retain a gener-

ally constant configuration. Impellers according to the present invention may be used in various applications, including improved blood pumps.

Blade Rows

Impellers according to embodiments of the present invention may include a plurality of blades which may be arranged in one or more blade rows positioned along the impeller hub. Figures 1A and 1B illustrate end and side views, respectively, of an impeller 100. The impeller includes a hub 10, and a plurality of blades 12 arranged in three blade rows. The first blade row includes blades 102 and 104, the second blade row includes blades 106 and 108, and the third blade row includes blades 110 and 112.

The provision of a plurality of blade rows facilitates the folding of the blades into a stored configuration as compared to the difficulty of folding a single helical blade extending a similar distance along the hub. Each blade row may include the same number of blades, for example, one to three blades. Alternatively, the number of blades in each blade row may differ. For embodiments in which there are more than two blade rows, the number of blades in at least one blade row may differ from the number of blades in other blade rows. The provision of a plurality of blade rows facilitates the achievement of larger values of fluid head or pressure rise than a single blade row, while allowing the impeller to be radially compressed while allowing into a stored configuration.

One approach to impeller design provides an impeller having a long helical blade exhibiting a significant degree of wrap around the central hub. However, the three-dimensional shape of long helical blades limits the degree to which they can be folded without breaking or permanently deforming. By dividing a single helical blade into a plurality (two, three or possibly more) of individual blades, arranged in blade rows, the blades in each, row exhibit less wrap around the hub. Therefore, the individual blades may have an essentially two-dimensional shape which allows easier deformation during the storage process. The combination of two or more blade rows can produce the same flow and pressure as a single helical blade of similar axial extent. For example, individual blades may have a height-to-chord length ratio in the range of about 0.5-1.5, and a plurality of blade rows of such more easily folded blades may combine to provide a similar hydraulic efficiency as a longer serpentine blade. Further, in blood pumping applications, the use of a long serpentine blade may lead to separated flows, leading to thrombosis, which can be avoided using multiple blade rows.

Hence, impellers according to some embodiments of the present invention may have multiple separate sets of blades, rather than a long, continuous helical blade. A continuous long helical blade is difficult to fold up against the hub, and by splitting a long blade into two or three shorter sections, the blade can be more easily folded into a cylindrical volume or space and subsequently deployed when desired.

An impeller according to the present invention may include at least two blades arranged about the circumference of the hub in a first blade row. The at least two blades may be positioned approximately $360/N$ ° apart from one another about the circumference of the hub, where N represents the total number of blades in the first blade row. The impeller alternatively may include a plurality of blades arranged in at least two blade rows, with each blade row including at least two blades. The at least two blades in the first row of blades may be positioned $360/N_1$ ° apart from one another about the circumference of the hub and the at least two blades in the second row of blades may be positioned $360/N_2$ ° apart from

5

one another about the circumference of the hub, where N_1 represents the total number of blades in the first row and N_2 represents the total number of blades in the second row. N_1 and N_2 may be the same or different. The first and second rows of blades may be circumferentially offset relative to one another by $360/2N_1^\circ$.

Preferably, the number of blade rows is two or three. The blade rows may be interleaved (overlapping along the axial direction), which can increase performance but may increase the diameter of the impeller in the stored configuration. If blades are interleaved, they will tend to fold on each other in the stored configuration, increasing the stored diameter of the impeller.

To minimize the stored diameter of the impeller, for example for blood pumping applications, the blade rows are preferably spaced apart along the hub, proximate to each other but not interleaved. For example, the spacing between blade rows may be less than the axial extent of each blade row along the hub axis. A larger blade row spacing allows shear wakes to decay between blade rows, but lengthens the impeller, making it more difficult to move the impeller along a curved path, such as along blood vessels to a desired location within a patient.

Blade rows may also be clocked relative to each other, the clocking being an angular displacement between corresponding blades of each blade row, and in particular, an angular displacement between the trailing edge of a blade in one row and the leading edge of the corresponding blade in the next row. For example, an impeller may have at least a first blade row and a second blade row, the first and second blade rows including a similar number of blades. The blades of the first blade row may be angularly offset (clocked) relative to the corresponding blades of the second blade row. In blood pumping applications, the angular offset can be adjusted to reduce hemolysis. Blades may be clocked so that the leading edge of a following blade does not reside in the wake from a leading blade, and the clocking may be clockwise or counterclockwise to achieve this. Blade rows may be clocked relative to each other to avoid tandem blade effects, where the following blade resides wholly in the boundary layer or wake of the leading blade, so as to reduce shear stresses.

Other Blade Parameters

The amount of lean or tilt of the blades may be adjusted according to the blade stiffness. The blade lean may be a forward lean (toward the pressure face) of between about 30° and about 60° . A forward lean blade tends to deform so as to increase the angle of incidence of the fluid at the tip, and thus increase the load on the blades. Conventional propellers use a backward lean blade which tends to unload the blade tip under structural deflection. Hence, the use of a forward lean is unusual. Forward lean of flexible blades may be used to minimize the gap between a blade tip and the inside surface of a conduit in which the impeller operates. A backward lean may make it more difficult to control the size of the gap.

The twist pitch angles of the blades also may be variable, for example for impeller operation in a conduit. The blade deviation angle may be in the range about 15 to 30 degrees to assist impeller operation at low Reynolds number operation (for example, less than 50,000 for the blade tip chord) within a conduit, hence reducing hemolysis in blood pump applications. The pitch at the tip, measured relative to a circumferential direction, can be appreciably less than the pitch at the root, to match slower fluid flow within a boundary layer near the inside surface of the conduit. Hence, the blade of an improved impeller for operation within a

6

laminar flow profile within a cannula has a blade twist, the pitch being approximately matched to the flow profile, the blade tip having a smaller pitch angle than the blade root. The blade may have a slightly humped appearance in a region of relatively rapid change in blade pitch.

For external flow applications (not in a conduit), the twist in the blade pitch may be in the opposite direction as any boundary layer will tend to be closer to the hub.

The root of a serpentine blade exhibits a geometric characteristic known as camber, the curvature of the airfoil section if laid out on a flat surface. By dividing a single long blade into two or more sections, the camber of the resulting partial sections can be limited, for example to values of less than 10%, for example 5%. In the latter case, the deviation of any section from a straight line will be on the order of five percent of the section chord length. For example, a serpentine blade having a camber of 15% may be divided into three sections, each of which will be substantially linear and more easily folded. The result would be an impeller having three blades, arranged in three blade rows, with performance similar to that of the serpentine blade. The wrap angle (from the leading edge to the trailing edge) can be limited to a maximum of about 30 degrees. Hence, blade rows may each contain substantially two-dimensional blades (compared with a single pumping efficiency), the blades serpentine blade of similar more readily folding against the hub than would a serpentine blade.

The modulus of the blades may be lower for deforming the blades to the stored configuration of the impeller, which may correspond, for example, to strains of 100% 200%. In representative examples, the modulus for the larger blade deformations in the stored configuration may be about ten times less than the modulus for operational stresses. For example, the impeller may include blades formed from a material having a flexural modulus of about 10,000 psi for operational stresses and about 1,000 psi for storage deformations.

FIG. 2A shows an impeller **200** in a deployed configuration, the impeller including a hub **210** and a plurality of blades **212**. Impeller **200** has a radius R_1 in the deployed configuration, as measured from the central longitudinal axis of hub **210** to the outermost blade tip. The deployed diameter is twice the deployed radius, and is the diameter of a circle described by the blade tip as impeller **200** rotates around the longitudinal axis of hub **210**. Also shown is a wall of a conduit **250** through which fluid flows relative to impeller **200**. Impeller **200** may be used as an axial pump, to pump fluid through conduit **250**. Alternatively, impeller **200** may be used as a motive force provider for a vehicle. For example, the impeller may power a boat, such as a jet-boat, or other water craft. In such example, conduit **250** could be a tube immersed in the water surrounding the vehicle, or there may be no conduit at all.

FIG. 2B shows impeller **200** in a stored configuration, with blades **212** folded or otherwise deformed towards hub **210** and held in this stored configuration by a storage housing **260**. The radius R_2 of storage housing **260**, which in this case defines the stored diameter of impeller **200**, is appreciably less than the radius R_1 of the deployed impeller shown in FIG. 2A. A deformed impeller blade **212** may contact the inside surface of storage housing **260** at one or more locations.

In embodiments of the present invention, the flexible blades **212** of impeller **200** can be folded or otherwise radially compressed such that the maximum diameter of the impeller in the stored configuration is approximately half, or less than half, the diameter of the impeller in the deployed

configuration. Referring to FIGS. 2A and 2B, this corresponds to $R_2 \approx (R_1/2)$. A ratio of $R_2 \approx (R_1/2)$ is useful for blood pump applications, allowing a blood pump to deploy to a diameter of between about 6 millimeters and about 7 millimeters within a human body, while being non-surgically inserted with a diameter of between about 3 millimeters and about 4 millimeters. Other diameter ratios are useful for other applications.

Impeller Deployment within a Conduit

FIG. 3 is a schematic view illustrating the deployment of impeller 200. Impeller 200 has hub 210 and blades 212, and is retained in the stored configuration by storage housing 260. Storage housing 260 may be a tube in which impeller 200 is stored prior to deployment. A drive shaft 230 is used to rotate impeller 200. The figure also shows an optional guide wire 280 within rotating drive shaft 230 which can be used to position impeller 200 at a desired location. The rotation of drive shaft 230 may also assist in deploying impeller 200, for example through twisting the impeller out of storage housing 260 if the inner surface of the storage housing has a threaded texture.

On the left of the figure, conduit 250 is shown into which impeller 200 is deployed for operation in its larger deployed configuration. Conduit 250 may represent any structure through which a fluid may flow relative to impeller 200, such as a tube, catheter, cannula, or body vessel such as a blood vessel.

Impeller 200 contained within its storage housing 260 may be deployed within a conduit, such as a utility pipe (water, gas, sewage, and the like), body vessel (such as a blood vessel), portion of a thrust unit for a vehicle, or other structure through which a fluid may flow. The impeller can be conveyed to a desired location within the conduit in a stored configuration, and then deployed to the deployed configuration. Impeller 200 can be deployed by urging the impeller axially out of storage housing 260, for example using drive shaft 230 attached to the impeller. The impeller then unfolds into the deployed configuration using the stored potential energy of blades 212 in the stored configuration.

The stored configuration facilitates conveyance of impeller 200 to the desired location, enabling it to be passed through openings which are smaller than the diameter of the impeller in the deployed configuration. To remove impeller 200 from the conduit after use, the impeller may be radially compressed back into the stored configuration, for example by urging the impeller back into storage housing 260, such as by re-folding flexible blades 212 against hub 210. The stored impeller may then be removed from the use location through an access hole having a dimension less than the diameter of the impeller in the deployed configuration. Hence, impeller 200 can be inserted in the stored configuration through a relatively small entrance hole into a conduit 250 of larger diameter.

Although storage housing 260 is described above as a tube from which impeller 200 may be deployed by axial movement out of the storage housing, that need not be the case. Rather, storage housing 260 may itself be expandable or have an expandable portion, as described below. Expansion of storage housing 260 would allow impeller 200 to deploy, such that the impeller would not need to be pushed axially out of the storage housing to achieve the deployed configuration. Thus, with reference to FIG. 3, conduit 250 may represent storage housing 260 in an expanded condition.

Winglets

Impellers according to the present invention may include at least one blade having a winglet. In some embodiments,

all blades within a blade row may include such a winglet; other blades in the impeller may or may not include a winglet. A winglet may improve hydrodynamic performance of the impeller in the operating state, and may also reduce shear stresses that exist within the fluid being pumped. As a result, when the fluid being pumped includes biological structures such as cells, the degradation of such structures by the pumping action may be reduced.

An impeller blade typically has a pair of opposed faces: a pressure face inducing relative motion of the fluid through pressure as the blade rotates through the fluid; and a suction face inducing fluid motion by suction. Typically, the pressure and suction faces are not planar, but rather are curved in the same general direction to define an airfoil shape. The blade also has a leading edge cutting through the fluid as the blade rotates, a trailing edge, and an outer edge (which may also be referred to as a blade tip or distal end of the blade). A winglet may extend in the direction of motion of the impeller (from the pressure face of the blade), in the direction opposite the direction of motion (from the suction face of the blade), or in both directions.

FIGS. 4A and 4B show perspective and cross-sectional views, respectively, of a blade 212 of impeller 200 having a winglet 222 at its distal end. FIG. 4A shows the cross-section of the blade 212 where it joins winglet 222 as a dashed line, the winglet significantly enlarging the cross-section of the blade at its distal end. FIG. 4B shows blade 212 and winglet 222 in cross-section, in which the winglet and blade form an approximate T-shape. As shown in FIG. 4B, the suction side of the blade is on the right, and the pressure side is on the left. If blade 212 has a thickness between the pressure face and the suction face at the distal end of the blade, winglet 222 may have a width between about 1 and 3 times the distal thickness of the blade, measured in a direction parallel to the blade rotation direction. If blade 212 has a chord length, winglet 222 may have a length approximately equal to the chord length.

Winglets 222 are preferably aerodynamically smooth shapes having leading edges where flows impact the edges of the winglets, and trailing edges where flow is discharged from the winglet surfaces. Winglets 222 preferably have smooth aerodynamic cross-sections, generally in the direction of the mean flow, which is parallel to the flow direction along the blade tip surfaces. FIG. 5D shows possible leading edge geometries, including a radius edge 240, a sharp edge 242, and chisel edges 244 and 246.

Where impeller 200 rotates within a conduit 250 for fluid flow, the distal end of blades 212 in the deployed configuration, either with or without a winglet 222, may be located proximate to the interior surface of the conduit, so as to define a tip gap 255 between the blade distal end and the inner surface of the conduit. The tip gap 255 may be about 10 to 50 percent of the maximum thickness of the distal end of the blade. In such circumstances, appropriately shaping the tips of the blades, such as by providing the tips with winglets 222, can improve the quality of the flow field and reduce shear stresses. As shown in FIG. 4B, the winglet 222 may be proximate to the inner surface of conduit 250, a configuration which may be used as a hydraulic bearing for an impeller 200.

For blood pump applications, simulations have shown that most hemolysis occurs at the blade tip, and that winglets lowered the hemolysis. Alternatively, the tip shape may be rounded to reduce hemolysis. A rounded blade tip reduces flow separation and turbulence at the tip compared with a squared-off tip, and winglets further reduce flow separation and hence turbulence in the wake of the blade tip as it moves

relative to the fluid. By using a winglet **222**, the gap **255** between the tip of blade **212** and the inner surface of conduit **250** may be increased while retaining the performance of an impeller having a smaller tip gap but no winglets. This effect is analogous to retained lift near the end of winglet-equipped airplane wings. Fluid head losses are minimized for a tip gap in the range of about 0.10-0.15 times the maximum thickness of the distal end of the blade for blades without winglets, this range being expected to reduce hemolysis by reducing shear stresses in gap flows. Winglet-equipped blades show minimum fluid head losses at a tip gap of about 0.25-0.30 times the maximum thickness of the distal end of the blade. An increased tip gap reduces shear stresses for impeller operation in a conduit, and for blood pumping applications hemolysis is reduced compared with the use of a smaller tip gap.

FIG. **5A** shows a suction side winglet **262** extending from the outer edge of the suction face of blade **212**. This is a view from the leading edge, in cross-section, so that the blade rotates towards the direction of viewing. FIG. **5B** shows a pressure side winglet **264** extending from the pressure face of blade **212**. The parameters may be similar to the suction side winglet. The function of the pressure side winglet is to reduce flow through the gap **255**. There is less effect of creating a hydrodynamic bearing, but the pressure side winglet "scrapes" low momentum fluid off the inner surface of the conduit **250** and prevents this fluid from entering gap **255** and subsequently being used in the core of a tip vortex. This can reduce shearing stresses in the bulk of the fluid flow.

FIG. **5C** illustrates a combined winglet **266** extending from the outer edge of both the pressure and suction faces of blade **212**. Embodiments of the present invention include the configurations shown in FIGS. **5A-5C**. Numerical methods can be used to design the winglet configurations. Where the blade chord lengths are long and the blade has a significant helical extent, the geometry and shape of the blade tip and the winglet can become complex.

FIGS. **6B-6D** further illustrate winglet configurations, the blade supporting the winglet retaining the same shape in these examples. FIG. **6A** illustrates the outer edge shape **270** of a blade **212** not having a winglet.

FIG. **6B** shows a pressure side winglet extending from the outer edge of the pressure face of blade **212**, extending over portion **272**. The portion **270** of the winglet corresponds to the original outer edge shape of the blade shown in FIG. **6A**.

FIG. **6C** shows a suction side winglet, the portion **274** extending from the outer edge of the suction face of the blade, and the portion **270** corresponding to the original outer edge shape of the blade. In some embodiments of the present invention, the pressure side of the blade will have a radius of approximately $\frac{1}{3}$ to $\frac{1}{2}$ the blade thickness or width. The extent of the winglet may be from $\frac{1}{2}$ to 3 times the blade thickness. A thickness approximately equal to the blade thickness is shown. The winglet is mostly positioned to the downstream half of the blade as shown. The purpose of this is to create a hydrodynamic bearing in which the outer face of the winglet is in close proximity to the inner surface of the conduit in which the blade is operating. With such configuration, the flow in the gap **255** between the winglet and the inner surface of conduit **250** is reduced in strength, and a tip vortex is less likely to form. This reduces shearing stresses in the fluid. Gap **255** can be between about 10 percent and about 25 percent of the base blade maximum thickness, and is an area that is mostly parallel to the fluid conduit **250**. It can be a cylindrical, conical or curved side cylinder where the radius is a function of the axial position

of the blade element. Parameters for pressure side winglets and combined winglets (described below) may be similar.

FIG. **6D** shows a combined pressure side and suction side winglet extending from both the pressure face and the suction face of the blade, the portion **276** extending from the suction face, the portion **278** extending from the pressure face, and the portion **270** corresponding to the original outer edge shape of the blade.

Features to Aid Stored Configuration

Referring to FIGS. **7A** and **7B**, impeller **200** may have one or more structural features which aid achievement of the stored configuration. Such structural features may include one or more indentations proximate to the blade root to facilitate deformation of the blade into the stored configuration. An elongated indentation, such as a trench **282**, may be formed in the hub **210** of impeller **200** proximate at least part of the proximal end of the blade (the blade root, where the blade joins the hub). Trench **282** may be formed adjacent to one or both of the suction face and the pressure face of blade **212**. Preferably, trench **282** is formed in hub **210** parallel with and adjacent to the proximal end of the blade.

The structural features may facilitate movement of the distal end of blade **212** towards hub **210**. For example, a trench **282** around some or all of the blade root can help reduce internal mechanical stresses in blade **212** when the blade is in the stored configuration, for example folded against hub **210**.

In some embodiments, blade **212** may have a cross-section in the shape of an airfoil and the indentation may be a curved trench formed in the impeller hub **210** parallel to the proximal end of the blade. Other structural features which may aid achievement of the stored configuration include hinges (such as living hinges) with one or more indentations or cuts, not shown but known in the art, in the blade **212** and/or hub **210**; forming a portion of impeller **200** proximate the blade root from a more easily deformable material; and the like.

The indentation may also be referred to as a "dillet," and may include any undermining of the blade root. A dillet may be a trench proximate the blade root, for example having a depth between about 0.5 and about 1.5 times the blade width, and/or a width of a similar size range. A dillet can facilitate folding of the impeller blade towards the hub to achieve the stored configuration.

The dillet may also reduce fluid shear stress and flow vortices in a fluid moving relative to impeller **200** as the impeller operates. In blood pumping applications, lower shear stresses lead to reduced hemolysis of the blood.

Hub **210** may have dillets proximate both faces of blade **212**, one dillet facilitating folding of the blade (depending on the direction the blade is folded towards the hub), both dillets reducing the formation of a root junction vortex and hence reducing hemolysis in blood pumping applications. The dillet may be a horseshoe dillet, for example approximating the shape of a horseshoe vortex that would otherwise form at the blade root. Hence, dillets may be provided to reduce shear stresses, even for impeller blades that are not folded.

Blade Materials and, Modulus

Impeller **200** may be in the form of impeller body, including hub **210** and one or more unitary blades **212** formed from a single material. Alternatively, blades **212** and hub **210** may be formed from different materials. Preferably, blades **212** are flexible so that they can be deformed towards hub **210** in the stored configuration. Blades **212** may be formed in any way that allows expansion from a stored

configuration to a deployed configuration, the deployed diameter of impeller **200** being larger than its stored diameter.

Blades **212** may be formed from any material that permits the achievement of a stored configuration in which the blades are folded toward hub **210**. In that regard, the blades may be formed from a rubbery, elastic or other material having sufficient resilience to expand when the blades are no longer held in the stored configuration, such as when impeller **200** is deployed from a storage housing. The blades may be formed from polymer materials, such as polyurethane or other polymers having suitable elasticity properties. For medical devices such as blood pumps, biocompatible polymers are preferred. The average molecular weight may be chosen within a given range to obtain desired properties. Alternatively, the blades may be formed from other flexible polymers, from expandable foam optionally with a skin, or from other compressible or deformable materials including shape-change or shape-memory materials, and metals. Blades **212** may be formed with both a substantially rigid portion and a flexible portion, the blades being deformed towards hub **210** by deformation of the flexible portion. The flexible portion may include a hinge, such as a living hinge, a narrowed region, a material which is different from the material of the rigid portion, or other configuration.

Blades **212** and (optionally) hub **210** may be constructed of a low modulus polymer, for example a low flexural modulus polyurethane (this term includes polyurethane ureas, which were used to form impellers according to the present invention). Impeller **200** may be a unitary structure, with the blades and hub formed as one from the same material, for example by molding a polymer.

In some examples, blades **212** may have a stiffness approximating that of a thick rubber band. In such embodiments, the blades will have some stiffness, but will deform under operating loads. For example, the material forming impeller **200** may be chosen so as to have a linear modulus at operational stresses, allowing predictable deformation of the blades under load, and a non-linear modulus at the higher stresses used to deform the blades into the stored configuration.

Impeller **200** may have blades **212** formed from a polymer, such as a polyurethane, having a flexural modulus (for operational stresses) between about 3,000 psi and about 30,000 psi, more preferably between about 5,000 psi and about 20,000 psi, and still more preferably between about 7,000 psi and about 10,000 psi. The modulus for operational stresses corresponds to deformations of the impeller during operation, which in some examples may correspond to strains of approximately 5%. The blade thickness may be reduced when using higher modulus materials to achieve the desired flexibility.

Impeller blades **212** may, for example, occupy as much as 95% of the compressed volume of impeller **200** when the impeller is in the stored configuration.

FIG. 8 is a stress-strain curve for a non-linear material that can be used to form impeller **200** according to the present invention. The left (low stress) filled circle corresponds to an impeller operating point (stress and strain under operating conditions) and the right (high stress) filled circle corresponds to the impeller stored configuration. The stress/strain relationship is approximately linear at the impeller operating point, so that deformations due to operational stresses can be accurately predicted by numerical modeling. The stored configuration, in which blades **212** are folded against hub **210**, is within a high strain non-linear portion of the curve. This allows the stored configuration to be achieved without

passing the material tensile failure point, and also reduces the stresses necessary to achieve the stored configuration. In example impellers **200**, the maximum material elongation in the stored configuration is about 75 percent.

Preferably, a non-linear material, such as one having the characteristics of FIG. 8, is used for blades **212**. This allows the blade material to be relatively stiff at operating loads, and relatively flexible at higher strains, such as when blades **212** are folded in the stored configuration. For example, the strain might be 1-10 percent at operating loads and 75 percent while folded, and the stress/strain curve may correspond to a higher modulus (e.g., 10,000 psi) at operating loads, and to a lower modulus (e.g., 1000 psi) at the higher loads associated with folding. The stress-strain curve may have two approximately linear regions with a sharp change in slope between the operating point strain and the folded strain.

Impellers **200** may be fabricated from commercially available polyurethane polymers, for example having a modulus between about 5,000 psi and about 10,000 psi. Example impellers according to the present invention were fabricated as unitary bodies (including hub and blades) from elastomeric polymers. Example materials used include Conathane™ TU-901 (Cytec Industries, Inc., West Paterson, NJ), which had a modulus of about 10,000 psi for operational deformations; Conathane™ TU-701 (modulus of about 7,000 psi), and Hapflex™ 560 (Hapco Inc, Hanover, Mass), which had a modulus of about 5,000 psi. However, other polyurethanes, other polymers or other materials may be used.

A polymer impeller **200** retained in the stored configuration for excessive time periods may not properly deploy, for example due to creep or electrostatic welding between adjacent polymer surfaces. Preferably, impeller **200** is retained in the stored configuration only as long as necessary to insert the impeller to a desired location. Hydrodynamic stress and forward lean may be helpful both to deployment and overcoming any hysteresis effect.

Impeller **200** may deploy from the stored configuration due to stored potential energy associated with blade deformation towards hub **210** in the stored state. However, other stored potential energy may be used (for example, using shape memory materials). Depending on the application, external energy may be conveyed to impeller **200**, such as heat (for example, electrical heating of a wire or other structure), centrifugal forces, electromagnetic forces, gas jets, and the like to assist the deployment of the impeller.

Impeller Fabrication

Impeller **200** may be fabricated using molding, investment casting (for example, using a hard wax master), stereolithography, milling, or other techniques. Impellers **200** of the present invention have been fabricated using a flexible mold to avoid the presence of significant mold part lines.

Very small impellers, approximately 6-7 mm in diameter in the deployed configuration, may be fabricated from a polymer (such as a polyurethane) and extracted from a precision mold. This allows production of impellers at very low cost. The flexible blades **212** allow the impeller to be extracted from a mold without becoming mold-locked, and allow the use of one-piece molds, instead of multi-part or split molds. This can be advantageous for producing impellers designed for pumping bio-fluids.

Impeller Optimization

Blade shapes can be optimized using standard computational fluid dynamics analysis (CFD). If the impeller material is not flexible, there is no deformation of the impeller

when rotating. An improved method of optimizing an impeller **200** formed of a flexible material is to optimize the deployed configuration under operational stress (which may be termed the operational configuration). The impeller can be designed so that the operational configuration is optimized, which is not necessarily the same as the deployed configuration under no loading. A structural computation allows the determination of deformation under the load of operational stresses. Hence, impeller **200** may have flexible blades **212** that deform into an optimized hydrodynamic shape when rotating and operating under design load conditions.

The impeller blade **212** can be designed so as to minimize destruction of delicate particles (such as emulsion droplets, suspensions, and the like) within a biological structures fluid. A CFD model such as cells, may be used to simulate the through an intermediate be used to destruction shear stresses experienced by particles passing simulated impeller. Time integrations of shear stresses experienced by the particles may provide an estimated probability of cell in a biomedical application. A split blade design, in which there are a plurality of blade rows such as discussed above, reduces the residence time in which cells remain in intermediate shear stress regions, allowing an advantageous reduction in cell or other particle destruction compared with a single long helical blade.

The impeller blade(s) **212** may deform during operation, and the optimum configuration of a blade may be achieved only upon deployment and rotation. For example, the optimal, design configuration of blade **212** may be achieved only with operational stresses. Hence, blade deformation in operation, due to flexibility of the blade, need not lead to reduced performance. Successful operation can occur even when impeller **200** exhibits significant deflections from a manufactured shape. The impeller can be manufactured with allowance for the deflection included in the design. The configuration of an impeller operating at a predetermined rotation rate, or within a predetermined operating range, can be optimized. Hence, in further embodiments of the present invention, the operational configuration of the impeller, including deformation due to operational stresses, is optimized.

For blood pump applications, CFD optimization may be used to minimize flow velocity over blade surfaces (Reynolds number), vortex formation, flow jets, root junction flows, and to avoid formation of separated flows that may lead to thrombosis.

Reynolds Number

An impeller **200** according to the present invention can operate in a low Reynolds number conduit flow, where the conduit boundary layer comprises a majority of the flow in the conduit. The Reynolds number is the product of blade velocity and chord length, divided by the fluid viscosity. The Reynolds number varies with radius, and generally refers to the tip (distal end) of the blade unless otherwise stated. For example, the Reynolds number for operation of a conventional propeller may be on the order of millions, so that there is a turbulent flow transition as the fluid passes over the blade.

Impellers **200** can be used with flows of small Reynolds number, less than 30,000 for the blade tip, for example the pumping of relatively viscous fluids at low velocity or flow rate. Impellers according to the present invention may operate with blade chord Reynolds numbers of between about 1,000 and about 30,000, preferably between about 2,000 and about 20,000, and more preferably between about 5,000 and about 20,000. The operation at such low Reynolds

numbers corresponds to substantially laminar flow of the fluid over the blades. The reduced turbulence leads to reduced shear stress, and reduces hemolysis in blood pumping applications.

Impellers **200** may also be used for flows of larger Reynolds numbers, such as from 100,000 to several million. Impeller diameters can be in the range of several millimeters (or less) to several meters, depending on the application.

For operation in a conduit **250**, impeller **200** may be located within a fully developed laminar flow profile, for example at a distance of about 10-15 times the conduit diameter from the conduit inlet.

A plurality of blade rows may be operated at a lower Reynolds number than a single longer serpentine blade having similar hydraulic efficiency. Each blade row may be separately optimized, for example to obtain substantially laminar flow. The blade rows may be clocked relative to one another to reduce hemolysis in blood pumping applications. Flow separations leading to thromboses may be avoided. Further, each blade row—may include a different number of blades, for example 1, 2 or 3 blades. Hence, a plurality of blade rows may be used to reduce hemolysis in blood pumping applications compared to a single serpentine blade, while retaining similar or improved efficiency. For example, axial heads may be obtained similar to mixed flow pump heads.

For example, in an impeller **200** operated within a cannula as a blood pump, such as blood pump **600** described below, the blade chord Reynolds number of the first blade row was about 12,600, and was about 15,800 for the second blade row. This suggests that the flow was substantially laminar. In this case, the blades **212** may not exhibit a transition to turbulent flow over the blade surface (where the shear stress suddenly jumps to a higher value), which for blood pump applications leads to lower hemolysis.

Hemolysis

Hemolysis refers to the breakdown or destruction of red blood cells, releasing the hemoglobin contained therein. For blood pumping applications, hemolysis for a given impeller **200** can be estimated using equations known in the literature, and parameters discussed herein adjusted to reduce hemolysis.

Both the hemolysis and platelet activation analyses can be conducted by combining the model of Garon and Farinas, "Fast Three-Dimensional Numerical Hemolysis Approximation," *Artificial Organs*, 28(11):1016-1025 (2004), with empirical correlations of Giersiepen et al., "Estimation of Shear Stress-related Blood Damage in Heart Valve Prostheses—In vitro Comparison of 25 Aortic Valves," *International Journal of Artificial Organs*, 13(5), 300-306 (1990). Giersiepen proposed empirical correlations for LDH-release by platelets:

$$\frac{\Delta LDH}{LDH} (\%) = 3.31 \times 10^{-6} t_{exp}^{0.77} \tau^{3.075},$$

and for Rb-release by red blood cells:

$$\frac{\Delta Hb}{Hb} (\%) = 3.62 \times 10^{-5} t_{exp}^{0.785} \tau^{2.416},$$

there t is in seconds and τ is in Pascals. The Garon and Farinas model, in short, provides a framework for any damage model of the form

15

$$D = \eta_{exp}^{\beta} \tau^{\alpha}$$

by calculating the net flux of the parameter through a volume. So

$$D = (D_I)^{\beta}$$

$$D_I = \frac{1}{Q} \int_V \sigma dV$$

and

$$\sigma = (\gamma)^{(1/\beta)} \tau^{(\alpha/\beta)}$$

The variable 1 is the scalar form of the stress tensor referred to as the Von Mises Criterion and is specified as:

$$\tau = \left[\frac{1}{2} [(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2] \right]$$

where σ_1 , σ_2 and σ_3 are the principal stresses.

$g/100$ L) is then: According to Garon and Farinas, the normalized index of hemolysis (in NUH=100 Hb D, where Hb is the hemoglobin concentration in grams per liter, and the modified index of hemolysis (in parts per million) is:

$$MIH = 10MIH = 10^6 D.$$

By analogy, the platelet activation rate would be given by $PIA = PID$, where both PI and PIA are platelet concentrations, for example in thousands per microliter.

Garon and Farinas only considered the hemolysis rate and only in laminar flow, and so the principal stresses are in reference to the laminar viscous stress tensor only. Garon and Farinas made no reference to how turbulent flow would be accommodated, but that discussion does take place in other research. Arvand et al. ["A Validated Computational Fluid Dynamics Model to Estimate Hemolysis in a Rotary Blood Pump," *Artificial Organs*, 29(7):531-540 (2005)] actually advocated neglecting the Reynolds stress term in turbulent flow simulations in order to avoid "numerically caused variety" of the scalar form of the shear stress used in the hemolysis regression model, but the more conventional approach has been to use the effective shear stress (laminar plus Reynolds stress). See, for example, Gu et al., "Evaluation of Computational Models for Hemolysis Estimation," *ASAIO Journal*, p. 202-207 (2005).

Both approaches may be taken and compared for both and platelet activation analyses. The flow is simulated using computational fluid dynamics, the shear stress is determined from the determined three-dimensional flow data, and the hemolysis may then be determined from the shear stress distribution.

Stored Configuration and Deployment

Impeller **200** may be stored in a storage housing, such as storage housing **260**, transported to a desired location in the stored configuration, and, once at the desired location, deployed into a deployed configuration. Rotation of the impeller then induces fluid flow at the location. For example, the impeller in the stored configuration may have a diameter approximately equal to or less than half the diameter of the impeller in the deployed configuration, the diameter of the stored configuration being generally defined by the inner diameter of the storage housing. The storage housing may be any assembly which acts to hold the impeller in the stored

16

configuration, and may comprise a tube, sleeve, or similar structure inside which the impeller is stored prior to deployment.

In the stored configuration of impeller **200**, blades **212** may be folded in towards hub **210**, or otherwise deformed or reconfigured so as to present a reduced diameter compared with the deployed configuration. Impeller **200** may be held in the stored configuration by storage housing **260**. In the stored configuration, the distal ends of blade **212** are closer to hub **210** than in the deployed configuration, and the stored diameter of the impeller can be significantly less than its deployed diameter. Stored diameters of about one-half of the deployed diameter or less are achievable.

The storage housing need not have a fixed diameter, as does storage housing **260**, but may include a non-expandable portion, in which impeller **200** is stored, and an expandable portion, into which the impeller can be moved for deployment. Impeller **200** may then deploy within the expanded portion of the storage housing. The expandable portion of the storage housing may also have a stored configuration. For example, the diameter of the expandable portion in the stored configuration may be approximately half or less of its diameter in the expanded state. Alternatively, the entirety of the storage housing may be expandable such that impeller **200** does not have to be moved axially for deployment.

Storage housings that are expandable or that include expandable portions may be held in a compressed state by a retainer sleeve, described below. Impeller **200** may be stored in a compressed configuration within the storage housing when the storage housing is retained in the compressed state. However, once the retainer sleeve is removed from that portion of the storage housing in which impeller **200** is located, the storage housing and impeller can expand to their expanded or deployed configurations. Impeller **200** may be deployed by urging the impeller blades **212** out of the confines of the storage housing, for example by pulling the retainer sleeve away from that portion of the storage housing overlying the impeller. In some embodiments, the retainer sleeve may be expanded in situ so as to allow impeller **200** to achieve the deployed configuration. Other methods of deploying impeller **200** will be clear to those skilled in the art.

Impeller **200** may be deployed by various methods. For example, the storage housing may be expandable so as to have an expanded configuration when impeller **200** is in the deployed configuration, and a compressed configuration when the impeller is in the stored configuration. In such embodiments, the storage housing acts to radially compress impeller **200** in the stored configuration and allows the impeller to deploy when the storage housing expands. Alternatively, for storage housings that do not expand, impeller **200** may move axially out of the storage housing.

FIG. 9A illustrates an impeller **300** in a stored configuration, showing blades **332** and **334** and hub **310**. Blades **332** and **334** are kept folded against hub **310** by the housing **360**. FIG. 9B shows impeller **300** pushed storage housing **360** and in the deployed configuration. Blades storage out of In the embodiment shown, impeller **300** has two rows of blades, as is seen more clearly in the deployed configuration, the first row including blades **332** and the second row including blades **334**.

FIG. 10 shows an impeller **400** including a hub **410** and a plurality of blades shown in both the deployed and operating configurations. The figure allows comparison of the deployed configuration under no load with the deployed configuration under operational stresses, when the impeller

rotates at the operational rotation speed. In the deployed configuration under no load, the blades assume a first shape indicated by reference numbers **462A**, **464A** and **466A**. When rotating in a fluid, the blades deform to an operational configuration indicated by reference numbers **462B**, **464B** and **466B**. Impeller **400** may be designed so that the flexible blades deform into an optimized hydrodynamic shape when rotating and operating under design load conditions.

In general, the blades deflect forward as the lift on the blades is such that they create thrust, a force directed towards the left side of the figure, moving the blades toward the left side of the picture. The leading edge of the second blade row is obscured. In this example, there are two blade rows, each with two identical blades. For example, the first blade row includes blade **462**, shown in an operating configuration at **462B** and under a no load condition at **462A**. The leading edge of each blade transitions smoothly into the trailing edge at the maximum blade radius. For a hub and blades formed from the same polymer, simulations showed that the hub also deflects slightly in a rotational manner, with the second blade row rotated at the root compared to the first blade row.

FIGS. **11** and **12** illustrate optimization for fluid shear stress for an example impeller having a design similar to impeller **400** shown in FIG. **10**. The distal ends of the impeller blades move proximate to the interior surface of a cylindrical conduit such that the tip gap between the blade distal end and the inner surface of the conduit is about 10 to 50 percent of the maximum thickness of the distal end of the blade.

The curves are double normalized, the design point values both being 1.0, the scales being read as percent of design flow and a factor times the value of stress at the design point. For example, FIG. **11** illustrates that at 70 percent of the design flow, the shear stress is 1.3 times the value at the design condition. FIG. **12** shows that making the tip gap smaller than the design value makes the shear stress higher, whereas making the gap bigger than the design value reduces the shear stress by a smaller factor. Therefore, the fluid shear stress can be reduced to lower hemolysis in blood pumping applications, without significantly compromising pumping efficiency.

FIG. **13** illustrates an impeller **500** including a hub **510**, and two rows of blades having two blades each. The first row includes blades **582** and **584**, and the second row includes blades **586** and **588**. Impeller **500** has highly curved leading and trailing edge lines where the blade pitch angles are adjusted for local values of relative flow angle. Impeller **500** was designed for operation inside a conduit, such as a cannula having a laminar flow profile, the flow rates being lower near the inside surface of the conduit. This illustration shows the design elements of a low Reynolds number impeller for use in a left ventricular assist device (LVAD) in which the thickness of the boundary layer on the fluid conduit walls approaches the diameter of the conduit. The Reynolds number for a blood pump application was determined to be in the range of 10,000-20,000.

Expandable Impeller and Expandable Cannula

In some embodiments, an expandable impeller is used together with a cannula which may or may not have an expandable portion. If the impeller is not stored in an expandable portion, the impeller must be moved axially for expansion to its deployed configuration. If the impeller is stored in an expandable cannula or in an expandable portion of a cannula, the impeller expands into its deployed configuration with the expansion of the cannula. This combi-

nation may be used in improved blood pumps, such as an improved left ventricular assist device (LVAD).

For example, a cannula may be provided that has expandable and non-expandable portions, and the impeller may be stored within, or proximate to, the non-expandable portion. The impeller can be urged out of the non-expandable portion of the cannula into an expanded portion of the cannula. The stored potential energy within the flexible blades of the impeller would then induce self-deployment of the impeller, and the cannula may also self-expand through stored potential energy. The expanded cannula then may have the role of a fluid conduit through which fluid flows when the impeller is rotated. An example of such system is blood pump **600** described below. An expandable cannula and impeller may both be stored within a retainer sheath and deployed together when urged out of the retainer sheath, as is also described below.

Applications

Impellers according to the present invention may be used for a variety of applications, including an axial pump for a fluid (gas or liquid), a motive force for a vehicle, or other applications. Applications of the improved impellers according to embodiments of the present invention include pumps for chemical engineering, propellers for airborne or maritime vessels, water pumps, and the like.

Impellers according to the present invention may be attached to one end of a flexible drive shaft. A torque applied to the other end of the drive shaft is then used to rotate the impeller. The torque may be applied by a rotating member, such as a motor.

Blood Pump

As noted above, impellers according to the present invention are well suited to blood pumping applications, including as a left ventricle assist device, as a right ventricle assist device, for pumping blood to other organs, and the like.

For blood pumping applications, the impeller may operate within the laminar flow profile of a cannula flow, so that the blade pitch preferably varies with radius to match the flow profile. An impeller with two blade rows, such as impeller **500** illustrated in FIG. **13**, feature in the second row blades that may have a groove-like takes a helical path from the leading edge to the trailing edge. This arises due to variations in the span wise loading, and allows an axial flow pump using this impeller to achieve a head rise similar to that of a mixed flow pump.

Computational fluid dynamics analysis shows that an axial blood pump—including an expandable impeller with two blade rows was suitable for use in a left ventricular assist device (LVAD). The impeller may be compressed and packaged into a storage housing, such as a tube, cannula, or other structure, for insertion into an object. For an object such as a living subject, the diameter of the storage housing can be about three to four millimeters or less. Having inserted the device, the impeller can be deployed in situ into a geometry that may be about six to seven millimeters in diameter. The impeller then can be rotated using a flexible drive shaft coupled to a drive motor external to the subject. Such impellers may be capable of pumping 4 Lpm (liters per minute), and more preferably 5 Lpm or greater, for example in a left ventricular assist device (LVAD).

In a representative example of such a device, the impeller may rotate at about 30,000 RPM. The impeller may include two or more airfoil shaped blades that form an axial flow pump, and may be positioned using a guide wire. The guide wire may run within a hollow center of the flexible drive shaft, and the hollow center may also convey saline solution or other fluid for infusion, cooling and/or lubrication pur-

poses. The guide wire may be removed, if desired. Implantation into a living subject may be achieved without surgical intervention through an insertion cannula having a diameter of about 3-4 mm. For example, a device including an impeller and a cannula may be inserted in a stored configuration through an insertion cannula in the femoral artery, the impeller and cannula then deploying (expanding radially) to approximately twice the stored configuration diameter when located at a desired location, such as proximate to the aortic valve.

For medical implantation, a drive shaft comprising a metal braid, a polymer braid or a composite material braid may be used, and the drive shaft diameter may be on the order of 1.1 to 2 millimeters, and may be hollow to allow the guide wire to pass therethrough.

An impeller according to the present invention can be operated within a cannula, and a flow of 5 Lpm at 100 mm Hg net pressure rise obtained (220 mm Hg across the impeller with pressure losses elsewhere). These parameters are well suited to blood pumping applications, for example within a left ventricular assist device (LVAD).

A blood pump **600** for use in blood pumping applications, such as those noted above, is shown in FIG. **14**. Blood pump **600** may be broken down into three main parts as shown in FIGS. **15A**, **15B** and **15C**. It should be noted, however, that these features may be combined to produce devices according to the present invention that are intended for applications other than blood pumping applications.

The first part, shown in FIG. **15A**, is an impeller **605** with a drive shaft **630** for implementing a rotational drive to the impeller. Impeller **605** includes a hub **610** and a plurality of blades **612**, and may have any or all of the features of the impellers described above. Hub **610** and drive shaft **630** may be hollow so as to define in blood pump **600** an internal lumen **670**.

The second part, shown in FIG. **15B**, is a housing or cannula **625** in which impeller **605** resides. Cannula **625** has a storage housing **660** for impeller **605** when the impeller is in a compressed state. Storage housing **660** may be nonexpandable. Alternatively, storage housing **660** may itself be expandable or cannula **625** may have an expandable portion for housing the impeller when the impeller is in its operational or deployed configuration. Whether there is a difference in the location of impeller **605** in its operating and stored configurations depends on whether the impeller is moved axially within cannula **625** for deployment or whether cannula **625** expands in the area in which the impeller is stored.

The third part, shown in FIG. **15C**, is a retainer sheath **700** which holds at least a portion of cannula **625** in a compressed state for insertion into a vessel of a patient. Each of these parts will be described more fully below.

The cannula **625** of blood pump **600** has a nonexpandable portion **623** at its proximal end and an expandable portion **626** at its distal end. The expandable portion **626** may be flared at one or both ends to aide in fluid flow. The nonexpandable portion **623** of cannula **625** may be formed from conventional biocompatible polymer tubing and the like. The expandable portion **626** of cannula **625**, on the other hand, may be formed from a mesh **631**, such as a metal or polymer mesh, and an elastomer coating **633**. The mesh predominantly defines the radial stiffness and bending characteristics of the cannula, while the elastomer coats the mesh to form a continuous duct having a fluid-carrying capability.

Mesh **631** may be in the form of a hexagonal cell matrix, or may include circumferential rings **692** and axial connectors **694**, as shown in FIG. **16**. The circumferential rings

predominantly control the radial characteristics while the axial connectors affect axial stiffness and bending performance.

Mesh **631** may be formed from a flexible material, such as a polymer, metal, any shape memory material, or other material, and may include a machined metal cylinder with laser cut voids, a matrix of woven wires, or other configuration. Where mesh **631** is made from a memory metal alloy, such as nitinol, a constant diameter tube of the metal, having a metal thickness on the order of thousandths of an inch, for example, a thickness in the range of 0.005-0.007 inch, may be cut using a laser so as to leave a mesh structure. The constant-diameter mesh may then be expanded/contracted radially to the desired shape using a mandrel, and optionally a clamping mechanism may be used to ensure the mesh conforms to the mandrel geometry. The material is "shape set" to this configuration using, for example, heat treatment. The mandrel, and hence the diameter profile of the expandable portion **626** of cannula **625**, optionally can be customized to a particular patient. Alternatively, mesh **631** may be formed from a polymer. Other suitable materials for mesh **631** include other metals (such as alloys, including other memory metal alloys), polymers, other shape memory materials, and the like.

Use of the laser-cutting and shape-setting steps enables complicated geometric patterns to be formed from the constant-diameter tube. An example cannula design may include a bell-mouth inlet (to minimize hydrodynamic losses), a hydrodynamic diffuser at the outlet (for pressure recovery from fluid velocity), a screen-like device at the inlet end (for avoidance of inlet flow obstructions), and additional material at the screen tip that serves as a dilator when the cannula is contracted.

Once mesh **631** has been formed, a coating, such as elastomer coating **633**, may be applied to the mesh inner surface, outer surface and/or interstitially. The coating (which may be, for example, biocompatible, corrosion resistant and/or flow improving) may be formed by a solution casting method or by other techniques known in the art, including forming the coating as a separate tube, fitting it over the mesh and heat shrinking it to produce a tight fit. An elastic polymer such as Elastane™ or Biospan™ may be used for coating **633**, as may other polyurethanes, or other polymers. Mesh **631** and coating **633** may provide a flexible, expandable portion **626** of cannula **625** that is a conduit for fluid flow. The expandable portion **626** of cannula **625** may be generally cylindrical with a flow inlet **642** at its distal end and a flow outlet **644** at its proximal end.

The mesh **631** is radially expansible in a way which imparts a minimal length change (along the axial direction) during radial expansion/contraction. The expandable portion **626** of cannula **625** may radially contract or expand using stored potential energy, and thus is preferably a self-expanding/self-contracting device.

The radial stiffness of the expandable portion **626** is controllable via the mesh thickness and the geometric density of the cell structure, which can vary along the cannula length. Such variability is useful to match the cannula stiffness with the imposed hydrodynamic loading, enabling a nearly constant radial deflection of the tube when operating as a flow duct (wherein the hydrodynamic pressure varies along the length). This is important in the region of the impeller to provide a constant operational tip gap.

Cannula bending stiffness is also a controllable parameter that may vary axially. For example, where circumferential rings **692** and axial connectors **694** are used to form mesh **631**, the bending stiffness is predominantly controlled by the

21

number and placement of the axial connectors, but also depends on the stiffness of the circumferential rings and the stiffness of the elastomer coating 633. The relative placement of the circumferential rings largely affects the radial stability of the cannula during bending. For example, as shown in FIG. 16, mesh 631 may have a substantial amount of interleaving of adjacent circumferential rings. This configuration yields a very stable cannula with respect to radial buckling caused by a bending deflection. Conversely, a mesh pattern with no interleaving yields a cannula that is prone to radial buckling during a bending deflection. Radial stiffness may be augmented via mesh thickness or mesh density. A dense mesh exhibits greater radial stability than a less dense mesh.

FIG. 17 depicts cannula 625 in the expanded state. The expanded portion 626 of cannula 625 includes a distal end 646 having an inlet 642 through which blood enters the cannula, and a proximal end 648 having an outlet 644 through which blood leaves the cannula. The portion between inlet 642 and outlet 644 is the expandable portion 626 of cannula 625. Inlet 642 may be provided with a plurality of inlet struts 652 which prevent obstructions from entering the cannula. Similarly, outlet 644 may be provided with a plurality of discharge struts 654 which act as stationary stator blades and remove swirl velocity from the discharge flow of impeller 605. Inlet struts 652 and discharge struts 654 may occupy a short section of the cannula assembly (such as 1 cm) and may be flat linear elements arranged in a uniform circular disposition about the central axis of the device or may be part of mesh 631. Alternatively, struts 654 may be formed with airfoil type cross-sections. Impeller 605 is located close to the proximal end 648, and a guide wire 680 extends through cannula 625 and through hub 610 of impeller 605. The blood flow through the expanded portion 626 of cannula 625 is from right to left (as shown in FIG. 17) for an LVAD, blood entering the device through the distal end 646 and leaving the device through the proximal end 648.

An uncut region of the original tube retains the original tube diameter and may be used as the storage housing 660 in the form of a cup for retaining impeller 605 in the stored configuration. Storage housing 660 may be referred to as a non-expanded portion of cannula 625 through which blood does not flow. In this case, the expandable portion 626 of cannula 625 is attached to storage housing 660 through the discharge struts 654. Alternatively, the cannula mesh 631, discharge struts 654 and storage housing 660 may be formed from a single tube formed from the same. Hence, the cannula can and also be various attached components can be manufactured from a single piece of tube, for example from a nitinol tube using laser cutting, with a mandrel used for shaping the mesh portion. Alternatively, various portions of the cannula can be manufactured separately and attached together using welding or other attachment techniques. Storage housing 660 may have a flared end 649 which may be defined by the shape of discharge struts 654 to aid in moving impeller 605 back to its stored position.

Impeller 605 may be held in storage housing 660 in the stored configuration and moved axially into expandable portion 626 for deployment, such as by using drive shaft 630 to urge the impeller out of the storage housing. Impeller 605 then unfolds into the deployed configuration using the stored potential energy of blades 612 in the stored configuration. Alternatively, impeller 605 may be held in the stored configuration in the expandable portion 626 while contracted, and may deploy automatically upon expansion of the expandable portion. In still other embodiments, storage

22

housing 660 itself may be expandable, allowing impeller 605 to expand to its deployed diameter without axial movement.

In an example of cannula 625 described above, the expandable portion 626 thereof was formed from a nitinol tube having an inner diameter of 2.62 mm, an outer diameter of 3.02 mm, and a length of 150 mm. In the expanded state, portion 626 had a nominal inner diameter of 6.6 mm in the expanded section, and a nominal length of 133 mm. The expandable portion included 35 circumferential rings 692, four axial connectors 694 per ring in a fully connected region (involving eight circumferential rings), and one axial connector 694 per ring in a minimally connected circumferential rings). Each region (involving twenty-eight circumferential ring 692 had four waves per ring, with a wave amplitude of 5.05 mm (at cut diameter). The interleaved fraction of the rings was $2.05/5$ (where the interleaved fraction for fully interleaved is 1, the interleaved parameter being the overlapped distance divided by the wave amplitude). Finally, the typical thickness of inlet struts 652 and discharge struts 654 was 0.2 mm.

A rotatable drive shaft 630 provides rotational coupling between a motor (not shown), located outside of the patient, and the impeller 605. Drive shaft 630 may have a substantially rigid portion 632 at its distal end which is connected to impeller 605, and a substantially flexible portion 634. The flexible portion 634 of the drive shaft may be housed within a flexible tube 638 which supports the flexible portion and maintains its shape as it is driven rotationally. The proximal end of drive shaft 630 may be connected to the motor for rotating the drive shaft and with it impeller 605. Alternatively, drive shaft 630 may be omitted, and the electric power may be provided through a proximal portion of the assembly to operate a pump motor and impeller 605.

Drive shaft 630 may have a diameter on the order of 1.1 to 2 mm, and may be hollow to allow guide wire 680 to pass therethrough. The flexible portion 634 of drive shaft 630 may be formed from a metal or polymer braid which is easily bendable so as to achieve a bend radius on the order of 1 cm. Commercially available flexible impeller drive shafts may be used in blood pump 600, such as those formed from metal wire construction. However, heating problems due to friction between rotating and non-rotating components may occur within any small radius bends required for operation. A composite flexible shaft may be used to reduce such heating problems. The heating problem also can be addressed by providing lubricating or low-friction films on one or both of adjacent surfaces with high relative rotational motion. In that regard, a preferred drive shaft 630 may be constructed from coiled stainless steel, with an optional polymer support tube. A particularly preferred polymer is polytetrafluorethylene.

The rigid portion 632 of drive shaft 630 may be supported by one or more bearings 672 retained in a bearing housing 675. A saline solution may be directed into bearing housing 675 through internal lumen 670, and the bearing unit end seal 674 may be dimensioned so that a of clean saline solution is infused (approximately 1-2 cc/hr). This fluid very small quantity into the patient flow helps clean impeller 605 and dampens drive shaft vibrations. The fluid flow may also prevent blood from entering bearing housing 675 and compromising its operation and life. If the density of drive shaft 630 is approximately the same as that of the saline solution or other introduced fluid, most of the vibration can be damped. Drive shaft 630 may be formed from carbon or other fiber and polymer composite which has a lower density than metal and more closely matches the density of the

saline solution. Other lower density draft shafts and/or higher density fluids may be used for vibration damping. The saline solution or other fluid may be introduced to bearing housing 675 through openings 678 in hollow drive shaft 630.

FIGS. 23A and 23B show an embodiment of blood pump 600 having an axially slidable storage housing 660. As can be seen in these figures, bearing housing 675 may have a reduced diameter portion 677 between its ends housing bearings 672. This reduced diameter portion thus defines a longitudinal space for sliding movement of an internal rib 662 defined by an indented annular channel 664 in storage housing 660. In the deployed condition shown in FIG. 23A, storage housing 660 has been moved proximally by the maximum extent permitted by the engagement of internal rib 662 with a proximal shoulder of bearing housing 675, thereby revealing blades 612 of impeller 605 for deployment. In the stored configuration shown in FIG. 23B, on the other hand, storage housing 660 has been moved distally to the maximum extent permitted by internal rib 662 contacting a distal shoulder of bearing housing 675. In this position, the distal end of storage housing 660 surrounds blades 612 of impeller 605, retaining them in the stored configuration.

The internal lumen 670 of blood pump 600 receives guide wire 680. Together, lumen 670 and guide wire 680 assist in positioning blood pump 600 within the patient. Guide wire 680 may have a two-part structure to assist in threading the guide wire through the bearing/seal assembly and through the hub 610 of impeller 605, since part of that threading may be accomplished under factory controlled conditions, rather than at the time of use. In one embodiment, guide wire 680 may have a J-tip 682 which facilitates navigation of the tortuous arterial pathway from the femoral insertion site to the cardiac left ventricle chamber. Guide wire 680 may include an optional device on its proximal end to allow the attachment of a similar diameter extension of the guide wire, residing in the collapsed cannula 625. The lumen 670 in blood pump 600 may have a relatively large diameter relative to the diameter of guide wire 680. Guide wire 680 may have one or more additional distal end features such as a spherical shape, or a valve plug 689 to plug a hole in impeller 605 after withdrawal of the guide wire (see FIG. 22A).

Optionally, the guide wire channel extending through impeller 605 and the bearing unit end seal 674 may have a valve action, as is known in the art, sealing the guide wire passage after removal of guide wire 680. Guide wire 680 may leave a mechanical seal, not shown, upon removal, or the material of impeller 605 may be designed so as to close the opening into lumen 670 upon removal of the guide wire. This avoids excess saline infusion into the patient.

Blood pump 600 may be inserted into the patient's body using a sheathless insertion procedure. Such procedure may employ a retainer sheath 700 having a distal portion 702 and a proximal portion 704, as shown in FIG. 15C. Distal portion 702 may be about 20 cm in length, and have an inner diameter of about 9 fr (3.0 mm) and an outer diameter of about 10.5 fr (3.5 mm). The inner diameter of the distal portion allows storage of the collapsed cannula/impeller assembly. The proximal portion 704 of retainer sheath may be about 1 meter in length with an outer diameter of about 9 fr. This proximal portion 704 may serve as the housing for the flexible portion 634 of drive shaft 630 and for the non-expandable portion 623 of cannula 625.

A "pre-parked" integrated insertion sheath 800 may slide over the proximal portion 704 of retainer sheath 700. The outer diameter of insertion sheath 800 is preferably about the

same as the outer diameter of the distal portion 702 of retainer sheath 700, in this example about 10.5 fr. When the proximal end of the distal portion 702 of retainer sheath 700 is pulled up against the distal end 802 of insertion sheath 800, a smooth transition is evident, as shown in FIG. 25. Thus, the combined retainer sheath 700 and insertion sheath 800 may be inserted into the patient's femoral artery as a single entity. After the insertion sheath is fully inserted into the femoral artery, the proximal portion 704 of retainer sheath 700 may be pushed into the patient, pushing the distal end of the retainer sheath and its contents into the patient's left ventricle.

The most distal end 710 of retainer sheath 700 may have a series of slots (not shown) that allow slight expansion of the retainer sheath distal end during removal of cannula 625. As the expandable portion 626 of cannula 625 must be collapsed during this process, the funnel shape created by these slots and the subsequent bending of the material of the sheath facilitates the collapse of the expandable portion of the cannula. Alternate means may be provided to facilitate the recollapse of the expandable portion 626 of cannula 625.

The drive motor rotates drive shaft 630 without rotating cannula 625 or retainer sheath 700. The operation of blood pump 600 is controlled and monitored by a control unit (not shown) which displays status and controls various functions. Sensors, such as a pressure sensor and a flow rate sensor, may be affixed to various regions of the patient and/or blood pump 600.

The control unit preferably displays rpm of the drive motor, patient blood pressures, blood flow rate, information as to the location of the blood pump in the left ventricle, saline infusion and discharge rates, saline infusion temperature, etc. A filter may also be provided to show the presence of debris or blood in the saline discharge stream. The heart rate and blood flow rate are useful to be able to reduce patient dependency on the machine during recovery. Detailed Description of the Deployment of the Heart Assist Device

Blood pump 600 may be percutaneously inserted through the femoral artery and threaded toward the heart for use, for example, as a left ventricular assist device. Blood pump 600 may be inserted into a patient using conventional cannula insertion methods. The impeller 605 of the device is then expandable in situ to enable an increased blood pumping capacity compared to conventional non-expandable devices. This can eliminate the requirement for surgical intervention.

Insertion may be accomplished using the Seldinger technique which is well known in the art and used daily by surgeons and interventional cardiologists. In such technique, an introducer needle (not shown) is inserted into the femoral artery and used to introduce guide wire 680. Once guide wire 680 is in place, the needle is withdrawn. An optional predilator (not shown) can be used over guide wire 680 to open up the arteriotomy (an opening in the femoral artery) to a size appropriate for insertion of blood pump 600.

A guide wire extension 685 (see FIG. 26) contained within the collapsed cannula 625 may be attached to the attachment device at the proximal end of guide wire 680. By fixing the guide wire extension at the proximal end of guide wire 680, the entire assembly is moved along the guide wire through the femoral artery opening previously created. During this process, the attachment of guide wire 680 to the guide wire extension 685 enters the body of the collapsed cannula assembly. To facilitate insertion without an introducer sheath, the distal end of the device may be provided with a tapered distal end dilator 688, shown in FIGS. 20 and 22. This may be a compressed form of the inlet 642 of

cannula **626** in which the inlet struts **652** fold down into a tapered, closed configuration similar in profile to a conventional dilator tip.

The blood pump **600**, in its collapsed state, is then threaded over guide wire **680** and inserted into the artery. Once blood pump **600** is positioned, guide wire **680** may be removed. A nose bearing or seal at the distal end of impeller **605** can then seal the guide wire opening through the hub **610** of the impeller. This allows saline solution to be injected into impeller **605** for cooling and lubrication purposes, as well as to prevent blood from entering the lumen **670** of blood pump **600**.

At the time of insertion, the expandable portion **626** of cannula **625** and impeller **605** are collapsed and may be contained within retainer sheath **700**. As described above, the proximal end **704** of retainer sheath **700** may include an optional second integrated insertion sheath **800** which replaces the function of the separate introducer sheath when the device is positioned in place. If there is no "introducer" sheath present during the insertion of the assist device into the femoral artery, the process is referred to as sheathless insertion. In FIG. 25, a smooth transition from the diameter of insertion sheath **800** to the maximum diameter of retainer sheath **700** is shown.

During the insertion of blood pump **600**, the transition region has effectively zero length and zero change in outer diameter. The transition region is the region where the enlarged distal portion **702** of retainer sheath **700** meets the end **802** of insertion sheath **800**. After the collapsed cannula assembly is inserted into the femoral artery, and the position of insertion sheath **800** is fixed at the patient boundary, the transition region will be located several centimeters past the femoral artery opening. At this point, the medical practitioner holds the exterior of insertion sheath **800** stationary and continues to push the retainer sheath assembly along guide wire **680**, until the distal end of the collapsed cannula and retainer sheath reside at the distal end of the guide wire, within the left ventricle cavity **900**.

A possible location of the device for LVAD use is shown in FIG. 26. The non-expandable portion **623** of cannula **625** extends from the descending aorta **905** and into the femoral artery, from which it exits the host body. Guide wire **680** may be advanced into the left ventricle **900** using an optional guiding catheter (not shown), and a fluoroscope may be used to establish proper positioning of the assist device.

Once blood pump **600** is properly positioned, retainer sheath **700** may be retracted, by a dimension of about 15 cm for some embodiments, allowing the expandable portion **626** of cannula **625** to expand to the deployed configuration.

The final step in deploying blood pump **600** involves pushing impeller **605** from its stored position within storage housing **660** and positioning it at a specified location within the expanded portion **626** of cannula **625**. This may be accomplished by applying a small force to drive shaft **630** while holding retainer sheath **700** at a fixed location. Retainer sheath **700**, in turn, holds the expanded portion **626** of cannula **625** at a previously fixed location. Once no longer restrained by storage housing **660**, impeller **605** expands to the deployed configuration due to the action of stored strain energy.

The expandable portion **626** of cannula **625** may attain its deployed configuration through the action of stored strain energy (potential energy). This process reveals the cannula inlet **642**, the expanded portion **626** of the cannula and the cannula outlet **644**. Successful operation requires that the cannula inlet **642** reside in the left ventricle **900** of the heart and that the outlet **644** reside in the aorta. A fluid seal must

exist where the cannula is proximate to the aortic valve, and the surface smoothness of the cannula is preferably such that clinically significant abrasion of the aortic valve is prevented. Also, distal end struts **652** form an inlet grid which prevents the inlet from becoming blocked by soft tissues within the left ventricle.

Impeller **605** may be moved toward the distal end of cannula **625** which curves around through a valve into the left ventricle **900** of the heart, while the flexible portion **634** of drive shaft **630**, coupled to impeller **605**, extends outside of the body of the patient and is rotated by a drive motor. The non-expanded portion **623** of cannula **625** similarly extends through the femoral artery and outside of the patient. The inlet **642** and struts **652** at the distal end of cannula portion **626** allow substantially unrestricted flow of blood into the device, where it is driven by impeller **605** outside of the device through a discharge mesh or struts **654** at the proximal end **648** of cannula portion **626**.

Other methods of expanding impeller **605** may be used. One possible alternative approach may include infusing a liquid or gas through a shaft to inflate impeller **605**. Another approach may use rotational forces to induce blades **612** to form a desired shape. The potential energy in the blades, particularly in the blade roots, may be used to deploy the blades into their unstrained position, and hydrodynamic forces may cause the blades to further deform into their operating configuration.

The retainer sheath **700**, previously retracted, serves to fix the position of the entire assembly within the patient. Blood pump **600** is now deployed and ready for connection to supporting equipment and use.

When the patient recovers and can be weaned from the necessity of using blood pump intervention, impeller **605** may be pulled back into an inactive compressed configuration in storage housing **660** or in a non-expandable portion **623** of cannula **625**, and the expandable portion **626** of the cannula may be pulled into retainer sheath **700**. Slots (not shown) or an outward flare **649** (see FIG. 9) may be provided at the distal end of storage housing **660** to assist in retraction of cannula **625**. Retainer sheath **700** may then be pulled into proximity with insertion sheath **800**, and the insertion sheath, retainer sheath and cannula within it may be removed from the patient through the original femoral artery site. Subsequently, the wound in the patient may be closed in a conventional fashion. The flow rate and pressure rise of blood pumped by blood pump **600** is greater than current non-collapsible devices of the same diameter, and the rate of blood damage (hemolysis) is maintained at a clinically acceptable low level. The use of the device as an RVAD is similar to that described above.

The expansion feature of blood pump **600** is an advantage over non-expandable prior art devices. If the device were non-expandable, the maximum cross-section would be limited to approximately 3 mm to allow for percutaneous insertion. However, this cross-section is insufficient to achieve sufficient blood flow to maintain the health of the patient.

Other applications of the device according to the present invention include providing additional blood flow to other organs, assisting the heart during operations, and the like.

The expandable portion **626** of cannula **625** may be expanded by any desired method. In one approach, mesh **631** may expand in a radial direction when the expandable portion **626** is contracted along an axial direction, for example using a mesh **631** having a hexagonal structure. In this approach, by applying tension to guide wire **680** through bull nose grommet **684**, shown in FIG. 21A, the expandable

27

portion 626 of cannula 625 can be shortened in the axial direction, providing radial expansion into the expanded state. Where a shape-memory material is used for the expandable portion of the cannula, the cannula will achieve its expanded state as the shape memory material reaches a predetermined temperature, such as when the cannula is inserted into a patient's blood vessel. Both impeller 605 and cannula 625 in their stored configurations may be held within retainer sheath 700, whereby both may be mechanically deployed or may self-deploy when removed from the retainer sheath.

Cannula 625 may have at least two configurations, including a stored configuration and an expanded (deployed) configuration. When used as part of a blood pump, cannula 625 in the deployed configuration may be about 20-30 cm long with a diameter of about 6-7 mm. In the stored configuration, cannula 625 may have a diameter of about 3 mm, allowing nonsurgical insertion of blood pump 600 into a human subject through a femoral artery. The larger deployed diameter allows for higher fluid flow rates after insertion, and reduced friction pressure losses compared with a non-surgically inserted blood pump having a non-expandable cannula.

An improved process for blood pumping within a living subject includes providing an expandable impeller, inserting the impeller into a patient in a stored configuration (for example, with a diameter of between about 3 mm and about 4 mm), positioning the impeller at a desired location within a blood vessel of the patient, deploying the impeller (for example, to a diameter of between about 6 mm and about 7 mm), and operating the impeller in an operating configuration at a Reynolds number of between about 1,000 and about 30,000, and preferably between about 2,000 and about 10,000. Higher Reynolds number operation and more efficient pump operation may be possible with higher rotation speeds, but may increase the destruction of structures within the pumped fluid, such as blood cells. The operating diameter of the impeller may be at least about 50% greater than the stored diameter. In other examples, the operating diameter may be at least about 100% greater than the stored diameter. For animals, components may be scaled according to the size of the animal.

Novel configurations and material choice allow the improved device to be compressed for cannula insertion into a patient. Representative devices include an expandable impeller, and a cannula that is at least in part expandable, inside of which the impeller rotates. Both the expandable impeller and the cannula have stored states that allow cannula insertion into a vein or artery using non-surgical methods. After insertion and location of the device, the expandable impeller and cannula expand into deployed states. The impeller can be driven through a flexible drive shaft from a drive motor external to the host, or using a motor proximate to the impeller, possibly integrated with a bearing system.

Other Pump Applications

Applications of the improved fluid pump designs described herein are not limited to ventricular assist devices. The improved cannula and impeller designs are useful for any application where a stored configuration having reduced diameter is useful for locating the pump at a desired location. For example, a fluid pump operating underground may be introduced into a pipe, channel, or cavity through an opening of lesser diameter, and operate at a diameter greater than that of the opening used. Applications of an impeller deploying within an expandable cannula include a collaps-

28

ible fire hose with an integral booster pump, a collapsible propeller, a biomedical pump for a biological fluid, and the like.

In other examples, impellers may also be formed from metal sheets, plastic and non-resilient materials, for example in foldable configurations. Deployment may include the use of motors or other mechanical devices to unfold blades, automatic deployment induced by centrifugal forces, and the like.

Although the invention herein has been described with reference understood that to particular embodiments, these embodiments are merely it is to be illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

What is claimed is:

1. A catheter blood pump comprising:
a cannula comprising:

a generally cylindrical, flexible, and expandable central portion having a proximal end, a distal end and a blood flow conduit extending between the proximal end and the distal end;

a flexible inlet portion at the distal end through which blood enters the blood flow conduit when the central portion is expanded, the flexible inlet portion provided with a plurality of spaced apart inlet struts configured to prevent an obstruction from entering the blood flow conduit; and

a flexible outlet portion at the proximal end through which blood leaves the blood flow conduit when the central portion is expanded, the flexible outlet portion provided with a plurality of spaced apart outlet struts operative to reduce a swirl velocity of blood leaving the blood flow conduit;

wherein the plurality of spaced apart inlet struts and the plurality of spaced apart outlet struts are fabricated from a collapsible shape memory material.

2. The catheter blood pump of claim 1, wherein the cylindrical, flexible, and expandable central portion defines a longitudinal axis between the proximal end and the distal end, and the plurality of spaced apart inlet struts and the plurality of spaced apart outlet struts respectively extend obliquely to the longitudinal axis.

3. The catheter blood pump of claim 2, wherein the cylindrical, flexible, and expandable central portion has a first longitudinal length, wherein the inlet portion has a second longitudinal length, wherein the outlet portion has a third longitudinal length, and where the second and third longitudinal length are each substantially smaller than the first axial length.

4. The catheter blood pump of claim 2, wherein the plurality of spaced apart inlet struts and the plurality of spaced apart outlet struts extend as flat linear elements arranged in a uniform circular disposition about the longitudinal axis.

5. The catheter blood pump of claim 1, wherein the generally cylindrical, flexible, and expandable central portion includes a nitinol tube, and wherein the plurality of spaced apart inlet struts are integrally formed with and extend from the nitinol tube.

6. The catheter blood pump of claim 1, wherein the generally cylindrical, flexible, and expandable central por-

29

tion includes a nitinol tube, the plurality of spaced apart outlet struts are integrally formed with and extend from the nitinol tube.

7. The catheter blood pump of claim 1, wherein the flexible, and expandable central portion is radially expandable from a reduced diameter to an expanded diameter defining the blood flow conduit.

8. The catheter blood pump of claim 7, wherein each of the reduced diameter and the expanded diameter extend uniformly between the proximal end and the distal end.

9. The catheter blood pump of claim 1, wherein the cannula further comprises a generally cylindrical, non-expandable portion attached to the generally cylindrical, flexible, and expandable central portion.

10. The catheter blood pump of claim 9, wherein no blood flows through the generally cylindrical, non-expandable portion in an operation of the pump.

11. The catheter blood pump of claim 9, wherein the outlet portion extends adjacent the generally cylindrical, non-expandable portion when the generally cylindrical, flexible, and expandable central portion is expanded.

12. The catheter blood pump of claim 1, further comprising an impeller disposed in the generally cylindrical, flexible, and expandable central portion of the cannula, the impeller operable to pump blood through the blood flow conduit between the inlet portion and the outlet portion.

13. The catheter blood pump of claim 12, wherein a combination of the impeller and cannula are sized to be inserted into a heart of a patient.

14. The catheter blood pump of claim 1, further comprising a retainer sheath operable to cause at least the plurality of spaced apart discharge struts to collapse.

15. A catheter blood pump comprising:

a cannula comprising:

a generally cylindrical, flexible, and expandable central portion having a first end, a second end opposing the first end, and a blood flow conduit extending between the first end and the second end;

a flexible inlet portion at the first end through which blood enters the blood flow conduit when the central portion is expanded, the flexible inlet portion pro-

30

vided with a plurality of separated inlet struts configured to prevent an obstruction from entering the blood flow conduit; and

a flexible outlet portion at the second end through which blood leaves the blood flow conduit when the central portion is expanded, the flexible outlet portion provided with a plurality of separated outlet struts operative to reduce a swirl velocity of blood leaving the blood flow conduit;

wherein the plurality of inlet struts and the plurality of outlet struts are fabricated from a collapsible shape memory material; and

an impeller disposed in the generally cylindrical, flexible, and expandable central portion of the cannula, the impeller operable to pump blood through the blood flow conduit between the inlet portion and the outlet portion.

16. The catheter blood pump of claim 15, wherein the generally cylindrical, flexible, and expandable central portion of the cannula includes a nitinol tube, and the plurality of inlet struts and the plurality of outlet struts are integrally formed with the nitinol tube.

17. The catheter blood pump of claim 15, wherein the cylindrical, flexible, and expandable central portion defines a longitudinal axis between the proximal end and the distal end, and the plurality of separated inlet struts and the plurality of separated outlet struts respectively extend obliquely to the longitudinal axis.

18. The catheter blood pump of claim 17, wherein the cylindrical, flexible, and expandable central portion has a first longitudinal length, wherein the inlet portion has a second longitudinal length, wherein the outlet portion has a third longitudinal length, and where the second and third longitudinal lengths are substantially smaller than the first longitudinal length.

19. The catheter blood pump of claim 17, wherein the plurality of inlet struts and the plurality of outlet struts extend as flat linear elements arranged in a uniform circular disposition about the longitudinal axis.

* * * * *